

MH1201: Linear Algebra II

Weeks 1–7 Lecture Notes

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Chapter 1

Algebraic Foundations: Fields and Vector Spaces

1.1 Fields

1.1.1 Definition of a Field

A field is the algebraic structure that provides the scalars for a vector space. The formal definition requires two binary operations satisfying nine axioms.

Definition 1.1.1 (Field). A *field* is a set $\mathbb{F}$ with at least two elements, equipped with two binary operations,

- $+\mathbb{F} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$, called *field addition for $\mathbb{F}$* , and
- $\times_{\mathbb{F}} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$, called *field multiplication for $\mathbb{F}$* ,

that satisfy the following nine properties:

1. **Associativity of field addition:** For all $a, b, c \in \mathbb{F}$,

$$(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c).$$

2. **Associativity of field multiplication:** For all $a, b, c \in \mathbb{F}$,

$$(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c).$$

3. **Commutativity of field addition:** For all $a, b \in \mathbb{F}$,

$$a +_{\mathbb{F}} b = b +_{\mathbb{F}} a.$$

4. **Commutativity of field multiplication:** For all $a, b \in \mathbb{F}$,

$$a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a.$$

5. **Left distributivity of field multiplication over field addition:** For all $a, b, c \in \mathbb{F}$,

$$a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = (a \times_{\mathbb{F}} b) +_{\mathbb{F}} (a \times_{\mathbb{F}} c).$$

6. **Existence of an additive identity:** $\mathbb{F}$ has an element $0_{\mathbb{F}}$, called the *additive identity*, such that for all $a \in \mathbb{F}$,

$$a +_{\mathbb{F}} 0_{\mathbb{F}} = a = 0_{\mathbb{F}} +_{\mathbb{F}} a.$$

The additive identity is unique: if $\tilde{0}$ also satisfies $a +_{\mathbb{F}} \tilde{0} = a$ for all a , then $\tilde{0} = \tilde{0} +_{\mathbb{F}} 0_{\mathbb{F}} = 0_{\mathbb{F}}$.

7. **Existence of additive inverses:** For every $a \in \mathbb{F}$, there exists $b \in \mathbb{F}$, called an *additive inverse of a* , such that

$$a +_{\mathbb{F}} b = 0_{\mathbb{F}} = b +_{\mathbb{F}} a.$$

The additive inverse of a is unique; it is denoted $-a$.

8. **Existence of a multiplicative identity:** $\mathbb{F}$ has an element $1_{\mathbb{F}}$, called the *multiplicative identity*, such that for all $a \in \mathbb{F}$,

$$a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 1_{\mathbb{F}} \times_{\mathbb{F}} a.$$

The multiplicative identity is unique.

9. **Existence of multiplicative inverses:** For every $a \in \mathbb{F} \setminus \{0_{\mathbb{F}}\}$, there exists $b \in \mathbb{F}$, called a *multiplicative inverse of a* , such that

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} = b \times_{\mathbb{F}} a.$$

The multiplicative inverse of a is unique; it is denoted a^{-1} .

Remark 1.1.2. When $\mathbb{F}$ is clear from context, we omit the subscript $\mathbb{F}$ and write $a + b$, ab , 0 , and 1 instead of $a +_{\mathbb{F}} b$, $a \times_{\mathbb{F}} b$, $0_{\mathbb{F}}$, and $1_{\mathbb{F}}$. Note that by the requirement that $\mathbb{F}$ has *at least two elements*, we have $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$.

1.1.2 Elementary Consequences of the Field Axioms

The following properties all follow directly from the nine axioms. They are used frequently in proofs.

Proposition 1.1.3. *Let $\mathbb{F}$ be a field. Then:*

1. *For all $a \in \mathbb{F}$, we have $-(-a) = a$.*
2. *For all $a \in \mathbb{F}$, we have $0_{\mathbb{F}} \times_{\mathbb{F}} a = 0_{\mathbb{F}} = a \times_{\mathbb{F}} 0_{\mathbb{F}}$.*
3. *For all $a \in \mathbb{F}$, we have $(-1_{\mathbb{F}}) \times_{\mathbb{F}} a = -a = a \times_{\mathbb{F}} (-1_{\mathbb{F}})$.*
4. *$(-1_{\mathbb{F}}) \times_{\mathbb{F}} (-1_{\mathbb{F}}) = 1_{\mathbb{F}}$.*
5. *$0_{\mathbb{F}} \neq 1_{\mathbb{F}}$.*

Proof. (1) Since $(-a) +_{\mathbb{F}} a = 0_{\mathbb{F}} = a +_{\mathbb{F}} (-a)$, the element a is the additive inverse of $-a$, so $-(-a) = a$.

(2) We have $0_{\mathbb{F}} \times_{\mathbb{F}} a = (0_{\mathbb{F}} +_{\mathbb{F}} 0_{\mathbb{F}}) \times_{\mathbb{F}} a = (0_{\mathbb{F}} \times_{\mathbb{F}} a) +_{\mathbb{F}} (0_{\mathbb{F}} \times_{\mathbb{F}} a)$. Adding $-(0_{\mathbb{F}} \times_{\mathbb{F}} a)$ to both sides gives $0_{\mathbb{F}} = 0_{\mathbb{F}} \times_{\mathbb{F}} a$.

(3) We compute $[(-1_{\mathbb{F}}) \times_{\mathbb{F}} a] +_{\mathbb{F}} a = [(-1_{\mathbb{F}}) +_{\mathbb{F}} 1_{\mathbb{F}}] \times_{\mathbb{F}} a = 0_{\mathbb{F}} \times_{\mathbb{F}} a = 0_{\mathbb{F}}$. So $(-1_{\mathbb{F}}) \times_{\mathbb{F}} a$ is the additive inverse of a , i.e., $(-1_{\mathbb{F}}) \times_{\mathbb{F}} a = -a$.

(4) By (3): $(-1_{\mathbb{F}}) \times_{\mathbb{F}} (-1_{\mathbb{F}}) = -(-1_{\mathbb{F}})$. By (1): $-(-1_{\mathbb{F}}) = 1_{\mathbb{F}}$.

(5) If $0_{\mathbb{F}} = 1_{\mathbb{F}}$, then for all $a \in \mathbb{F}$: $a = 1_{\mathbb{F}} \times_{\mathbb{F}} a = 0_{\mathbb{F}} \times_{\mathbb{F}} a = 0_{\mathbb{F}}$, so $\mathbb{F} = \{0_{\mathbb{F}}\}$, contradicting the requirement that $\mathbb{F}$ has at least two elements. $\square$

1.1.3 Field Subtraction and Division

Definition 1.1.4 (Field subtraction). *Field subtraction* for $\mathbb{F}$ is the binary operation $-_{\mathbb{F}} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$ defined by

$$\forall a, b \in \mathbb{F} : \quad a -_{\mathbb{F}} b \stackrel{\text{df}}{=} a +_{\mathbb{F}} (-b).$$

Definition 1.1.5 (Field division). *Field division* for $\mathbb{F}$ is the binary operation $/_{\mathbb{F}} : \mathbb{F} \times (\mathbb{F} \setminus \{0_{\mathbb{F}}\}) \rightarrow \mathbb{F}$ defined by

$$\forall a \in \mathbb{F}, \forall b \in \mathbb{F} \setminus \{0_{\mathbb{F}}\} : \quad a /_{\mathbb{F}} b \stackrel{\text{df}}{=} a \times_{\mathbb{F}} b^{-1}.$$

1.1.4 Examples of Fields

Example 1.1.6 (Infinite fields). The following are the most commonly encountered fields:

- The field $\mathbb{Q}$ of rational numbers (countably infinite).
- The field $\mathbb{R}$ of real numbers (uncountably infinite).
- The field $\mathbb{C}$ of complex numbers (uncountably infinite).

These are all *infinite fields* since they have infinitely many elements.

Example 1.1.7 (A finite field: $\mathbb{F}_2$). The field $\mathbb{F}_2$ has precisely two elements $0_{\mathbb{F}_2}$ and $1_{\mathbb{F}_2}$. Its addition and multiplication tables are:

$+_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$	$1_{\mathbb{F}_2}$
$0_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$	$1_{\mathbb{F}_2}$
$1_{\mathbb{F}_2}$	$1_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$

$\times_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$	$1_{\mathbb{F}_2}$
$0_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$
$1_{\mathbb{F}_2}$	$0_{\mathbb{F}_2}$	$1_{\mathbb{F}_2}$

Finite fields form the foundation of coding theory.

Theorem 1.1.8 (Size of a finite field). *Let $n \in \mathbb{N}$. There exists a field with exactly n elements if and only if n is a power of a prime number.*

Remark 1.1.9. This theorem (not assessed) tells us, for instance, that a field with $3^4 = 81$ elements exists, but no field with exactly 15 elements exists (since $15 = 3 \times 5$ is not a prime power).

Key Result: Field Axioms Summary

A field $\mathbb{F}$ has **9 axioms**: associativity and commutativity of both operations (4), left distributivity (1), existence of additive and multiplicative identities (2), existence of additive inverses for all elements (1), and existence of multiplicative inverses for all nonzero elements (1). Always recall that $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$.

1.2 Vector Spaces

1.2.1 Definition of a Vector Space

Definition 1.2.1 (Vector space). A *vector space over a field $\mathbb{F}$* is a set V equipped with two binary operations,

- $+_V : V \times V \rightarrow V$, called *vector addition for V* , and
- $\cdot_V : \mathbb{F} \times V \rightarrow V$, called *scalar multiplication for V* ,

that satisfy the following eight properties:

1. **Associativity of vector addition:** For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$,

$$(\mathbf{u} +_V \mathbf{v}) +_V \mathbf{w} = \mathbf{u} +_V (\mathbf{v} +_V \mathbf{w}).$$

2. **Commutativity of vector addition:** For all $\mathbf{u}, \mathbf{v} \in V$,

$$\mathbf{u} +_V \mathbf{v} = \mathbf{v} +_V \mathbf{u}.$$

3. **Left distributivity of scalar multiplication over vector addition:** For all $c \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$,

$$c \cdot_V (\mathbf{u} +_V \mathbf{v}) = (c \cdot_V \mathbf{u}) +_V (c \cdot_V \mathbf{v}).$$

4. **Right distributivity of scalar multiplication over field addition:** For all $c, d \in \mathbb{F}$ and $\mathbf{u} \in V$,

$$(c +_{\mathbb{F}} d) \cdot_V \mathbf{u} = (c \cdot_V \mathbf{u}) +_V (d \cdot_V \mathbf{u}).$$

5. **Compatibility of scalar multiplication with field multiplication:** For all $c, d \in \mathbb{F}$ and $\mathbf{u} \in V$,

$$c \cdot_V (d \cdot_V \mathbf{u}) = (c \times_{\mathbb{F}} d) \cdot_V \mathbf{u}.$$

6. **Existence of an identity for scalar multiplication:** For all $\mathbf{u} \in V$,

$$1_{\mathbb{F}} \cdot_V \mathbf{u} = \mathbf{u}.$$

7. **Existence of an identity for vector addition:** V has an element $\mathbf{0}_V$, called the *additive identity*, such that for all $\mathbf{u} \in V$,

$$\mathbf{u} +_V \mathbf{0}_V = \mathbf{u} = \mathbf{0}_V +_V \mathbf{u}.$$

The additive identity is unique.

8. **Existence of inverses for vector addition:** For every $\mathbf{u} \in V$, there exists $\mathbf{v} \in V$ such that

$$\mathbf{u} +_V \mathbf{v} = \mathbf{0}_V = \mathbf{v} +_V \mathbf{u}.$$

The additive inverse of $\mathbf{u}$ is unique; it is denoted $-\mathbf{u}$.

Remark 1.2.2. When V (and $\mathbb{F}$) are clear from context, we drop subscripts and write $\mathbf{u} + \mathbf{v}$, $c \cdot \mathbf{u}$ (or simply $c\mathbf{u}$), and $\mathbf{0}$. In particular, the compatibility axiom becomes $c \cdot (d \cdot \mathbf{u}) = cd \cdot \mathbf{u}$.

Definition 1.2.3 (Vector, rational/real/complex vector space). An element of a vector space V is called a V -*vector* or simply a *vector*. A vector space over $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ is called a *rational*, *real*, or *complex vector space*, respectively. In MH1201, we mostly study real and complex vector spaces.

1.2.2 Elementary Consequences of the Vector Space Axioms

Proposition 1.2.4. *Let V be a vector space over a field $\mathbb{F}$. Then:*

1. *For all $\mathbf{u} \in V$: $-(-\mathbf{u}) = \mathbf{u}$.*
2. *For all $\mathbf{u} \in V$: $0_{\mathbb{F}} \cdot_V \mathbf{u} = \mathbf{0}_V$.*
3. *For all $\mathbf{u} \in V$: $(-1_{\mathbb{F}}) \cdot_V \mathbf{u} = -\mathbf{u}$.*
4. *For all $c \in \mathbb{F}$: $c \cdot_V \mathbf{0}_V = \mathbf{0}_V$.*

Proof. (1) Since $(-\mathbf{u}) +_V \mathbf{u} = \mathbf{0}_V$, $\mathbf{u}$ is the additive inverse of $-\mathbf{u}$, giving $-(-\mathbf{u}) = \mathbf{u}$.

(2) We compute $0_{\mathbb{F}} \cdot_V \mathbf{u} = (0_{\mathbb{F}} +_{\mathbb{F}} 0_{\mathbb{F}}) \cdot_V \mathbf{u} = (0_{\mathbb{F}} \cdot_V \mathbf{u}) +_V (0_{\mathbb{F}} \cdot_V \mathbf{u})$. Adding $-(0_{\mathbb{F}} \cdot_V \mathbf{u})$ to both sides gives $\mathbf{0}_V = 0_{\mathbb{F}} \cdot_V \mathbf{u}$.

(3) We compute $[(-1_{\mathbb{F}}) \cdot_V \mathbf{u}] +_V \mathbf{u} = [(-1_{\mathbb{F}}) +_{\mathbb{F}} 1_{\mathbb{F}}] \cdot_V \mathbf{u} = 0_{\mathbb{F}} \cdot_V \mathbf{u} = \mathbf{0}_V$ by (2). So $(-1_{\mathbb{F}}) \cdot_V \mathbf{u} = -\mathbf{u}$.

(4) We compute $c \cdot_V \mathbf{0}_V = c \cdot_V (0_{\mathbb{F}} \cdot_V \mathbf{0}_V) = (c \times_{\mathbb{F}} 0_{\mathbb{F}}) \cdot_V \mathbf{0}_V = 0_{\mathbb{F}} \cdot_V \mathbf{0}_V = \mathbf{0}_V$ by (2). $\square$

1.2.3 Vector Subtraction

Definition 1.2.5 (Vector subtraction). *Vector subtraction* for V is the binary operation $-_V : V \times V \rightarrow V$ defined by

$$\forall \mathbf{u}, \mathbf{v} \in V : \quad \mathbf{u} -_V \mathbf{v} \stackrel{\text{df}}{=} \mathbf{u} +_V (-\mathbf{v}).$$

1.2.4 Important Examples of Vector Spaces

Example 1.2.6 (A field is a vector space over itself). Let $\mathbb{F}$ be a field. Define $+$: $\mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$ by $+$ and $\cdot$: $\mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$ by $\times_{\mathbb{F}}$. Then $\mathbb{F}$ equipped with $+$ and $\cdot$ is a vector space over itself, since the field axioms directly imply all eight vector space axioms.

Example 1.2.7 (Finite sequences: $\mathbb{F}^n$). Let $\mathbb{F}$ be a field and $n \in \mathbb{N}$. Let $\mathbb{F}^n$ denote the set of all sequences of length n with entries in $\mathbb{F}$. Define *entry-wise addition* and *entry-wise scalar multiplication* by

$$(a_1, \dots, a_n) + (b_1, \dots, b_n) \stackrel{\text{df}}{=} (a_1 + b_1, \dots, a_n + b_n),$$

$$c \cdot (a_1, \dots, a_n) \stackrel{\text{df}}{=} (ca_1, \dots, ca_n).$$

Then $\mathbb{F}^n$ equipped with $+$ and $\cdot$ is a vector space over $\mathbb{F}$.

Example 1.2.8 (Matrices: $\mathbb{F}^{m,n}$). Let $\mathbb{F}$ be a field and $m, n \in \mathbb{N}$. Let $\mathbb{F}^{m,n}$ denote the set of all m -by- n matrices with entries in $\mathbb{F}$. Define *matrix addition* ($+$) and *matrix scalar multiplication* ($\cdot$) entry-wise:

$$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix} + \begin{bmatrix} b_{1,1} & \cdots & b_{1,n} \\ \vdots & \ddots & \vdots \\ b_{m,1} & \cdots & b_{m,n} \end{bmatrix} \stackrel{\text{df}}{=} \begin{bmatrix} a_{1,1} + b_{1,1} & \cdots & a_{1,n} + b_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} + b_{m,1} & \cdots & a_{m,n} + b_{m,n} \end{bmatrix},$$

$$c \cdot \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix} \stackrel{\text{df}}{=} \begin{bmatrix} ca_{1,1} & \cdots & ca_{1,n} \\ \vdots & \ddots & \vdots \\ ca_{m,1} & \cdots & ca_{m,n} \end{bmatrix}.$$

Then $\mathbb{F}^{m,n}$ is a vector space over $\mathbb{F}$.

Example 1.2.9 (Degree-bounded polynomials: $\mathcal{P}_n(\mathbb{F})$). Let $\mathbb{F}$ be a field and $n \in \mathbb{N}$. Let $\mathcal{P}_n(\mathbb{F})$ denote the set of polynomials with coefficients in $\mathbb{F}$ of degree $\leq n$. Define polynomial addition and scalar multiplication by

$$\sum_{k=0}^n a_k X^k + \sum_{k=0}^n b_k X^k \stackrel{\text{df}}{=} \sum_{k=0}^n (a_k + b_k) X^k, \quad c \cdot \sum_{k=0}^n a_k X^k \stackrel{\text{df}}{=} \sum_{k=0}^n ca_k X^k.$$

Then $\mathcal{P}_n(\mathbb{F})$ is a vector space over $\mathbb{F}$.

Example 1.2.10 (All polynomials: $\mathcal{P}(\mathbb{F})$). Let $\mathbb{F}$ be a field. Let $\mathcal{P}(\mathbb{F})$ denote the set of all polynomials with coefficients in $\mathbb{F}$ (the degree is finite but may be arbitrarily large). Using the same operations as in Example 1.2.9 but with sums running to ∞ (noting that each polynomial has only finitely many nonzero coefficients), $\mathcal{P}(\mathbb{F})$ is a vector space over $\mathbb{F}$.

Example 1.2.11 (Function spaces: $\mathbb{F}^S$). Let S be a set and $\mathbb{F}$ a field. Let $\mathbb{F}^S$ denote the set of all functions $f : S \rightarrow \mathbb{F}$. Define *pointwise addition* and *pointwise scalar multiplication* by

$$\forall s \in S : \quad (f + g)(s) \stackrel{\text{df}}{=} f(s) + g(s), \quad (c \cdot f)(s) \stackrel{\text{df}}{=} c \cdot f(s).$$

Then $\mathbb{F}^S$ is a vector space over $\mathbb{F}$. This is an abstract vector space whose vectors are functions rather than tuples.

Key Result: Recognising Vector Spaces

To verify that a set V is a vector space over $\mathbb{F}$, one must specify the operations $+_V$ and $\cdot_V$ explicitly, then check all 8 axioms. The most important and commonly used examples are $\mathbb{F}^n$, $\mathbb{F}^{m,n}$, $\mathcal{P}_n(\mathbb{F})$, and $\mathbb{F}^S$.

1.3 Vector Subspaces

1.3.1 Definition of a Vector Subspace

Informally, a vector subspace is a vector space sitting inside an existing vector space.

Definition 1.3.1 (Vector subspace). Let V be a vector space over a field $\mathbb{F}$. A *vector subspace* of V is a subset W of V for which there exist two binary operations $+_W : W \times W \rightarrow W$ and $\cdot_W : \mathbb{F} \times W \rightarrow W$ satisfying:

1. W equipped with $+_W$ and $\cdot_W$ is a vector space over $\mathbb{F}$ (consequently $W \neq \emptyset$).
2. For all $\mathbf{u}, \mathbf{v} \in W$: $\mathbf{u} +_W \mathbf{v} = \mathbf{u} +_V \mathbf{v}$.
3. For all $c \in \mathbb{F}$ and $\mathbf{u} \in W$: $c \cdot_W \mathbf{u} = c \cdot_V \mathbf{u}$.

1.3.2 The Subspace Test

The definition requires checking eight axioms for W , which is cumbersome. The Subspace Test reduces this to just three conditions.

Theorem 1.3.2 (The Subspace Test). *A subset W of a vector space V is a vector subspace of V if and only if the following three statements hold:*

1. W is **non-empty**.
2. W is **closed under vector addition for V** : for all $\mathbf{u}, \mathbf{v} \in W$, we have $\mathbf{u} +_V \mathbf{v} \in W$.
3. W is **closed under scalar multiplication for V** : for all $c \in \mathbb{F}$ and $\mathbf{u} \in W$, we have $c \cdot_V \mathbf{u} \in W$.

Key Result: The Subspace Test

To show $W \subseteq V$ is a subspace, verify: (1) $W \neq \emptyset$, (2) $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$, (3) $c \in \mathbb{F}, \mathbf{u} \in W \Rightarrow c\mathbf{u} \in W$.

Theorem 1.3.3 (Additive identity of a subspace). *If W is a vector subspace of V , then $\mathbf{0}_W = \mathbf{0}_V$.*

Corollary 1.3.4 (The Non-Subspace Test). *Let V be a vector space, and let $W \subseteq V$. If $\mathbf{0}_V \notin W$, then W is **not** a vector subspace of V .*

Warning: Non-Subspace Test is One-Directional

The Non-Subspace Test (Corollary 1.3.4) can only be used to show a subset is *not* a subspace. You **cannot** conclude that a subset *is* a subspace merely because $\mathbf{0}_V \in W$; you must apply the full Subspace Test.

1.3.3 Examples of Subspaces

Example 1.3.5 (Trivial subspaces). Let V be a vector space. Then $\{\mathbf{0}_V\}$ and V are both vector subspaces of V (verified immediately via the Subspace Test).

Example 1.3.6 ($\mathbb{Q}$ is not a subspace of $\mathbb{R}$). $\mathbb{Q}$ is *not* a vector subspace of $\mathbb{R}$ (viewed as a vector space over $\mathbb{R}$) because it is not closed under scalar multiplication by the base field $\mathbb{R}$. For instance $1 \in \mathbb{Q}$, but $\sqrt{2} \cdot 1 = \sqrt{2} \notin \mathbb{Q}$.

1.3.4 Worked Examples Using the Subspace Test

Example 1.3.7. Determine whether $A = \{(a, b, c) \in \mathbb{R}^3 \mid a + 2b + 3c = 0\}$ is a vector subspace of $\mathbb{R}^3$.

Solution. We apply the Subspace Test.

1. **Non-empty:** $(0, 0, 0) \in A$ since $0 + 0 + 0 = 0$, so $A \neq \emptyset$.
2. **Closed under addition:** Let $\mathbf{v} = (a, b, c)$ and $\tilde{\mathbf{v}} = (\tilde{a}, \tilde{b}, \tilde{c})$ be arbitrary elements of A . Then

$$(a + \tilde{a}) + 2(b + \tilde{b}) + 3(c + \tilde{c}) = (a + 2b + 3c) + (\tilde{a} + 2\tilde{b} + 3\tilde{c}) = 0 + 0 = 0,$$

so $\mathbf{v} + \tilde{\mathbf{v}} \in A$.

3. **Closed under scalar multiplication:** Let $\lambda \in \mathbb{R}$ and $\mathbf{v} = (a, b, c) \in A$. Then

$$\lambda a + 2(\lambda b) + 3(\lambda c) = \lambda(a + 2b + 3c) = \lambda \cdot 0 = 0,$$

so $\lambda \mathbf{v} \in A$.

Since all three conditions hold, A is a vector subspace of $\mathbb{R}^3$. $\square$

Example 1.3.8. Determine whether $A = \{(a, b, c) \in \mathbb{R}^3 \mid a + 2b + 3c = 4\}$ is a vector subspace of $\mathbb{R}^3$.

Solution. Note that $\mathbf{0}_{\mathbb{R}^3} = (0, 0, 0)$ satisfies $0 + 0 + 0 = 0 \neq 4$, so $\mathbf{0}_{\mathbb{R}^3} \notin A$. By the Non-Subspace Test, A is **not** a vector subspace of $\mathbb{R}^3$. $\square$

Example 1.3.9. Determine whether $A = \{(a, b, c) \in \mathbb{R}^3 \mid abc = 0\}$ is a vector subspace of $\mathbb{R}^3$.

Solution. Note $\mathbf{0} \in A$, so the Non-Subspace Test does not apply. However, $(1, 1, 0) \in A$ and $(0, 0, 1) \in A$, but

$$(1, 1, 0) + (0, 0, 1) = (1, 1, 1) \notin A$$

since $1 \cdot 1 \cdot 1 = 1 \neq 0$. Therefore A is **not** closed under vector addition and is hence **not** a vector subspace of $\mathbb{R}^3$. (Note: A is closed under scalar multiplication, illustrating that *both* closure conditions must hold.) $\square$

1.3.5 Affine Subspaces and When They Are Vector Subspaces

Theorem 1.3.10 (Affine subspace criterion). Let $n \in \mathbb{N}$, let V be a vector space over $\mathbb{F}$, and let $\mathbf{a}_1, \dots, \mathbf{a}_n, \mathbf{b} \in V$. Then

$$W = \{\lambda_1 \cdot \mathbf{a}_1 + \dots + \lambda_n \cdot \mathbf{a}_n + \mathbf{b} \in V \mid \lambda_1, \dots, \lambda_n \in \mathbb{F}\}$$

(called an affine subspace) is a vector subspace of V if and only if $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Remark 1.3.11. Linear combinations will be formally defined in Week 2. The theorem is proved by showing: if W is a subspace then $\mathbf{0}_V \in W$, which forces $\mathbf{b} = (-\lambda_1)\mathbf{a}_1 + \cdots + (-\lambda_n)\mathbf{a}_n$ for some scalars; and conversely, if $\mathbf{b}$ is already a linear combination of the $\mathbf{a}_i$, one verifies closure under addition and scalar multiplication directly.

Example 1.3.12. Determine whether

$$W = \left\{ \lambda \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \mathbb{R}^{3,1} \mid \lambda, \mu \in \mathbb{R} \right\}$$

is a vector subspace of $\mathbb{R}^{3,1}$.

Solution. By Theorem 1.3.10, it suffices to check whether $\mathbf{b} = [0, 1, 0]^\top$ is a linear combination of $\mathbf{a}_1 = [1, 0, -1]^\top$ and $\mathbf{a}_2 = [1, 2, 3]^\top$. Row-reducing the augmented matrix:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ -1 & 3 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 4 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}.$$

The system is **inconsistent** (the last row gives $0 = -2$), so $\mathbf{b}$ is *not* a linear combination of $\mathbf{a}_1, \mathbf{a}_2$. Therefore W is **not** a vector subspace of $\mathbb{R}^{3,1}$. $\square$

Exam Tip: Strategy for Subspace Problems

- **To show a subset IS a subspace:** use the Subspace Test (3 conditions).
- **To show a subset is NOT a subspace:** find a single counterexample violating non-emptiness, closure under addition, or closure under scalar multiplication. The Non-Subspace Test ($\mathbf{0}_V \notin W$) is the quickest check, but finding a pair $\mathbf{u}, \mathbf{v} \in W$ with $\mathbf{u} + \mathbf{v} \notin W$ also works.
- For affine subspaces of the form $\lambda_1 \mathbf{a}_1 + \cdots + \lambda_n \mathbf{a}_n + \mathbf{b}$, apply Theorem 1.3.10 and row-reduce.

Chapter 2

Subspaces, Sums, and Span

2.1 Vector Subspaces: Further Examples

Building on the Subspace Test and the Affine Subspace Criterion from Week 1 (Theorem 1.3.10), we work through two further examples.

Example 2.1.1. Determine whether

$$W = \left\{ \lambda \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + \mu \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \in \mathbb{R}^{3,1} \mid \lambda, \mu \in \mathbb{R} \right\}$$

is a vector subspace of $\mathbb{R}^{3,1}$.

Solution. By Theorem 1.3.10 it suffices to check whether $\mathbf{b} = [1, 2, 4]^\top$ is a linear combination of $\mathbf{a}_1 = [2, -1, 2]^\top$ and $\mathbf{a}_2 = [-3, 4, 0]^\top$. We row-reduce the augmented matrix $[\mathbf{a}_1 \mid \mathbf{a}_2 \mid \mathbf{b}]$:

$$\begin{aligned} & \begin{bmatrix} 2 & -3 & 1 \\ -1 & 4 & 2 \\ 2 & 0 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 0 & 4 \\ -1 & 4 & 2 \\ 2 & -3 & 1 \end{bmatrix} \xrightarrow{R_1 \mapsto \frac{1}{2}R_1} \begin{bmatrix} 1 & 0 & 2 \\ -1 & 4 & 2 \\ 2 & -3 & 1 \end{bmatrix} \\ & \xrightarrow{\substack{R_2 \mapsto R_2 + R_1 \\ R_3 \mapsto R_3 - 2R_1}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 4 & 4 \\ 0 & -3 & -3 \end{bmatrix} \xrightarrow{R_2 \mapsto \frac{1}{4}R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & -3 & -3 \end{bmatrix} \xrightarrow{R_3 \mapsto R_3 + 3R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

The system is **consistent**, so $\mathbf{b}$ is indeed a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$. Therefore W is a **vector subspace** of $\mathbb{R}^{3,1}$. $\square$

Example 2.1.2. Determine whether $A = \{P \in \mathcal{P}(\mathbb{R}) \mid P'' + 7P' - 3P = 0\}$ is a vector subspace of $\mathcal{P}(\mathbb{R})$.

Solution. We apply the Subspace Test.

1. **Non-empty:** The zero polynomial satisfies $0'' + 7 \cdot 0' - 3 \cdot 0 = 0$, so $0 \in A$ and $A \neq \emptyset$.

2. **Closed under addition:** Let $P, Q \in A$ be arbitrary. Then

$$(P + Q)'' + 7(P + Q)' - 3(P + Q) = (P'' + 7P' - 3P) + (Q'' + 7Q' - 3Q) = 0 + 0 = 0,$$

so $P + Q \in A$.

3. **Closed under scalar multiplication:** Let $\lambda \in \mathbb{R}$ and $P \in A$. Then

$$(\lambda P)'' + 7(\lambda P)' - 3(\lambda P) = \lambda P'' + 7\lambda P' - 3\lambda P = \lambda(P'' + 7P' - 3P) = \lambda \cdot 0 = 0,$$

so $\lambda P \in A$.

Since all three conditions hold, A is a **vector subspace** of $\mathcal{P}(\mathbb{R})$. $\square$

Exam Tip: Polynomial and Function Subspaces

For subsets of $\mathcal{P}(\mathbb{R})$ or function spaces defined by a linear equation (e.g. $P'' + 7P' - 3P = 0$), the Subspace Test is usually the cleanest approach. The key observation is that differentiation is linear, so a linear ODE always defines a subspace.

2.2 Sums of Vector Subspaces

2.2.1 Definition

Definition 2.2.1 (Sum of vector subspaces). Let $n \in \mathbb{N}$, let V be a vector space, and let $V_1, \dots, V_n$ be vector subspaces of V (not necessarily distinct). The *sum* of $V_1, \dots, V_n$, denoted $V_1 + \dots + V_n$, is defined by

$$V_1 + \dots + V_n \stackrel{\text{df}}{=} \{\mathbf{v}_1 + \dots + \mathbf{v}_n \in V \mid \mathbf{v}_1 \in V_1, \dots, \mathbf{v}_n \in V_n\}.$$

Remark 2.2.2. The subspaces $V_1, \dots, V_n$ need not be distinct; repetition is allowed.

2.2.2 The Sum of Subspaces is a Subspace

Theorem 2.2.3. Let V be a vector space and let $V_1, \dots, V_n$ be vector subspaces of V . Then $V_1 + \dots + V_n$ is also a vector subspace of V .

Proof. We apply the Subspace Test.

1. **Non-empty:** Since each V_i is a subspace, $\mathbf{0}_V \in V_i$ for all i . Hence $\mathbf{0}_V = \mathbf{0}_V + \dots + \mathbf{0}_V \in V_1 + \dots + V_n$.
2. **Closed under addition:** Let $\mathbf{v}, \tilde{\mathbf{v}} \in V_1 + \dots + V_n$ be arbitrary. Then there exist $\mathbf{v}_i, \tilde{\mathbf{v}}_i \in V_i$ such that $\mathbf{v} = \mathbf{v}_1 + \dots + \mathbf{v}_n$ and $\tilde{\mathbf{v}} = \tilde{\mathbf{v}}_1 + \dots + \tilde{\mathbf{v}}_n$. Thus

$$\mathbf{v} + \tilde{\mathbf{v}} = (\mathbf{v}_1 + \tilde{\mathbf{v}}_1) + \dots + (\mathbf{v}_n + \tilde{\mathbf{v}}_n) \in V_1 + \dots + V_n,$$

since $\mathbf{v}_i + \tilde{\mathbf{v}}_i \in V_i$ (each V_i is closed under addition).

3. **Closed under scalar multiplication:** Let $\lambda \in \mathbb{F}$ and $\mathbf{v} \in V_1 + \dots + V_n$, so $\mathbf{v} = \mathbf{v}_1 + \dots + \mathbf{v}_n$ with $\mathbf{v}_i \in V_i$. Then

$$\lambda \cdot \mathbf{v} = \lambda \cdot \mathbf{v}_1 + \dots + \lambda \cdot \mathbf{v}_n \in V_1 + \dots + V_n,$$

since $\lambda \cdot \mathbf{v}_i \in V_i$ for each i .

Therefore $V_1 + \dots + V_n$ is a vector subspace of V . □

2.2.3 Worked Examples

Example 2.2.4. Define $U \stackrel{\text{df}}{=} \{(x, 0, 0) \in \mathbb{R}^3 \mid x \in \mathbb{R}\}$ and $V \stackrel{\text{df}}{=} \{(0, y, 0) \in \mathbb{R}^3 \mid y \in \mathbb{R}\}$. Give a simplified description of $U + V$.

Solution.

$$\begin{aligned} U + V &= \{(x, 0, 0) + (0, y, 0) \in \mathbb{R}^3 \mid x, y \in \mathbb{R}\} \\ &= \{(x, y, 0) \in \mathbb{R}^3 \mid x, y \in \mathbb{R}\} \\ &= \{x \cdot (1, 0, 0) + y \cdot (0, 1, 0) \in \mathbb{R}^3 \mid x, y \in \mathbb{R}\}. \quad \square \end{aligned}$$

Example 2.2.5. Define $U \stackrel{\text{df}}{=} \{(x, x, y, y) \in \mathbb{R}^4 \mid x, y \in \mathbb{R}\}$ and $V \stackrel{\text{df}}{=} \{(x, x, x, y) \in \mathbb{R}^4 \mid x, y \in \mathbb{R}\}$. Give a simplified description of $U + V$.

Solution. To avoid confusion between the parameters of U and V , rewrite $U = \{(r, r, s, s) \in \mathbb{R}^4 \mid r, s \in \mathbb{R}\}$. Then

$$\begin{aligned} U + V &= \{(r, r, s, s) + (x, x, x, y) \in \mathbb{R}^4 \mid r, s, x, y \in \mathbb{R}\} \\ &= \{(r + x, r + x, s + x, s + y) \in \mathbb{R}^4 \mid r, s, x, y \in \mathbb{R}\}. \end{aligned}$$

To identify this more cleanly, we form the matrix whose columns are $(1, 1, 0, 0)$, $(0, 0, 1, 1)$, $(1, 1, 1, 0)$, $(0, 0, 0, 1)$ and row-reduce to find the number of independent spanning vectors:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are 3 pivot columns, so

$$U + V = \{(r + x, r + x, s + x, s) \in \mathbb{R}^4 \mid r, s, x \in \mathbb{R}\}. \quad \square$$

2.3 Direct Sums

2.3.1 Definition

Definition 2.3.1 (Direct sum of vector subspaces). Let $n \in \mathbb{N}$, let V be a vector space, let $V_1, \dots, V_n$ be vector subspaces of V , and let W be another vector subspace of V . Then W is said to be the *direct sum* of $V_1, \dots, V_n$, written

$$W = V_1 \oplus \cdots \oplus V_n,$$

if and only if the following two conditions both hold:

1. $W = V_1 + \cdots + V_n$.
2. Every $\mathbf{w} \in W$ can be written in **precisely one** way as $\mathbf{v}_1 + \cdots + \mathbf{v}_n$ with $\mathbf{v}_i \in V_i$.

Warning: Sum $\neq$ Direct Sum in General

It is **generally false** that $V_1 + \cdots + V_n = V_1 \oplus \cdots \oplus V_n$. The direct sum imposes an additional uniqueness condition on representations.

2.3.2 Characterising Direct Sums

The following two theorems give practical criteria for deciding when a sum is direct.

Theorem 2.3.2 (Necessary and sufficient condition). *Let $V_1, \dots, V_n$ be vector subspaces of a vector space V . Then*

$$V_1 + \cdots + V_n = V_1 \oplus \cdots \oplus V_n$$

if and only if whenever $\mathbf{v}_1 \in V_1, \dots, \mathbf{v}_n \in V_n$ satisfy

$$\mathbf{0} = \mathbf{v}_1 + \cdots + \mathbf{v}_n,$$

we must have $\mathbf{v}_1 = \cdots = \mathbf{v}_n = \mathbf{0}$.

Theorem 2.3.3 (The Direct-Sum Criterion). *Let $n \in \mathbb{N}$, let V be a vector space, let $V_1, \dots, V_n$ be vector subspaces of V , and let W be another vector subspace of V . Then $W = V_1 \oplus \cdots \oplus V_n$ if and only if the following two conditions both hold:*

1. $W = V_1 + \cdots + V_n$.
2. For every $i \in \{1, \dots, n\}$,

$$V_i \cap (V_1 + \cdots + V_{i-1} + V_{i+1} + \cdots + V_n) = \{\mathbf{0}\}.$$

In particular, for $n = 2$: $V_1 + V_2 = V_1 \oplus V_2$ if and only if $V_1 \cap V_2 = \{\mathbf{0}_V\}$.

Proof. (Forward direction.) Suppose $W = V_1 \oplus \cdots \oplus V_n$. By definition, $W = V_1 + \cdots + V_n$ holds. Fix $i \in \{1, \dots, n\}$ and let $\mathbf{v} \in V_i \cap (V_1 + \cdots + V_{i-1} + V_{i+1} + \cdots + V_n)$. Then there exist $\mathbf{v}_j \in V_j$ (for $j \neq i$) such that $\mathbf{v} = \mathbf{v}_1 + \cdots + \mathbf{v}_{i-1} + \mathbf{v}_{i+1} + \cdots + \mathbf{v}_n$. Rearranging,

$$\mathbf{0} = \mathbf{v}_1 + \cdots + \mathbf{v}_{i-1} + (-\mathbf{v}) + \mathbf{v}_{i+1} + \cdots + \mathbf{v}_n,$$

where $-\mathbf{v} \in V_i$. Since $W = V_1 \oplus \cdots \oplus V_n$, by Theorem 2.3.2 all components must be $\mathbf{0}$, so in particular $\mathbf{v} = \mathbf{0}$.

(Backward direction.) Suppose conditions (1) and (2) hold. Let $\mathbf{v} \in V_1 + \cdots + V_n$ and suppose $\mathbf{v} = \mathbf{v}_1 + \cdots + \mathbf{v}_n = \tilde{\mathbf{v}}_1 + \cdots + \tilde{\mathbf{v}}_n$ are two representations. For each i ,

$$\mathbf{v}_i - \tilde{\mathbf{v}}_i = (\tilde{\mathbf{v}}_1 - \mathbf{v}_1) + \cdots + (\tilde{\mathbf{v}}_{i-1} - \mathbf{v}_{i-1}) + (\tilde{\mathbf{v}}_{i+1} - \mathbf{v}_{i+1}) + \cdots + (\tilde{\mathbf{v}}_n - \mathbf{v}_n).$$

The left side lies in V_i and the right side lies in $V_1 + \cdots + V_{i-1} + V_{i+1} + \cdots + V_n$, so by condition (2), $\mathbf{v}_i - \tilde{\mathbf{v}}_i = \mathbf{0}$, i.e. $\mathbf{v}_i = \tilde{\mathbf{v}}_i$. Hence the representation is unique. $\square$

Key Result: The Direct-Sum Criterion

For two subspaces: $V_1 + V_2 = V_1 \oplus V_2 \iff V_1 \cap V_2 = \{\mathbf{0}\}$. For three subspaces the conditions $V_i \cap (V_j + V_k) = \{\mathbf{0}\}$ for all distinct i, j, k must all hold simultaneously.

2.3.3 Worked Example

Example 2.3.4. Define subspaces of $\mathbb{R}^3$ by

$$U \stackrel{\text{df}}{=} \{(x, y, 0) \mid x, y \in \mathbb{R}\}, \quad V \stackrel{\text{df}}{=} \{(0, 0, z) \mid z \in \mathbb{R}\}, \quad W \stackrel{\text{df}}{=} \{(0, y, y) \mid y \in \mathbb{R}\}.$$

(1) Show that $\mathbb{R}^3 = U + V + W$. (2) Show that $\mathbb{R}^3 \neq U \oplus V \oplus W$.

Solution of (1). Let $\mathbf{v} = (a, b, c) \in \mathbb{R}^3$. Then

$$(a, b, c) = \underbrace{(a, b, 0)}_{\in U} + \underbrace{(0, 0, c)}_{\in V} + \underbrace{(0, 0, 0)}_{\in W},$$

so $\mathbb{R}^3 = U + V + W$.

Solution of (2). It suffices to exhibit two distinct representations of the zero vector:

$$(0, 0, 0) = \underbrace{(0, 0, 0)}_{\in U} + \underbrace{(0, 0, 0)}_{\in V} + \underbrace{(0, 0, 0)}_{\in W}$$

and

$$(0, 0, 0) = \underbrace{(0, 1, 0)}_{\in U} + \underbrace{(0, 0, 1)}_{\in V} + \underbrace{(0, -1, -1)}_{\in W}.$$

Since the zero vector has more than one representation, by Theorem 2.3.2 the sum is not direct. Hence $\mathbb{R}^3 \neq U \oplus V \oplus W$. $\square$

2.4 Linear Combinations and Span

2.4.1 Definitions

Definition 2.4.1 (Linear combination). Let $n \in \mathbb{N}_0$, let V be a vector space over a field $\mathbb{F}$, and let $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$ (repetition allowed). A *linear combination* of $\mathbf{v}_1, \dots, \mathbf{v}_n$ is a vector $\mathbf{v} \in V$ for which there exist scalars $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ such that

$$\mathbf{v} = \lambda_1 \cdot \mathbf{v}_1 + \cdots + \lambda_n \cdot \mathbf{v}_n.$$

Definition 2.4.2 (Span). Let V be a vector space and let S be a set of V -vectors (not necessarily finite). The *span* of S , denoted $\text{Span}(S)$, is defined as the set of all linear combinations of *finitely many* vectors in S . By convention, $\text{Span}(\emptyset) = \{\mathbf{0}\}$ (empty sums equal $\mathbf{0}$).

Remark 2.4.3. S itself may be infinite, but each individual linear combination involves only finitely many vectors. Also, S need not be a vector subspace.

2.4.2 Properties of Span

Theorem 2.4.4 (Span is a vector subspace). Let V be a vector space and S a set of V -vectors. Then $\text{Span}(S)$ is a vector subspace of V . Moreover, $\text{Span}(S)$ is the **smallest** vector subspace of V containing S as a subset.

Theorem 2.4.5 (Properties of spans). Let V be a vector space and let $A, B \subseteq V$. Then:

1. $A \subseteq \text{Span}(A)$.
2. If $A \subseteq B$, then $\text{Span}(A) \subseteq \text{Span}(B)$.
3. $\text{Span}(A \cup B) = \text{Span}(A) + \text{Span}(B)$.
4. $\text{Span}(\text{Span}(A)) = \text{Span}(A)$ (*span stabilises after one application*).

Key Result: Span Stabilises

Property (4) says that taking the span of a span changes nothing: $\text{Span}(\text{Span}(A)) = \text{Span}(A)$. Property (3) links span to the sum of subspaces: $\text{Span}(A \cup B) = \text{Span}(A) + \text{Span}(B)$.

2.4.3 Worked Example: Spanning a Polynomial Space

Example 2.4.6. Determine whether $\{4 - X + X^3, 1 + 2X - 3X^2, 3 - 3X + 3X^2 + X^3, X - X^3\}$ spans $\mathcal{P}_3(\mathbb{R})$.

Solution. Every polynomial in $\mathcal{P}_3(\mathbb{R})$ has the form $a + bX + cX^2 + dX^3$, which we identify with the column vector $[a, b, c, d]^\top$. The four given polynomials correspond to the columns of the matrix

$$M = \begin{bmatrix} 4 & 1 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 0 & -3 & 3 & 0 \\ 1 & 0 & 1 & -1 \end{bmatrix}.$$

We row-reduce M to REF:

$$M \xrightarrow{R_1 \leftrightarrow R_4} \cdots \rightarrow \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{REF})$$

There are only 3 pivot columns, so the column vectors fail to span $\mathbb{R}^4$. Therefore the given set of polynomials **fails to span** $\mathcal{P}_3(\mathbb{R})$. $\square$

Exam Tip: Testing Whether Vectors Span a Space

To test whether $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ spans a finite-dimensional vector space V of dimension n : form the matrix $[\mathbf{v}_1 \mid \cdots \mid \mathbf{v}_k]$ (using a basis to coordinatise), row-reduce, and check whether there are n pivots. Fewer than n pivots means the set does **not** span V .

2.5 Linear Independence

2.5.1 Definitions

Definition 2.5.1 (Linear independence). Let V be a vector space over a field $\mathbb{F}$, and let S be a set of V -vectors (not necessarily finite). Then S is said to be *linearly independent* if and only if for any $n \in \mathbb{N}$, any $\lambda_1, \dots, \lambda_n \in \mathbb{F}$, and any **distinct** $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$,

$$\lambda_1 \cdot \mathbf{v}_1 + \dots + \lambda_n \cdot \mathbf{v}_n = \mathbf{0} \implies \lambda_1 = \dots = \lambda_n = 0.$$

Definition 2.5.2 (Linear dependence). A set S is said to be *linearly dependent* if and only if it is not linearly independent.

Remark 2.5.3. By convention, the empty set $\emptyset$ is linearly independent (there are no vectors to form a non-trivial zero sum).

2.5.2 Linear Independence of a Finite Set

The next theorem clarifies that for finite sets the general definition reduces to the familiar criterion from MH201 Part I.

Theorem 2.5.4 (Linear independence of a finite set). *Let $n \in \mathbb{N}$, let V be a vector space, and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be **distinct** V -vectors. Then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent if and only if whenever $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ satisfy*

$$\lambda_1 \cdot \mathbf{v}_1 + \dots + \lambda_n \cdot \mathbf{v}_n = \mathbf{0},$$

we must have $\lambda_1 = \dots = \lambda_n = 0$.

Proof. The forward direction is immediate from the definition. For the backward direction, suppose the condition holds for all n scalars applied to all n vectors simultaneously. Let $m \leq n$ and pick any m distinct vectors from $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$; relabel them as $\mathbf{v}_1, \dots, \mathbf{v}_m$. If $\lambda_1 \mathbf{v}_1 + \dots + \lambda_m \mathbf{v}_m = \mathbf{0}$, pad with zero coefficients:

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_m \mathbf{v}_m + 0 \cdot \mathbf{v}_{m+1} + \dots + 0 \cdot \mathbf{v}_n = \mathbf{0}.$$

By assumption $\lambda_1 = \dots = \lambda_m = 0$. Hence the set is linearly independent. $\square$

2.5.3 The Three-Concepts-in-One Theorem

The following theorem is a cornerstone result connecting linear independence, direct sums, and span.

Theorem 2.5.5 (Three-Concepts-in-One Theorem). *Let V be a vector space, let S be a linearly independent subset of V , and let $\{A, B\}$ be a partition of S (i.e., $A \cup B = S$ and $A \cap B = \emptyset$). Then*

$$\text{Span}(S) = \text{Span}(A) \oplus \text{Span}(B).$$

Proof. Step 1: $\text{Span}(S) = \text{Span}(A) + \text{Span}(B)$. Any $\mathbf{v} \in \text{Span}(S)$ is a linear combination of finitely many vectors $\mathbf{a}_1, \dots, \mathbf{a}_m \in A$ and $\mathbf{b}_1, \dots, \mathbf{b}_n \in B$. Hence $\mathbf{v} = \underbrace{(\lambda_1 \mathbf{a}_1 + \dots + \lambda_m \mathbf{a}_m)}_{\in \text{Span}(A)} + \underbrace{(\mu_1 \mathbf{b}_1 + \dots + \mu_n \mathbf{b}_n)}_{\in \text{Span}(B)}$, so $\text{Span}(S) \subseteq \text{Span}(A) + \text{Span}(B)$. The reverse

inclusion follows from $A, B \subseteq S$ and Theorem 2.4.5(2–3).

Step 2: $\text{Span}(A) \cap \text{Span}(B) = \{\mathbf{0}\}$. Let $\mathbf{v} \in \text{Span}(A) \cap \text{Span}(B)$. Then there exist distinct $\mathbf{a}_1, \dots, \mathbf{a}_m \in A$, distinct $\mathbf{b}_1, \dots, \mathbf{b}_n \in B$, and scalars $\lambda_i, \mu_j \in \mathbb{F}$ such that

$$\mathbf{v} = \lambda_1 \mathbf{a}_1 + \dots + \lambda_m \mathbf{a}_m = \mu_1 \mathbf{b}_1 + \dots + \mu_n \mathbf{b}_n.$$

Rearranging: $\lambda_1 \mathbf{a}_1 + \cdots + \lambda_m \mathbf{a}_m + (-\mu_1) \mathbf{b}_1 + \cdots + (-\mu_n) \mathbf{b}_n = \mathbf{0}$. Since $A \cap B = \emptyset$, the vectors $\mathbf{a}_1, \dots, \mathbf{a}_m, \mathbf{b}_1, \dots, \mathbf{b}_n$ are all *distinct* elements of S . By linear independence of S , all coefficients are zero, so $\mathbf{v} = \mathbf{0}$.

By the Direct-Sum Criterion (Theorem 2.3.3), $\text{Span}(S) = \text{Span}(A) \oplus \text{Span}(B)$. $\square$

Key Result: Three-Concepts-in-One

If S is linearly independent and $S = A \sqcup B$ (disjoint union), then partitioning S induces a direct-sum decomposition: $\text{Span}(S) = \text{Span}(A) \oplus \text{Span}(B)$. This is the bridge between linear independence, span, and direct sums.

2.5.4 Application: Decomposing a Subspace

Example 2.5.6. Define $U \stackrel{\text{df}}{=} \{(x, y, x + y, x - y) \in \mathbb{R}^4 \mid x, y \in \mathbb{R}\}$.

1. Find vector subspaces V and W of $\mathbb{R}^4$ such that $U = V \oplus W$.
2. Find a vector subspace X of $\mathbb{R}^4$ such that $\mathbb{R}^4 = U \oplus X$.

Solution of (1). We rewrite U :

$$\begin{aligned} U &= \{(x, 0, x, x) + (0, y, y, -y) \in \mathbb{R}^4 \mid x, y \in \mathbb{R}\} \\ &= \{x \cdot (1, 0, 1, 1) + y \cdot (0, 1, 1, -1) \in \mathbb{R}^4 \mid x, y \in \mathbb{R}\} \\ &= \text{Span}(\{(1, 0, 1, 1), (0, 1, 1, -1)\}). \end{aligned}$$

Since $(1, 0, 1, 1)$ and $(0, 1, 1, -1)$ are not scalar multiples of each other, the set $S = \{(1, 0, 1, 1), (0, 1, 1, -1)\}$ is linearly independent. Partitioning S into $A = \{(1, 0, 1, 1)\}$ and $B = \{(0, 1, 1, -1)\}$, the Three-Concepts-in-One Theorem gives

$$U = \underbrace{\text{Span}(\{(1, 0, 1, 1)\})}_V \oplus \underbrace{\text{Span}(\{(0, 1, 1, -1)\})}_W. \quad \square$$

Solution of (2). We extend S to a linearly independent subset of $\mathbb{R}^4$ by row-reducing the matrix with columns $(1, 0, 1, 1)$, $(0, 1, 1, -1)$, and the standard basis vectors $\mathbf{e}_3, \mathbf{e}_4$:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_3 \mapsto R_3 - R_1 \\ R_4 \mapsto R_4 - R_1}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_3 \mapsto R_3 - R_2 \\ R_4 \mapsto R_4 + R_2}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (\text{RREF})$$

All four columns are pivots, so $\{(1, 0, 1, 1), (0, 1, 1, -1), (0, 0, 1, 0), (0, 0, 0, 1)\}$ is a linearly independent subset of $\mathbb{R}^4$ that spans $\mathbb{R}^4$. By the Three-Concepts-in-One Theorem,

$$\mathbb{R}^4 = \text{Span}(\{(1, 0, 1, 1), (0, 1, 1, -1)\}) \oplus \text{Span}(\{(0, 0, 1, 0), (0, 0, 0, 1)\}) = U \oplus X,$$

where $X \stackrel{\text{df}}{=} \text{Span}(\{(0, 0, 1, 0), (0, 0, 0, 1)\})$. $\square$

Chapter 3

Linear Independence, Bases, and Dimension

3.1 Linear Independence: Further Examples

The following examples illustrate the Linear Independence Test in several different vector spaces encountered so far.

Example 3.1.1 (Matrices in $\mathbb{R}^{2,2}$). Determine whether or not

$$\left\{ \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix} \right\}$$

is a linearly independent subset of $\mathbb{R}^{2,2}$.

Solution. We use the fact (to be made precise later) that $\mathbb{R}^{2,2}$ and $\mathbb{R}^{4,1}$ are isomorphic vector spaces: the map $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$ converts the linear independence question in $\mathbb{R}^{2,2}$ into one in $\mathbb{R}^{4,1}$. We therefore instead determine whether

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \\ -2 \end{bmatrix} \right\}$$

is a linearly independent subset of $\mathbb{R}^{4,1}$.

We form the matrix with these as columns and row-reduce to REF:

$$\begin{aligned} & \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & -1 & 2 & 1 \\ -2 & 1 & 1 & 2 \\ 1 & 1 & 0 & -2 \end{bmatrix} \xrightarrow{\substack{R_3 \mapsto R_3 + 2R_1 \\ R_4 \mapsto R_4 - R_1}} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & -1 & 2 & 1 \\ 0 & 1 & -1 & 6 \\ 0 & 1 & 1 & -4 \end{bmatrix} \\ & \xrightarrow{\substack{R_3 \mapsto R_3 + R_2 \\ R_4 \mapsto R_4 + R_2}} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & -1 & 2 & 1 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 3 & -3 \end{bmatrix} \xrightarrow{R_4 \mapsto R_4 - 3R_3} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & -1 & 2 & 1 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & -24 \end{bmatrix}. \quad (\text{REF}) \end{aligned}$$

There are four pivot columns, so the four column vectors in $\mathbb{R}^{4,1}$ are linearly independent. Therefore the given set of matrices is a **linearly independent** subset of $\mathbb{R}^{2,2}$. $\square$

Warning: The Zero Function vs. the Number Zero

When working in a function space such as $\mathbb{R}^{\mathbb{R}}$, the zero vector is the *zero function* $0_{\mathbb{R}^{\mathbb{R}}}$, the function that maps every real number to 0. Writing $\lambda f + \mu g = 0$ in this context means the function $x \mapsto \lambda f(x) + \mu g(x)$ is identically zero — not that a single numerical value equals zero.

Example 3.1.2 (Functions in $\mathbb{R}^{\mathbb{R}}$). Define $f, g : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) \stackrel{\text{df}}{=} e^x$ and $g(x) \stackrel{\text{df}}{=} e^{-x}$ for all $x \in \mathbb{R}$. Determine whether $\{f, g\}$ is a linearly independent subset of $\mathbb{R}^{\mathbb{R}}$.

Solution. Let $\lambda, \mu \in \mathbb{R}$ be scalars for which $\lambda \cdot f + \mu \cdot g = 0_{\mathbb{R}^{\mathbb{R}}}$. This means, for all $x \in \mathbb{R}$:

$$\lambda e^x + \mu e^{-x} = 0.$$

Multiplying both sides by e^x gives:

$$(\star) \quad \lambda e^{2x} + \mu = 0.$$

Substituting $x = 0$: $\lambda + \mu = 0$, so $\mu = -\lambda$. Substituting $x = 1$: $\lambda e^2 + \mu = 0$, so $\mu = -\lambda e^2$. Hence $-\lambda = -\lambda e^2$, which gives

$$0 = \lambda(e^2 - 1).$$

Since $e^2 - 1 \neq 0$, we conclude $\lambda = 0$, and consequently $\mu = -\lambda = 0$. Therefore the only linear combination of f and g yielding the zero function is the trivial one, so $\{f, g\}$ is a **linearly independent** subset of $\mathbb{R}^{\mathbb{R}}$. $\square$

Exam Tip: Testing Linear Independence of Functions

To show $\{f_1, \dots, f_n\}$ is linearly independent in a function space, assume $\lambda_1 f_1 + \dots + \lambda_n f_n = 0_V$ and derive the equation pointwise. Then substitute carefully chosen values of x (e.g. $x = 0$, $x = 1$, or $x = \ln 2$) to generate a system of equations forcing all $\lambda_i = 0$.

Example 3.1.3 (Abstract differences of vectors). Let V be a vector space and let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ be distinct V -vectors such that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is linearly independent. Show that $\{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \mathbf{v}_3 - \mathbf{v}_4, \mathbf{v}_4\}$ is a linearly independent set of four V -vectors.

Proof. Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \mathbb{F}$ be scalars such that

$$\lambda_1(\mathbf{v}_1 - \mathbf{v}_2) + \lambda_2(\mathbf{v}_2 - \mathbf{v}_3) + \lambda_3(\mathbf{v}_3 - \mathbf{v}_4) + \lambda_4 \mathbf{v}_4 = \mathbf{0}_V.$$

Expanding:

$$\lambda_1 \mathbf{v}_1 + (\lambda_2 - \lambda_1) \mathbf{v}_2 + (\lambda_3 - \lambda_2) \mathbf{v}_3 + (\lambda_4 - \lambda_3) \mathbf{v}_4 = \mathbf{0}_V.$$

By the linear independence of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, all coefficients must be zero:

$$\lambda_1 = 0, \quad \lambda_2 - \lambda_1 = 0, \quad \lambda_3 - \lambda_2 = 0, \quad \lambda_4 - \lambda_3 = 0.$$

From the first equation $\lambda_1 = 0$; then $\lambda_2 = \lambda_1 = 0$; then $\lambda_3 = \lambda_2 = 0$; then $\lambda_4 = \lambda_3 = 0$. Hence all scalars are zero, so $\{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \mathbf{v}_3 - \mathbf{v}_4, \mathbf{v}_4\}$ is linearly independent. $\square$

3.2 Bases

3.2.1 Definition and Standard Examples

A basis for a vector space is a set of vectors that is simultaneously linearly independent and spanning — the smallest possible spanning set, or equivalently the largest possible linearly independent set.

Definition 3.2.1 (Basis). A *basis* for a vector space V is a set B of V -vectors with the following two properties:

1. B is linearly independent.
2. $\text{Span}(B) = V$ (the *spanning property*).

Example 3.2.2 (Standard basis for $\mathbb{F}^n$). For each $n \in \mathbb{N}$, a basis for $\mathbb{F}^n$ is the *standard basis* $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, where

$$\forall i \in \{1, \dots, n\} : \quad \mathbf{e}_i \stackrel{\text{df}}{=} (0_{\mathbb{F}}, \dots, 0_{\mathbb{F}}, \underbrace{1_{\mathbb{F}}}_{i\text{-th}}, 0_{\mathbb{F}}, \dots, 0_{\mathbb{F}}).$$

Example 3.2.3 (Standard basis for $\mathcal{P}_n(\mathbb{F})$). For each $n \in \mathbb{N}$, a basis for $\mathcal{P}_n(\mathbb{F})$ is $\{X^k\}_{k=0}^n = \{1, X, X^2, \dots, X^n\}$.

Example 3.2.4 (Basis for $\mathcal{P}(\mathbb{F})$). A basis for $\mathcal{P}(\mathbb{F})$, the space of all polynomials, is the countably infinite set $\{X^k\}_{k=0}^{\infty} = \{1, X, X^2, X^3, \dots\}$.

Example 3.2.5 (Standard basis for $\mathbb{F}^{m,n}$). For each $m, n \in \mathbb{N}$, a basis for $\mathbb{F}^{m,n}$ (the vector space of $m \times n$ matrices with entries in $\mathbb{F}$) consists of mn matrices and is given by $\{\mathbf{E}^{a,b}\}_{a=1, b=1}^{a=m, b=n}$, where $\mathbf{E}^{a,b}$ is the $m \times n$ matrix with a $1_{\mathbb{F}}$ in the (a, b) -entry and $0_{\mathbb{F}}$ everywhere else.

3.2.2 Worked Example: Finding a Basis for a Subspace

Example 3.2.6 (Basis for the trace-zero subspace). Find a basis for the vector subspace

$$V \stackrel{\text{df}}{=} \{\mathbf{M} \in \mathbb{R}^{2,2} \mid \text{trace}(\mathbf{M}) = 0\}.$$

Solution. For a matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2,2}$, the trace condition gives $a + d = 0$, i.e. $d = -a$. Hence

$$\begin{aligned} V &= \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \in \mathbb{R}^{2,2} \mid a, b, c \in \mathbb{R} \right\} \\ &= \left\{ a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\} \\ &= \text{Span} \left(\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\} \right). \end{aligned}$$

It remains to verify linear independence. Suppose $a, b, c \in \mathbb{R}$ satisfy

$$a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then $\begin{bmatrix} a & b \\ c & -a \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, so $a = b = c = 0$. This confirms linear independence. Therefore

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

is a basis for V . $\square$

3.2.3 Uniqueness of Basis Representations

A key property of bases is that they provide *coordinates*: every vector has a unique representation as a linear combination of basis vectors.

Theorem 3.2.7 (Every vector is a unique linear combination of basis vectors). *Let V be a vector space over a field $\mathbb{F}$, and let B be a basis for V . Then for every $\mathbf{v} \in V$, there exists a family $(\lambda_{\mathbf{b}})_{\mathbf{b} \in B}$ of scalars in $\mathbb{F}$, indexed by B and uniquely determined by $\mathbf{v}$, with the following two properties:*

1. There are only finitely many $\mathbf{b} \in B$ for which $\lambda_{\mathbf{b}} \neq 0_{\mathbb{F}}$.

2. $\mathbf{v} = \sum_{\mathbf{b} \in B} \lambda_{\mathbf{b}} \cdot \mathbf{b}$ (a finite sum by property (1)).

For each $\mathbf{b} \in B$, the scalar $\lambda_{\mathbf{b}} \in \mathbb{F}$ is called the $\mathbf{b}$ -coordinate of $\mathbf{v}$ with respect to B .

Proof. Existence. Since $V = \text{Span}(B)$, every $\mathbf{v} \in V$ is a linear combination of finitely many vectors in B , which furnishes a family $(\lambda_{\mathbf{b}})_{\mathbf{b} \in B}$ with properties (1) and (2).

Uniqueness. Suppose $(\mu_{\mathbf{b}})_{\mathbf{b} \in B}$ is any other family of scalars satisfying (1) and (2), so that $\mathbf{v} = \sum_{\mathbf{b} \in B} \mu_{\mathbf{b}} \cdot \mathbf{b}$ as well. Then

$$\sum_{\mathbf{b} \in B} \lambda_{\mathbf{b}} \cdot \mathbf{b} = \sum_{\mathbf{b} \in B} \mu_{\mathbf{b}} \cdot \mathbf{b},$$

which gives $\sum_{\mathbf{b} \in B} (\lambda_{\mathbf{b}} - \mu_{\mathbf{b}}) \cdot \mathbf{b} = \mathbf{0}$. Both families are finitely supported, so this is a finite linear combination. Since B is linearly independent, we get $\lambda_{\mathbf{b}} - \mu_{\mathbf{b}} = 0_{\mathbb{F}}$ for all $\mathbf{b} \in B$, i.e. $\lambda_{\mathbf{b}} = \mu_{\mathbf{b}}$ for all $\mathbf{b}$. $\square$

Key Result: Coordinates with Respect to a Basis

A basis B for V gives every vector $\mathbf{v} \in V$ a unique *coordinate representation*: $\mathbf{v} = \sum_{\mathbf{b} \in B} \lambda_{\mathbf{b}} \cdot \mathbf{b}$. This is the foundation for coordinatising abstract vector spaces.

3.2.4 Existence of Bases

Theorem 3.2.8 (Basis Existence Theorem). *Every vector space has a basis.*

Remark 3.2.9. The proof requires the Axiom of Choice (via Zorn's Lemma), which is a pure existence argument. As a consequence, for some vector spaces — such as $\mathbb{R}^{\mathbb{R}}$ or $\mathbb{R}$ as a vector space over $\mathbb{Q}$ — we know a basis must exist but cannot write one down explicitly. We will not prove this theorem in MH1201.

3.2.5 The Size-Limitation Lemma

The following lemma is a key stepping stone toward the main structural theorem of the course.

Lemma 3.2.10 (Size-Limitation Lemma). *Let V be a vector space that has a finite basis with size n . Then any linearly independent subset of V is finite with size $\leq n$.*

Proof. Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for V . Suppose for contradiction that there exists a linearly independent subset S of V with size $> n$. Pick any $n+1$ distinct vectors $\mathbf{v}_1, \dots, \mathbf{v}_{n+1}$ from S ; then $\{\mathbf{v}_1, \dots, \mathbf{v}_{n+1}\}$ is linearly independent (as a subset of a linearly independent set).

Since B is a basis, each $\mathbf{v}_i$ is a linear combination of $\mathbf{b}_1, \dots, \mathbf{b}_n$:

$$\mathbf{v}_i = \sum_{j=1}^n \lambda_{i,j} \cdot \mathbf{b}_j, \quad i = 1, \dots, n+1.$$

For scalars $\mu_1, \dots, \mu_{n+1} \in \mathbb{F}$ we then have

$$\mu_1 \mathbf{v}_1 + \dots + \mu_{n+1} \mathbf{v}_{n+1} = \sum_{j=1}^n \left(\sum_{i=1}^{n+1} \lambda_{i,j} \mu_i \right) \mathbf{b}_j.$$

For this to equal $\mathbf{0}$, the linear independence of B forces

$$(\star) \quad \begin{cases} \lambda_{1,1}\mu_1 + \cdots + \lambda_{n+1,1}\mu_{n+1} = 0_{\mathbb{F}}, \\ \vdots \\ \lambda_{1,n}\mu_1 + \cdots + \lambda_{n+1,n}\mu_{n+1} = 0_{\mathbb{F}}. \end{cases}$$

The system $(\star)$ is a homogeneous system of n equations in $n+1$ unknowns $\mu_1, \dots, \mu_{n+1}$. Being under-determined, it has a non-trivial solution $(\mu_1^*, \dots, \mu_{n+1}^*) \neq \mathbf{0}$. This means

$$\mu_1^* \mathbf{v}_1 + \cdots + \mu_{n+1}^* \mathbf{v}_{n+1} = \mathbf{0}$$

is a non-trivial linear combination equal to $\mathbf{0}$, contradicting the linear independence of $\{\mathbf{v}_1, \dots, \mathbf{v}_{n+1}\}$. Therefore any linearly independent subset of V has size $\leq n$. $\square$

3.2.6 The Fundamental Theorem of MH1201

Theorem 3.2.11 (Fundamental Theorem of MH1201). *Let V be a vector space. If V has a finite basis with size n , then all other bases for V are finite with size n .*

Proof. Let B be a finite basis for V with size n , and let $\tilde{B}$ be any other basis for V . Since $\tilde{B}$ is linearly independent, the Size-Limitation Lemma (Lemma 3.2.10) applied with B as the finite basis tells us that $\tilde{B}$ has size $m \leq n$. However, swapping roles — now applying the lemma with $\tilde{B}$ as the finite basis of size m and B as the linearly independent set — gives $n \leq m$. Together, $m = n$. $\square$

Key Result: All Finite Bases Have the Same Size

If a vector space V has one finite basis of size n , then *every* basis of V has exactly n elements. This is what makes the notion of dimension well-defined.

3.3 Dimension

3.3.1 The Fundamental Theorem of Vector Spaces

The Fundamental Theorem of MH1201 can be extended beyond finite-dimensional spaces.

Theorem 3.3.1 (Fundamental Theorem of Vector Spaces). *All bases of a vector space have the same size (even if they are infinite).*

Remark 3.3.2. The infinite case of this theorem is considerably harder and requires set-theoretic tools beyond the scope of MH1201. We accept it without proof.

3.3.2 Definition of Dimension

The Basis Existence Theorem (Theorem 3.2.8) guarantees every vector space has a basis, and the Fundamental Theorem of Vector Spaces (Theorem 3.3.1) guarantees all bases of a given space have the same size. Together, these results allow us to assign a well-defined size to every vector space.

Definition 3.3.3 (Dimension). Let V be a vector space.

- The *dimension* of V , denoted $\dim(V)$, is the size of any of its bases.
- V is said to be *finite-dimensional* if and only if $\dim(V)$ is finite.

- V is said to be *infinite-dimensional* if and only if $\dim(V)$ is infinite.

Example 3.3.4 (Dimensions of standard spaces). The dimensions of the standard vector spaces are:

- $\dim(\mathbb{F}^n) = n$, since the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ has n elements.
- $\dim(\mathbb{F}^{m,n}) = mn$, since the standard basis $\{\mathbf{E}^{a,b}\}$ has mn elements.
- $\dim(\mathcal{P}_n(\mathbb{F})) = n + 1$, since $\{1, X, \dots, X^n\}$ has $n + 1$ elements.
- $\dim(\mathcal{P}(\mathbb{F})) = \aleph_0$ (countably infinite), since $\{1, X, X^2, \dots\}$ is countably infinite.
- $\dim(V) = 3$ for the trace-zero subspace $V \subset \mathbb{R}^{2,2}$ from Example 3.2.6, since the basis found there has 3 elements.

Key Result: Dimension as a Basis-Independent Invariant

$\dim(V)$ is an intrinsic property of the vector space V — it does not depend on which basis you choose. This gives a powerful tool: to show two vector spaces are structurally different, it suffices to show they have different dimensions.

3.3.3 Worked Examples: Computing Dimensions

We now apply the definition of dimension to several concrete examples.

Example 3.3.5 (Dimension of a linear plane in $\mathbb{R}^3$). Find the dimension of the linear plane

$$\Pi \stackrel{\text{df}}{=} \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y + 3z = 0\}.$$

Solution. For any $(x, y, z) \in \mathbb{R}^3$,

$$\begin{aligned} (x, y, z) \in \Pi &\iff x + 2y + 3z = 0 \\ &\iff x = -2y - 3z \\ &\iff (x, y, z) = (-2y - 3z, y, z) \\ &\iff (x, y, z) = y(-2, 1, 0) + z(-3, 0, 1) \\ &\iff (x, y, z) \in \text{Span}(\{(-2, 1, 0), (-3, 0, 1)\}). \end{aligned}$$

Hence $\Pi = \text{Span}(\{(-2, 1, 0), (-3, 0, 1)\})$. Since neither vector is a scalar multiple of the other, $\{(-2, 1, 0), (-3, 0, 1)\}$ is linearly independent and therefore a basis for Π . Consequently, $\dim(\Pi) = 2$.

Example 3.3.6 (Dimension of the space of upper-triangular matrices). Let $n \in \mathbb{N}$. Find the dimension of

$$V \stackrel{\text{df}}{=} \{\mathbf{M} \in \mathbb{R}^{n,n} \mid \mathbf{M} \text{ is upper-triangular}\}.$$

Solution. For any upper-triangular matrix $\mathbf{M} \in \mathbb{R}^{n,n}$,

$$\mathbf{M} = \sum_{1 \leq i \leq j \leq n} M[i, j] \cdot \mathbf{E}^{ij},$$

where $\mathbf{E}^{ij}$ is the $n \times n$ matrix with a 1 in the (i, j) -entry and 0 elsewhere. Hence $V \subseteq \text{Span}(\{\mathbf{E}^{ij}\}_{1 \leq i \leq j \leq n})$. Conversely, since each $\mathbf{E}^{ij}$ (with $i \leq j$) is itself upper-triangular, we have $\text{Span}(\{\mathbf{E}^{ij}\}_{1 \leq i \leq j \leq n}) \subseteq V$. Therefore $V = \text{Span}(\{\mathbf{E}^{ij}\}_{1 \leq i \leq j \leq n})$.

The set $\{\mathbf{E}^{ij}\}_{1 \leq i \leq j \leq n}$ is a subset of the standard basis $\{\mathbf{E}^{ij}\}_{i,j=1}^n$ for $\mathbb{R}^{n,n}$, which is linearly independent; hence it is linearly independent. It is therefore a basis for V , with cardinality $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. Thus

$$\dim(V) = \frac{n(n+1)}{2}.$$

Sanity check: for $n = 1$, $\dim(V) = 1$; for $n = 2$, $\dim(V) = 3$; for $n = 3$, $\dim(V) = 6$. $\square$

Exam Tip: Finding Dimensions via Parametric Forms

To find $\dim(V)$ for a subspace defined by linear equations, express membership in V as a parametric condition. The number of free parameters equals $\dim(V)$, and the vectors corresponding to setting each free parameter to 1 and the rest to 0 form a basis.

3.3.4 Exam Problem: A Lower Bound on Dimension

Example 3.3.7 (AY2023–24 Final Exam, Problem 6). Let V be a vector space and let $\mathbf{v}_1, \dots, \mathbf{v}_{2024}$ be distinct V -vectors such that $\{\mathbf{v}_1, \dots, \mathbf{v}_{2024}\}$ is linearly independent. Show that for any V -vector $\mathbf{w}$,

$$\dim(\text{Span}(\{\mathbf{v}_1 + \mathbf{w}, \dots, \mathbf{v}_{2024} + \mathbf{w}\})) \geq 2023.$$

Solution. Consider the 2023 vectors

$$\mathbf{v}_1 - \mathbf{v}_2, \quad \mathbf{v}_1 - \mathbf{v}_3, \quad \dots, \quad \mathbf{v}_1 - \mathbf{v}_{2024}.$$

Since the span is a vector subspace, it is closed under subtraction. As both $\mathbf{v}_k + \mathbf{w}$ and $\mathbf{v}_1 + \mathbf{w}$ lie in the span, we have

$$\mathbf{v}_1 - \mathbf{v}_k = (\mathbf{v}_1 + \mathbf{w}) - (\mathbf{v}_k + \mathbf{w}) \in \text{Span}(\{\mathbf{v}_1 + \mathbf{w}, \dots, \mathbf{v}_{2024} + \mathbf{w}\})$$

for each $k = 2, \dots, 2024$. It therefore suffices to show that $\{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_3, \dots, \mathbf{v}_1 - \mathbf{v}_{2024}\}$ is linearly independent.

Let $\lambda_2, \lambda_3, \dots, \lambda_{2024} \in \mathbb{F}$ be scalars such that

$$\sum_{k=2}^{2024} \lambda_k (\mathbf{v}_1 - \mathbf{v}_k) = \mathbf{0}_V.$$

Expanding,

$$\left(\sum_{k=2}^{2024} \lambda_k \right) \mathbf{v}_1 - \lambda_2 \mathbf{v}_2 - \lambda_3 \mathbf{v}_3 - \cdots - \lambda_{2024} \mathbf{v}_{2024} = \mathbf{0}_V.$$

Since $\mathbf{v}_1, \dots, \mathbf{v}_{2024}$ are distinct and $\{\mathbf{v}_1, \dots, \mathbf{v}_{2024}\}$ is linearly independent, all coefficients must be zero:

$$\lambda_2 + \lambda_3 + \cdots + \lambda_{2024} = 0, \quad \lambda_2 = 0, \quad \lambda_3 = 0, \quad \dots, \quad \lambda_{2024} = 0.$$

Hence all scalars are zero, confirming linear independence. Therefore

$$\dim(\text{Span}(\{\mathbf{v}_1 + \mathbf{w}, \dots, \mathbf{v}_{2024} + \mathbf{w}\})) \geq 2023. \quad \square$$

Chapter 4

Dimension Theory and Subspace Sums

4.1 Further Dimension Theory

4.1.1 The Add-One-More Lemma

Lemma 4.1.1 (Add-One-More Lemma). *Let V be a vector space, let S be a linearly independent subset of V , and let $\mathbf{v} \in V$. Then the following two statements are equivalent:*

1. $\mathbf{v} \notin S$ and $S \cup \{\mathbf{v}\}$ is linearly independent.
2. $\mathbf{v} \notin \text{Span}(S)$.

Proof. **(1) $\Rightarrow$ (2).** Suppose $\mathbf{v} \notin S$ and $S \cup \{\mathbf{v}\}$ is linearly independent. If $\mathbf{v} \in \text{Span}(S)$, then we can find distinct $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$ and scalars $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ such that $\mathbf{v} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n$. But then

$$1_{\mathbb{F}} \cdot \mathbf{v} + (-\lambda_1) \mathbf{v}_1 + \dots + (-\lambda_n) \mathbf{v}_n = \mathbf{0},$$

a non-trivial linear combination of distinct vectors in $S \cup \{\mathbf{v}\}$ that equals $\mathbf{0}$ — contradicting the linear independence of $S \cup \{\mathbf{v}\}$. Hence $\mathbf{v} \notin \text{Span}(S)$.

(2) $\Rightarrow$ (1). Suppose $\mathbf{v} \notin \text{Span}(S)$. Since $S \subseteq \text{Span}(S)$, we immediately have $\mathbf{v} \notin S$. Now suppose $\lambda \cdot \mathbf{v} + (\text{a linear combination of distinct vectors in } S) = \mathbf{0}$. If $\lambda \neq 0_{\mathbb{F}}$, then $\mathbf{v} = \lambda^{-1} \cdot (\text{a linear combination in } S) \in \text{Span}(S)$, a contradiction. Hence $\lambda = 0_{\mathbb{F}}$, and the linear independence of S then forces all remaining coefficients to be zero. Thus $S \cup \{\mathbf{v}\}$ is linearly independent. $\square$

Key Result: The Add-One-More Lemma

To augment a linearly independent set S without losing independence, the only requirement is $\mathbf{v} \notin \text{Span}(S)$. Equivalently, $S \cup \{\mathbf{v}\}$ fails to be linearly independent if and only if $\mathbf{v}$ is already expressible in terms of S .

4.1.2 Finite Dimensionality is Hereditary

Theorem 4.1.2 (Finite dimensionality is a hereditary property). *Let V be a finite-dimensional vector space, and let W be a vector subspace of V . Then W is also finite-dimensional.*

Proof. Since V is finite-dimensional, the Size-Limitation Lemma (Lemma 3.2.10) tells us every linearly independent subset of V has size $\leq \dim(V)$. In particular, every linearly independent subset of W (which is also a subset of V) has size $\leq \dim(V)$, so it is finite.

Since $\emptyset$ is a linearly independent subset of W , linearly independent subsets of W do exist. By the finiteness bound, we can find a *largest* linearly independent subset B of W . We claim $\text{Span}(B) = W$.

We already have $\text{Span}(B) \subseteq W$ (since each vector in B belongs to W and W is a subspace). If $\text{Span}(B) \neq W$, there exists $\mathbf{v} \in W \setminus \text{Span}(B)$. By the Add-One-More Lemma (Lemma 4.1.1), $B \cup \{\mathbf{v}\}$ would then be a linearly independent subset of W strictly larger than B — contradicting the maximality of B . Hence $\text{Span}(B) = W$, so B is a finite basis for W , and $\dim(W)$ is finite. $\square$

4.1.3 The Basis Extension Theorem

Theorem 4.1.3 (Basis Extension Theorem). *Let V be a finite-dimensional vector space, and let S be a linearly independent subset of V . Then S can be extended to a basis for V .*

Proof. Since V is finite-dimensional, the Size-Limitation Lemma guarantees all linearly independent subsets of V have size $\leq \dim(V)$. The collection of linearly independent subsets of V that contain S is non-empty (it contains S itself). We can therefore find a *largest* such subset B .

We claim $\text{Span}(B) = V$. If not, there would exist $\mathbf{v} \in V \setminus \text{Span}(B)$. By the Add-One-More Lemma, $B \cup \{\mathbf{v}\}$ is linearly independent and contains S , contradicting the maximality of B . Hence $\text{Span}(B) = V$, making B a basis for V that extends S . $\square$

Exam Tip: Extending a Basis Algorithmically

To extend a linearly independent set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ to a basis for V :

1. Choose a known basis $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ for V (e.g. the standard basis).
2. Form the combined list $\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{b}_1, \dots, \mathbf{b}_n$ (as column vectors if necessary).
3. Row-reduce. The pivot columns identify which vectors to keep; discard any $\mathbf{b}_j$ corresponding to a non-pivot column.
4. The resulting set $\{\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{b}_{i_1}, \dots, \mathbf{b}_{i_{n-m}}\}$ is a basis for V , and moreover $V = \text{Span}(S) \oplus \text{Span}(\{\mathbf{b}_{i_1}, \dots, \mathbf{b}_{i_{n-m}}\})$.

Example 4.1.4 (Basis extension for a polynomial subspace). Let $V \stackrel{\text{df}}{=} \{P \in \mathcal{P}_4(\mathbb{R}) \mid P(6) = 0\}$.

1. Find a basis for V and determine $\dim(V)$.

Writing $P = a + bX + cX^2 + dX^3 + eX^4$, the condition $P(6) = 0$ gives $a + 6b + 36c + 216d + 1296e = 0$, i.e. $a = -6b - 36c - 216d - 1296e$. Substituting:

$$\begin{aligned} P &= (-6b - 36c - 216d - 1296e) + bX + cX^2 + dX^3 + eX^4 \\ &= b(X - 6) + c(X^2 - 36) + d(X^3 - 216) + e(X^4 - 1296). \end{aligned}$$

Hence $V = \text{Span}(\{X - 6, X^2 - 36, X^3 - 216, X^4 - 1296\})$. To check linear independence, note that $\{1, X, X^2, X^3, X^4\}$ is a basis for $\mathcal{P}_4(\mathbb{R})$: if $\lambda_1(X - 6) + \lambda_2(X^2 - 36) + \lambda_3(X^3 - 216) + \lambda_4(X^4 - 1296) = 0$, then comparing coefficients of $1, X, X^2, X^3, X^4$ forces all $\lambda_i = 0$. Thus $\{X - 6, X^2 - 36, X^3 - 216, X^4 - 1296\}$ is a basis for V and $\dim(V) = 4$.

2. Extend the basis for V to a basis for $\mathcal{P}_4(\mathbb{R})$.

Append the standard basis vector $\{1\}$ to the basis for V . Row reduction of the coefficient matrix (with columns corresponding to $X - 6, X^2 - 36, X^3 - 216, X^4 - 1296, 1$ relative to the basis $\{1, X, X^2, X^3, X^4\}$) yields full rank, confirming that

$$\{1, X - 6, X^2 - 36, X^3 - 216, X^4 - 1296\}$$

is a basis for $\mathcal{P}_4(\mathbb{R})$ extending the basis for V .

3. **Find W such that $\mathcal{P}_4(\mathbb{R}) = V \oplus W$.**

By the Three-Concepts-in-One Theorem (Theorem 2.5.5),

$$\mathcal{P}_4(\mathbb{R}) = \text{Span}(\{X - 6, X^2 - 36, X^3 - 216, X^4 - 1296\}) \oplus \text{Span}(\{1\}) = V \oplus W,$$

where $W = \text{Span}(\{1\}) = \{a \in \mathcal{P}_4(\mathbb{R}) \mid a \in \mathbb{R}\}$ is the subspace of constant polynomials.

4.1.4 Dimension of the Sum of Two Subspaces

Theorem 4.1.5 (Dimension of a sum of subspaces). *Let V be a finite-dimensional vector space, and let V_1 and V_2 be vector subspaces of V . Then V_1 , V_2 , $V_1 + V_2$, and $V_1 \cap V_2$ are all finite-dimensional, and*

$$\dim(V_1 + V_2) = \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2).$$

Remark 4.1.6. This formula is the vector-space analogue of the inclusion-exclusion principle for finite sets: $\#(A \cup B) = \#(A) + \#(B) - \#(A \cap B)$.

Proof. Since V is finite-dimensional, all subspaces are finite-dimensional by Theorem 4.1.2. Let A be a basis for $V_1 \cap V_2$. By the Basis Extension Theorem (Theorem 4.1.3):

- Extend A to a basis $A \cup B$ for V_1 (with $A \cap B = \emptyset$), so $V_1 = \text{Span}(A) \oplus \text{Span}(B)$.
- Extend A to a basis $A \cup C$ for V_2 (with $A \cap C = \emptyset$), so $V_2 = \text{Span}(A) \oplus \text{Span}(C)$.

Since $V_1 = \text{Span}(A \cup B)$ and $V_2 = \text{Span}(A \cup C)$,

$$V_1 + V_2 = \text{Span}(A \cup B) + \text{Span}(A \cup C) = \text{Span}(A \cup B \cup C).$$

We claim $A \cup B \cup C$ is linearly independent (hence a basis for $V_1 + V_2$). Suppose

$$\sum_{\mathbf{a} \in A} \lambda_{\mathbf{a}} \cdot \mathbf{a} + \sum_{\mathbf{b} \in B} \mu_{\mathbf{b}} \cdot \mathbf{b} + \sum_{\mathbf{c} \in C} \kappa_{\mathbf{c}} \cdot \mathbf{c} = \mathbf{0}.$$

Then $\sum_{\mathbf{b} \in B} \mu_{\mathbf{b}} \cdot \mathbf{b} = -\sum_{\mathbf{a} \in A} \lambda_{\mathbf{a}} \cdot \mathbf{a} - \sum_{\mathbf{c} \in C} \kappa_{\mathbf{c}} \cdot \mathbf{c}$. The left side lies in $\text{Span}(B) \subseteq V_1$ and the right side lies in V_2 , so both belong to $V_1 \cap V_2 = \text{Span}(A)$. But $\text{Span}(B) \cap \text{Span}(A) = \{\mathbf{0}\}$ (from the direct-sum decomposition of V_1), so $\sum_{\mathbf{b}} \mu_{\mathbf{b}} \cdot \mathbf{b} = \mathbf{0}$; linear independence of B forces all $\mu_{\mathbf{b}} = 0$. Similarly all $\kappa_{\mathbf{c}} = 0$, and then all $\lambda_{\mathbf{a}} = 0$. Hence A, B, C are mutually disjoint and $A \cup B \cup C$ is linearly independent, giving

$$\begin{aligned} \dim(V_1 + V_2) &= \#(A \cup B \cup C) = \#(A) + \#(B) + \#(C) \\ &= [\#(A) + \#(B)] + [\#(A) + \#(C)] - \#(A) \\ &= \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2). \quad \square \end{aligned}$$

□

Key Result: Dimension Formula for Subspace Sums

$\dim(V_1 + V_2) = \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2)$. In particular, $V_1 + V_2$ is a direct sum (i.e. $V_1 \cap V_2 = \{\mathbf{0}\}$) if and only if $\dim(V_1 + V_2) = \dim(V_1) + \dim(V_2)$.

4.1.5 Worked Examples: Dimension Formula

Example 4.1.7 (Intersection of two linear planes in $\mathbb{R}^3$). Let Π_1 and Π_2 be linear planes in $\mathbb{R}^3$ (i.e. $\dim(\Pi_i) = 2$). What are the possible dimensions of $\Pi_1 \cap \Pi_2$?

Solution. Since $\Pi_1 + \Pi_2$ is a subspace of $\mathbb{R}^3$, we have $\dim(\Pi_1 + \Pi_2) \leq 3$. By the dimension formula,

$$\dim(\Pi_1 + \Pi_2) = 2 + 2 - \dim(\Pi_1 \cap \Pi_2) = 4 - \dim(\Pi_1 \cap \Pi_2).$$

The bound $4 - \dim(\Pi_1 \cap \Pi_2) \leq 3$ gives $\dim(\Pi_1 \cap \Pi_2) \geq 1$. Also, $\Pi_1 \cap \Pi_2 \subseteq \Pi_1$, so $\dim(\Pi_1 \cap \Pi_2) \leq 2$.

Therefore $\dim(\Pi_1 \cap \Pi_2) \in \{1, 2\}$. Both values are realised:

- *Dimension 1:* Let Π_1 be the xy -plane and Π_2 the xz -plane; then $\Pi_1 \cap \Pi_2$ is the x -axis, which has dimension 1.
- *Dimension 2:* Take $\Pi_1 = \Pi_2$; then $\Pi_1 \cap \Pi_2 = \Pi_1$ has dimension 2. $\square$

Example 4.1.8 (Showing a direct-sum decomposition of $\mathbb{R}^8$). Let U and V be vector subspaces of $\mathbb{R}^8$ satisfying $\dim(U) = 3$, $\dim(V) = 5$, and $\dim(U + V) = 8$. Show that $\mathbb{R}^8 = U \oplus V$.

Solution. Since $U + V$ is a subspace of $\mathbb{R}^8$ and $\dim(U + V) = 8 = \dim(\mathbb{R}^8)$, we must have $U + V = \mathbb{R}^8$. By the dimension formula,

$$8 = \dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V) = 3 + 5 - \dim(U \cap V),$$

so $\dim(U \cap V) = 0$, i.e. $U \cap V = \{\mathbf{0}_{\mathbb{R}^8}\}$. Therefore, by the Direct-Sum Criterion (Theorem 2.3.3), $\mathbb{R}^8 = U + V = U \oplus V$. $\square$

Warning: Inclusion-Exclusion for Three Subspaces

Unlike the two-subspace case, the naive three-set inclusion-exclusion formula does *not* generalise to three vector subspaces. Specifically, for subspaces V_1, V_2, V_3 of a finite-dimensional vector space, it is *not* necessarily true that

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - [\dim(V_1 \cap V_2) + \dim(V_2 \cap V_3) + \dim(V_3 \cap V_1)] + \dim(V_1 \cap V_2 \cap V_3). \end{aligned} \quad (4.1)$$

Counterexamples can be constructed in $\mathbb{R}^3$ using three lines through the origin.

Chapter 5

Linear Transformations and Matrix Representations

5.1 Linear Transformations

5.1.1 Definition and First Examples

Definition 5.1.1 (Linear transformation). Let V and W be vector spaces over the same field $\mathbb{F}$. A **linear transformation** (or *linear map*) from V to W is a function $T : V \rightarrow W$ satisfying:

1. **Additivity.** For all $\mathbf{v}, \tilde{\mathbf{v}} \in V$,

$$T(\mathbf{v} +_V \tilde{\mathbf{v}}) = T(\mathbf{v}) +_W T(\tilde{\mathbf{v}}).$$

2. **Homogeneity.** For all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$,

$$T(\lambda \cdot_V \mathbf{v}) = \lambda \cdot_W T(\mathbf{v}).$$

We write $\mathcal{L}(V, W)$ for the set of all linear transformations from V to W , and $\mathcal{L}(V) \stackrel{\text{df}}{=} \mathcal{L}(V, V)$.

Example 5.1.2 (Zero and identity operators). Let V and W be vector spaces over the same field.

- The **zero operator** $\mathbf{0}_{V \rightarrow W} : V \rightarrow W$, defined by $\mathbf{0}_{V \rightarrow W}(\mathbf{v}) = \mathbf{0}_W$ for all $\mathbf{v} \in V$, is a linear transformation.
- The **identity operator** $I_V : V \rightarrow V$, defined by $I_V(\mathbf{v}) \stackrel{\text{df}}{=} \mathbf{v}$ for all $\mathbf{v} \in V$, is a linear transformation.

Example 5.1.3 (Derivative and multiplication operators on polynomials). Let $\mathbb{F}$ be a field.

- The **derivative operator** $D : \mathcal{P}(\mathbb{F}) \rightarrow \mathcal{P}(\mathbb{F})$, defined by $D(P) \stackrel{\text{df}}{=} P'$, is a linear transformation. For example, $D(X^2 + 3X) = 2X + 3$.
- For any fixed polynomial $Q \in \mathcal{P}(\mathbb{F})$, the **Q -multiplication operator** $M_Q : \mathcal{P}(\mathbb{F}) \rightarrow \mathcal{P}(\mathbb{F})$, defined by $M_Q(P) = QP$, is a linear transformation. For example, $M_{X^2+1}(3X) = (X^2 + 1)(3X) = 3X^3 + 3X$.

Example 5.1.4 (Shift operators on sequence spaces). Let $\mathbb{F}$ be a field.

- The **backward-shift operator** $\sigma_{\leftarrow} : \mathbb{F}^{\infty} \rightarrow \mathbb{F}^{\infty}$, defined by $\sigma_{\leftarrow}(x_1, x_2, x_3, \dots) \stackrel{\text{df}}{=} (x_2, x_3, x_4, \dots)$, is a linear transformation.
- The **forward-shift operator** $\sigma_{\rightarrow} : \mathbb{F}^{\infty} \rightarrow \mathbb{F}^{\infty}$, defined by $\sigma_{\rightarrow}(x_1, x_2, x_3, \dots) \stackrel{\text{df}}{=} (0_{\mathbb{F}}, x_1, x_2, \dots)$, is a linear transformation.

Note that $\sigma_{\leftarrow} \circ \sigma_{\rightarrow} = I$, but $\sigma_{\rightarrow} \circ \sigma_{\leftarrow} \neq I$, so $\sigma_{\rightarrow}$ has a left inverse but no right inverse.

Example 5.1.5 (Matrix operators). Let $\mathbb{F}$ be a field, $m, n \in \mathbb{N}$, and $\mathbf{A} \in \mathbb{F}^{m,n}$.

- The **A-matrix operator on column vectors**, $T_{\mathbf{A}} : \mathbb{F}^{n,1} \rightarrow \mathbb{F}^{m,1}$, defined by $T_{\mathbf{A}}(\mathbf{x}) \stackrel{\text{df}}{=} \mathbf{A}\mathbf{x}$, is a linear transformation.
- For $\mathbf{A} \in \mathbb{F}^{l,m}$, the **A-matrix operator on matrices**, $T_{\mathbf{A}} : \mathbb{F}^{m,n} \rightarrow \mathbb{F}^{l,n}$, defined by $T_{\mathbf{A}}(\mathbf{M}) \stackrel{\text{df}}{=} \mathbf{A}\mathbf{M}$, is also a linear transformation.

Example 5.1.6 (Synthesis operator). Let V be a vector space over $\mathbb{F}$, $n \in \mathbb{N}$, and $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$. The **synthesis operator** $\mathcal{S}_{\mathbf{v}_1, \dots, \mathbf{v}_n} : \mathbb{F}^{n,1} \rightarrow V$ is defined by

$$\mathcal{S}_{\mathbf{v}_1, \dots, \mathbf{v}_n} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \stackrel{\text{df}}{=} \sum_{i=1}^n \lambda_i \cdot \mathbf{v}_i.$$

This is a linear transformation that forms a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$ with the given coefficients.

Example 5.1.7 (Composition of linear transformations). Let U, V, W be vector spaces over the same field, $S \in \mathcal{L}(U, V)$, and $T \in \mathcal{L}(V, W)$. Then the composition $T \circ S : U \rightarrow W$ is also a linear transformation.

Proof. For all $\mathbf{u}, \tilde{\mathbf{u}} \in U$ and $\lambda \in \mathbb{F}$:

$$\begin{aligned} (T \circ S)(\mathbf{u} + \tilde{\mathbf{u}}) &= T(S(\mathbf{u} + \tilde{\mathbf{u}})) = T(S(\mathbf{u}) + S(\tilde{\mathbf{u}})) \\ &= T(S(\mathbf{u})) + T(S(\tilde{\mathbf{u}})) = (T \circ S)(\mathbf{u}) + (T \circ S)(\tilde{\mathbf{u}}), \\ (T \circ S)(\lambda \cdot \mathbf{u}) &= T(S(\lambda \cdot \mathbf{u})) = T(\lambda \cdot S(\mathbf{u})) \\ &= \lambda \cdot T(S(\mathbf{u})) = \lambda \cdot (T \circ S)(\mathbf{u}). \quad \square \end{aligned}$$

5.1.2 Key Theorems about Linear Transformations

Theorem 5.1.8 (Linear transformations preserve zero vectors). Let V and W be vector spaces over the same field, and let $T \in \mathcal{L}(V, W)$. Then $T(\mathbf{0}_V) = \mathbf{0}_W$.

Proof. By homogeneity, $T(\mathbf{0}_V) = T(0_{\mathbb{F}} \cdot_V \mathbf{0}_V) = 0_{\mathbb{F}} \cdot_W T(\mathbf{0}_V) = \mathbf{0}_W$. $\square$

Corollary 5.1.9 (The Non-Linearity Test). Let V and W be vector spaces, and let $T : V \rightarrow W$ be a function. If $T(\mathbf{0}_V) \neq \mathbf{0}_W$, then T is not a linear transformation.

Warning: Non-Linearity Test is one-directional

The converse of the Non-Linearity Test is *not* necessarily true: even if $T(\mathbf{0}_V) = \mathbf{0}_W$, the function T need not be linear. Always verify both additivity and homogeneity to confirm linearity. Do **not** use the Non-Linearity Test to conclude that T is linear.

Theorem 5.1.10 (Concise characterisation of linearity). Let V and W be vector spaces over a field $\mathbb{F}$. A function $T : V \rightarrow W$ belongs to $\mathcal{L}(V, W)$ if and only if for all $\mathbf{v}, \tilde{\mathbf{v}} \in V$ and $\lambda \in \mathbb{F}$,

$$T(\mathbf{v} + \lambda \cdot \tilde{\mathbf{v}}) = T(\mathbf{v}) + \lambda \cdot T(\tilde{\mathbf{v}}).$$

Proof. ($\Rightarrow$) If $T \in \mathcal{L}(V, W)$, then by additivity and homogeneity, $T(\mathbf{v} + \lambda \tilde{\mathbf{v}}) = T(\mathbf{v}) + T(\lambda \tilde{\mathbf{v}}) = T(\mathbf{v}) + \lambda T(\tilde{\mathbf{v}})$.

($\Leftarrow$) Suppose the single condition holds. Additivity: set $\lambda = 1_{\mathbb{F}}$. Homogeneity: $T(\lambda \cdot \mathbf{v}) = T(\mathbf{0}_V + \lambda \cdot \mathbf{v}) = T(\mathbf{0}_V) + \lambda T(\mathbf{v})$. One shows $T(\mathbf{0}_V) = \mathbf{0}_W$ by applying the condition with $\mathbf{v} = \tilde{\mathbf{v}} = \mathbf{0}_V$ and $\lambda = 1$, yielding $T(\mathbf{0}_V) = 2T(\mathbf{0}_V)$, so $T(\mathbf{0}_V) = \mathbf{0}_W$. Hence $T(\lambda \cdot \mathbf{v}) = \lambda T(\mathbf{v})$. $\square$

Key Result: Concise Linearity Check

To verify $T \in \mathcal{L}(V, W)$, it suffices to check the single combined condition

$$T(\mathbf{v} + \lambda \cdot \tilde{\mathbf{v}}) = T(\mathbf{v}) + \lambda \cdot T(\tilde{\mathbf{v}}) \quad \forall \mathbf{v}, \tilde{\mathbf{v}} \in V, \lambda \in \mathbb{F}.$$

This simultaneously captures additivity (set $\lambda = 1$) and homogeneity (set $\mathbf{v} = \mathbf{0}_V$).

Theorem 5.1.11 (Linear transformations form a vector space). *Let V and W be vector spaces over a field $\mathbb{F}$. Define pointwise addition and scalar multiplication on $\mathcal{L}(V, W)$:*

$$(S + T)(\mathbf{v}) \stackrel{\text{df}}{=} S(\mathbf{v}) +_W T(\mathbf{v}), \quad (\lambda \cdot T)(\mathbf{v}) \stackrel{\text{df}}{=} \lambda \cdot_W T(\mathbf{v}).$$

Then $\mathcal{L}(V, W)$ equipped with these operations is a vector space over $\mathbb{F}$, with the zero vector being the zero operator $\mathbf{0}_{V \rightarrow W}$.

If $\dim(V) = m$ and $\dim(W) = n$, then $\mathcal{L}(V, W)$ and $\mathbb{F}^{n,m}$ are essentially identical vector spaces (this will be made precise via matrix representations in Section 5.4).

5.1.3 The Null Space and Range

Definition 5.1.12 (Null space and range). Let V and W be vector spaces over the same field and $T \in \mathcal{L}(V, W)$.

- The **null space** (or *kernel*) of T , denoted $\text{Null}(T)$, is

$$\text{Null}(T) \stackrel{\text{df}}{=} \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W\} \subseteq V.$$

- The **range** of T , denoted $\text{Range}(T)$, is the image of V under T :

$$\text{Range}(T) \stackrel{\text{df}}{=} \{T(\mathbf{v}) \in W \mid \mathbf{v} \in V\} \subseteq W.$$

Theorem 5.1.13 (Null space and range are vector subspaces). *Let V and W be vector spaces over $\mathbb{F}$ and $T \in \mathcal{L}(V, W)$. Then $\text{Null}(T)$ is a vector subspace of V , and $\text{Range}(T)$ is a vector subspace of W .*

Proof. Both contain $\mathbf{0}$ (by Theorem 5.1.8), so they are non-empty. For closure of $\text{Null}(T)$: if $\mathbf{v}, \tilde{\mathbf{v}} \in \text{Null}(T)$ and $\lambda \in \mathbb{F}$, then $T(\mathbf{v} + \tilde{\mathbf{v}}) = T(\mathbf{v}) + T(\tilde{\mathbf{v}}) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$ and $T(\lambda \cdot \mathbf{v}) = \lambda \cdot T(\mathbf{v}) = \mathbf{0}_W$. For closure of $\text{Range}(T)$: if $\mathbf{w} = T(\mathbf{v})$ and $\tilde{\mathbf{w}} = T(\tilde{\mathbf{v}})$, then $\mathbf{w} + \tilde{\mathbf{w}} = T(\mathbf{v} + \tilde{\mathbf{v}}) \in \text{Range}(T)$ and $\lambda \cdot \mathbf{w} = T(\lambda \cdot \mathbf{v}) \in \text{Range}(T)$. $\square$

An important fact connecting ranges to bases: if $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for V , then $\text{Range}(T) = \text{Span}(\{T(\mathbf{b}_1), \dots, T(\mathbf{b}_n)\})$.

5.1.4 Worked Example: Linearity with Parameters

Example 5.1.14. Let $a, b \in \mathbb{R}$. Show that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) \stackrel{\text{df}}{=} (x - y, axy, 2x + y + b)$ is a linear transformation if and only if $a = b = 0$.

Solution. ($\Leftarrow$) Suppose $a = b = 0$, so $T(x, y) = (x - y, 0, 2x + y)$. Let $x, y, \tilde{x}, \tilde{y} \in \mathbb{R}$ and $\lambda \in \mathbb{R}$.

$$\begin{aligned} T((x, y) + (\tilde{x}, \tilde{y})) &= ((x + \tilde{x}) - (y + \tilde{y}), 0, 2(x + \tilde{x}) + (y + \tilde{y})) \\ &= (x - y, 0, 2x + y) + (\tilde{x} - \tilde{y}, 0, 2\tilde{x} + \tilde{y}) = T(x, y) + T(\tilde{x}, \tilde{y}). \end{aligned}$$

Similarly, $T(\lambda(x, y)) = (\lambda(x - y), 0, \lambda(2x + y)) = \lambda T(x, y)$. So T is linear.

($\Rightarrow$) Suppose T is linear. By Theorem 5.1.8, $T(0, 0) = (0, 0, 0)$. But $T(0, 0) = (0, 0, b)$, so $b = 0$. Then $T(1, 1) = (0, a, 3)$ and $T(2, 2) = (0, 4a, 6)$. By linearity, $T(2, 2) = 2 \cdot T(1, 1) = (0, 2a, 6)$. Comparing: $4a = 2a$, so $a = 0$. $\square$

5.2 The Rank-Nullity Theorem

5.2.1 Definitions

Definition 5.2.1 (Nullity and rank). Let V and W be vector spaces over the same field and $T \in \mathcal{L}(V, W)$.

- The **nullity** of T is $\text{Nullity}(T) \stackrel{\text{df}}{=} \dim(\text{Null}(T))$.
- The **rank** of T is $\text{Rank}(T) \stackrel{\text{df}}{=} \dim(\text{Range}(T))$.

5.2.2 Statement and Proof

Theorem 5.2.2 (The Rank-Nullity Theorem). *Let V and W be vector spaces over the same field, with V finite-dimensional, and $T \in \mathcal{L}(V, W)$. Then $\text{Nullity}(T)$ and $\text{Rank}(T)$ are both finite, and*

$$\dim(V) = \text{Rank}(T) + \text{Nullity}(T).$$

Proof. Let $n \stackrel{\text{df}}{=} \dim(V)$ and $m \stackrel{\text{df}}{=} \text{Nullity}(T)$ (finite since $\text{Null}(T) \subseteq V$). Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ be a basis for $\text{Null}(T)$. By the Basis Extension Theorem 4.1.3, extend B to a basis $B \cup C = \{\mathbf{b}_1, \dots, \mathbf{b}_m, \mathbf{v}_1, \dots, \mathbf{v}_{n-m}\}$ for V , where $C = \{\mathbf{v}_1, \dots, \mathbf{v}_{n-m}\}$ consists of $n - m$ new vectors. By the Three-Concepts-in-One Theorem 2.5.5, $V = \text{Span}(B) \oplus \text{Span}(C) = \text{Null}(T) \oplus \text{Span}(C)$.

We claim: $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_{n-m})\}$ is a basis for $\text{Range}(T)$.

Linear independence. Suppose $\lambda_1 T(\mathbf{v}_1) + \dots + \lambda_{n-m} T(\mathbf{v}_{n-m}) = \mathbf{0}_W$. By linearity, $T(\lambda_1 \mathbf{v}_1 + \dots + \lambda_{n-m} \mathbf{v}_{n-m}) = \mathbf{0}_W$, so $\lambda_1 \mathbf{v}_1 + \dots + \lambda_{n-m} \mathbf{v}_{n-m} \in \text{Null}(T) \cap \text{Span}(C) = \{\mathbf{0}_V\}$. Since C is linearly independent, all $\lambda_i = 0$.

Spanning. Let $\mathbf{w} \in \text{Range}(T)$; pick $\mathbf{v} \in V$ with $T(\mathbf{v}) = \mathbf{w}$. Write $\mathbf{v} = \sum_{i=1}^m \mu_i \mathbf{b}_i + \sum_{j=1}^{n-m} \nu_j \mathbf{v}_j$. Then $\mathbf{w} = T(\mathbf{v}) = \sum_i \mu_i T(\mathbf{b}_i) + \sum_j \nu_j T(\mathbf{v}_j) = \sum_j \nu_j T(\mathbf{v}_j)$.

Therefore $\text{Rank}(T) = n - m$, i.e. $\dim(V) = \text{Rank}(T) + \text{Nullity}(T)$. $\square$

Key Result: Rank-Nullity Theorem

For any $T \in \mathcal{L}(V, W)$ with V finite-dimensional:

$$\underbrace{\dim(V)}_{\text{domain}} = \underbrace{\text{Rank}(T)}_{\dim(\text{Range}(T))} + \underbrace{\text{Nullity}(T)}_{\dim(\text{Null}(T))}.$$

This is one of the most important theorems in linear algebra.

5.2.3 Worked Example: Rank and Nullity of the Trace

Example 5.2.3. Let $n \in \mathbb{N}$. The trace map $\text{trace} : \mathbb{R}^{n,n} \rightarrow \mathbb{R}$ is defined by $\text{trace}(\mathbf{M}) = \sum_{i=1}^n \mathbf{M}[i, i]$. Find $\text{Rank}(\text{trace})$ and $\text{Nullity}(\text{trace})$.

Solution. The standard basis $\{E^{i,j}\}_{i,j=1}^n$ for $\mathbb{R}^{n,n}$ satisfies $\text{trace}(E^{i,j}) = 1$ if $i = j$ and 0 otherwise. Hence $\text{Range}(\text{trace}) = \text{Span}(\{1\}) = \mathbb{R}$, so $\text{Rank}(\text{trace}) = 1$. By the Rank-Nullity Theorem:

$$\text{Nullity}(\text{trace}) = \dim(\mathbb{R}^{n,n}) - \text{Rank}(\text{trace}) = n^2 - 1. \quad \square$$

5.3 Ordered Bases and Coordinate Vectors

5.3.1 Ordered Bases

Definition 5.3.1 (Ordered basis). Let V be a vector space. An **ordered basis** for V is a family $(\mathbf{b}_i)_{i \in I}$ of V -vectors, indexed by an ordered set I , such that the $\mathbf{b}_i$ are all distinct and $\{\mathbf{b}_i\}_{i \in I}$ is

a basis for V .

For a finite-dimensional space with $\dim(V) = n$, we take $I = \{1, \dots, n\}$ and write the ordered basis as $(\mathbf{b}_1, \dots, \mathbf{b}_n)$.

Warning: Order Matters

An ordered basis $(\mathbf{b}_1, \mathbf{b}_2)$ and $(\mathbf{b}_2, \mathbf{b}_1)$ consist of the same vectors but are *different* ordered bases. Greek letters α, β, γ are commonly used to denote ordered bases.

The standard ordered bases for common vector spaces over $\mathbb{F}$ are:

Vector space	Standard ordered basis
$\mathbb{F}$	$(1_{\mathbb{F}})$
$\mathbb{F}^n$	$(\mathbf{e}_1, \dots, \mathbf{e}_n)$
$\mathcal{P}_n(\mathbb{F})$	$(1, X, X^2, \dots, X^n)$
$\mathbb{F}^{2,2}$	$\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$

5.3.2 Coordinate Vectors

Definition 5.3.2 (Coordinate vector). Let V be a finite-dimensional vector space over $\mathbb{F}$ with $\dim(V) = n$, and let $\beta = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be an ordered basis for V . For each $\mathbf{v} \in V$, write $\mathbf{v} = \lambda_1 \mathbf{b}_1 + \dots + \lambda_n \mathbf{b}_n$ (unique by Theorem 3.2.7). The **coordinate vector** of $\mathbf{v}$ with respect to β , denoted $[\mathbf{v}]_{\beta}$, is

$$[\mathbf{v}]_{\beta} \stackrel{\text{df}}{=} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{bmatrix} \in \mathbb{F}^{n,1}.$$

Theorem 5.3.3 (Coordinatisation is a bijection). *The coordinate map $\mathbf{v} \mapsto [\mathbf{v}]_{\beta}$ is a bijection from V to $\mathbb{F}^{n,1}$. Different vectors in V yield different coordinate vectors, and every column vector in $\mathbb{F}^{n,1}$ is $[\mathbf{v}]_{\beta}$ for some $\mathbf{v} \in V$.*

Theorem 5.3.4 (Linearity of the coordinate map). *Let V be finite-dimensional with ordered basis β . The coordinate map $\mathbf{v} \mapsto [\mathbf{v}]_{\beta}$ is a bijective linear transformation from V to $\mathbb{F}^{n,1}$:*

1. For all $\mathbf{v}, \tilde{\mathbf{v}} \in V$: $[\mathbf{v} +_V \tilde{\mathbf{v}}]_{\beta} = [\mathbf{v}]_{\beta} + [\tilde{\mathbf{v}}]_{\beta}$.
2. For all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$: $[\lambda \cdot_V \mathbf{v}]_{\beta} = \lambda \cdot [\mathbf{v}]_{\beta}$.

The coordinate map for V with respect to β is called the **coordinate map for V w.r.t. β** .

Key Result: Abstract $\leftrightarrow$ Concrete

The coordinate map provides a bijective linear correspondence between an abstract n -dimensional space V and the concrete space $\mathbb{F}^{n,1}$:

$$\underbrace{V}_{\text{abstract}} \xrightarrow{\mathbf{v} \mapsto [\mathbf{v}]_{\beta}} \underbrace{\mathbb{F}^{n,1}}_{\text{concrete}}.$$

Computations can be carried out in $\mathbb{F}^{n,1}$ and translated back to V .

5.3.3 Specifying a Linear Transformation by Its Values on a Basis

Theorem 5.3.5 (Specifying a linear transformation by values on a basis). *Let V be a finite-dimensional vector space with $\dim(V) = n$, and let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Let W be any vector space over the same field, and let $\mathbf{w}_1, \dots, \mathbf{w}_n$ be any W -vectors (not necessarily distinct). Then there exists a unique $T \in \mathcal{L}(V, W)$ such that $T(\mathbf{v}_i) = \mathbf{w}_i$ for all $i \in \{1, \dots, n\}$.*

Proof. Existence. Define $T \stackrel{\text{df}}{=} \mathcal{S}_{\mathbf{w}_1, \dots, \mathbf{w}_n} \circ [\cdot]_\beta \in \mathcal{L}(V, W)$ (a composition of linear maps). For each i , $[\mathbf{v}_i]_\beta = \mathbf{e}_i$, so $T(\mathbf{v}_i) = \mathcal{S}_{\mathbf{w}_1, \dots, \mathbf{w}_n}(\mathbf{e}_i) = \mathbf{w}_i$.

Uniqueness. If $S \in \mathcal{L}(V, W)$ also satisfies $S(\mathbf{v}_i) = \mathbf{w}_i$ for all i , then for any $\mathbf{v} = \sum_i \lambda_i \mathbf{v}_i$: $S(\mathbf{v}) = \sum_i \lambda_i S(\mathbf{v}_i) = \sum_i \lambda_i \mathbf{w}_i = T(\mathbf{v})$. Hence $S = T$. $\square$

Exam Tip: A linear transformation is determined by its values on a basis

To uniquely define a linear transformation $T : V \rightarrow W$, it suffices to specify $T(\mathbf{v}_i) = \mathbf{w}_i$ for each vector $\mathbf{v}_i$ in a basis for V . The values on all other vectors are then forced by linearity. For instance, if $T(1, 1) = (3, -1)$ and $T(1, 2) = (0, 4)$ and $\{(1, 1), (1, 2)\}$ is a basis for $\mathbb{R}^2$, then $T(4, 6) = T(2(1, 1) + 2(1, 2)) = 2(3, -1) + 2(0, 4) = (6, 6)$.

5.3.4 Worked Example: Coordinate Vectors in $\mathbb{R}^{2,2}$

Example 5.3.6. Let $\beta = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right)$ and $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & -1 \end{bmatrix}$. Find $[\mathbf{A}]_\beta$.

Solution. Solve $\mathbf{A} = a\beta_1 + b\beta_2 + c\beta_3 + d\beta_4$. Row-reducing the augmented system yields $a = 1, b = 2, c = 0, d = -1$, so $[\mathbf{A}]_\beta = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$. $\square$

5.4 Matrix Representations of Linear Transformations

5.4.1 The Matrix Representation Theorem

It can be difficult to work directly with linear transformations between abstract vector spaces. By using coordinate vectors, we can view any such transformation as multiplication by a matrix.

Theorem 5.4.1 (Matrix representation of a linear transformation). *Let V and W be finite-dimensional vector spaces over $\mathbb{F}$, with $\dim(V) = m$ and $\dim(W) = n$. Let β and γ be ordered bases for V and W , respectively, and let $T \in \mathcal{L}(V, W)$. Then there exists a unique $n \times m$ matrix $\mathbf{M} \in \mathbb{F}^{n,m}$, called the **matrix representation of T with respect to (β, γ)** and denoted $[T]_\beta^\gamma$, such that*

$$[T(\mathbf{v})]_\gamma = \mathbf{M} \cdot [\mathbf{v}]_\beta \quad \text{for all } \mathbf{v} \in V.$$

Different linear transformations yield different matrix representations (with respect to the same pair of bases). Different choices of (β, γ) yield different matrices for the same T , but all representations have the same dimensions $n \times m$.

Key Result: Abstract $\rightarrow$ Concrete via Matrix Multiplication

The diagram

$$\underbrace{V}_{\text{abstract}} \xrightarrow{T} W \iff \underbrace{\mathbb{F}^{m,1}}_{\text{concrete}} \xrightarrow{\cdot [T]_{\beta}^{\gamma}} \mathbb{F}^{n,1}$$

commutes: $[T(\mathbf{v})]_{\gamma} = [T]_{\beta}^{\gamma} \cdot [\mathbf{v}]_{\beta}$. Do all computations in the concrete coordinate setting and translate back to V .

5.4.2 A Formula for the Matrix Representation

Theorem 5.4.2 (Formula for the matrix representation). *Under the hypotheses of Theorem 5.4.1, let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$. Then*

$$[T]_{\beta}^{\gamma} = \begin{bmatrix} [T(\mathbf{v}_1)]_{\gamma} & [T(\mathbf{v}_2)]_{\gamma} & \cdots & [T(\mathbf{v}_m)]_{\gamma} \end{bmatrix}.$$

That is, the j -th column of $[T]_{\beta}^{\gamma}$ is the coordinate vector of $T(\mathbf{v}_j)$ with respect to γ .

Exam Tip: Computing a Matrix Representation

To find $[T]_{\beta}^{\gamma}$:

1. Enumerate the ordered basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$.
2. For each $\mathbf{v}_j$, compute $T(\mathbf{v}_j)$ explicitly.
3. Express each $T(\mathbf{v}_j)$ as a linear combination of the γ -basis to get $[T(\mathbf{v}_j)]_{\gamma}$.
4. Stack these as columns: $[T]_{\beta}^{\gamma} = [[T(\mathbf{v}_1)]_{\gamma} \cdots [T(\mathbf{v}_m)]_{\gamma}]$.

5.4.3 Worked Example: Matrix Representation of a Differential Operator

Example 5.4.3. Define $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}), \mathbb{R}^{2,2})$ by

$$T(P) \stackrel{\text{df}}{=} \begin{bmatrix} P'(0) & 2P(1) \\ 0 & P''(3) \end{bmatrix}.$$

Let $\alpha = (1, X, X^2)$ be the standard ordered basis for $\mathcal{P}_2(\mathbb{R})$, and β the standard ordered basis for $\mathbb{R}^{2,2}$. Find $[T]_{\alpha}^{\beta}$.

Solution. Apply T to each α -basis element:

$$T(1) = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad T(X) = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, \quad T(X^2) = \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix}.$$

(Note: $1' = 0$, $1'' = 0$; $X' = 1$, $X'' = 0$; $(X^2)' = 2X$, so $(X^2)'(0) = 0$; $(X^2)'' = 2$.)

Express each in the standard basis β for $\mathbb{R}^{2,2}$:

$$[T(1)]_{\beta} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \quad [T(X)]_{\beta} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \quad [T(X^2)]_{\beta} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}.$$

Therefore,

$$[T]_{\alpha}^{\beta} = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}. \quad \square$$

Chapter 6

Composition, Invertibility, and Further Matrix Theory

6.1 Further Matrix Representation Theory

6.1.1 Recovering a Linear Transformation from a Matrix

The two theorems stated in Week 5 (Theorem 5.4.1 and Theorem 5.4.2) show how to go from an abstract linear transformation T to a concrete matrix $[T]_\beta^\gamma$. A natural question is whether the correspondence runs the other way: does every matrix arise as the matrix representation of some linear transformation?

Theorem 6.1.1 (Recovering an abstract linear transformation from a given matrix). *Let V and W be finite-dimensional vector spaces over a field $\mathbb{F}$, with $\dim(V) = m$ and $\dim(W) = n$. Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ be ordered bases for V and W , respectively. Then for every $\mathbf{M} \in \mathbb{F}^{n,m}$ there exists a unique $T \in \mathcal{L}(V, W)$ such that $\mathbf{M} = [T]_\beta^\gamma$.*

Proof. By Theorem 5.3.5, there is precisely one $T \in \mathcal{L}(V, W)$ satisfying

$$\forall j \in \{1, \dots, m\} : \quad T(\mathbf{v}_j) = \sum_{i=1}^n \mathbf{M}[i, j] \mathbf{w}_i,$$

where $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$. By Theorem 5.4.2, this is exactly the condition $[T]_\beta^\gamma = \mathbf{M}$. $\square$

Together, Theorems 5.4.1, 5.4.2, and 6.1.1 establish a *bijection* between $\mathcal{L}(V, W)$ and $\mathbb{F}^{n,m}$: every abstract linear transformation corresponds to exactly one concrete matrix (once ordered bases are fixed), and every matrix comes from exactly one linear transformation.

Key Result: Abstract $\leftrightarrow$ Concrete (both directions)

Fix ordered bases β for V and γ for W .

- Every $T \in \mathcal{L}(V, W)$ gives a unique $\mathbf{M} = [T]_\beta^\gamma \in \mathbb{F}^{n,m}$ (Theorems 5.4.1–5.4.2).
- Every $\mathbf{M} \in \mathbb{F}^{n,m}$ comes from a unique $T \in \mathcal{L}(V, W)$ (Theorem 6.1.1).

6.1.2 The Identity Transformation and the Identity Matrix

Example 6.1.2 (Matrix representation of the identity). Let V be a finite-dimensional vector space over $\mathbb{F}$ with $\dim(V) = n$, and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be any ordered basis of V . Then

$$[I_V]_\beta^\beta = I_n \in \mathbb{F}^{n,n}.$$

Proof. By Theorem 5.4.2, the i -th column of $[I_V]_\beta^\beta$ is $[I_V(\mathbf{v}_i)]_\beta = [\mathbf{v}_i]_\beta$. Since $\mathbf{v}_i = 0 \cdot \mathbf{v}_1 + \dots + 1 \cdot \mathbf{v}_i + \dots + 0 \cdot \mathbf{v}_n$, we have $[\mathbf{v}_i]_\beta = \mathbf{e}_i$. Therefore $[I_V]_\beta^\beta = [\mathbf{e}_1 \ \dots \ \mathbf{e}_n] = I_n$. $\square$

Warning: Mixed Bases and the Identity

If β and γ are *different* ordered bases for V , then in general $[I_V]_{\beta}^{\gamma} \neq I_n$. However, $[I_V]_{\beta}^{\gamma}$ is always an invertible matrix (it is a *change-of-basis matrix*).

6.2 Composition of Linear Transformations and Matrix Multiplication

6.2.1 The Composition Theorem

Recall from Week 5 that composition of linear transformations is generally *non-commutative*: there exist $S, T \in \mathcal{L}(V)$ with $S \circ T \neq T \circ S$. Matrix multiplication (over $\mathbb{R}$) is likewise non-commutative. The following theorem makes this parallel precise.

Theorem 6.2.1 (Composition of linear transformations and matrix multiplication). *Let U, V , and W be finite-dimensional vector spaces over a field $\mathbb{F}$. Let α, β , and γ be ordered bases for U, V , and W , respectively. Let $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$. Then*

$$[T \circ S]_{\alpha}^{\gamma} = [T]_{\beta}^{\gamma} [S]_{\alpha}^{\beta}.$$

Proof. For any $\mathbf{u} \in U$,

$$[T \circ S]_{\alpha}^{\gamma}[\mathbf{u}]_{\alpha} = [(T \circ S)(\mathbf{u})]_{\gamma} = [T(S(\mathbf{u}))]_{\gamma} = [T]_{\beta}^{\gamma} [S(\mathbf{u})]_{\beta} = [T]_{\beta}^{\gamma} [S]_{\alpha}^{\beta} [\mathbf{u}]_{\alpha}.$$

Since this holds for every $\mathbf{u} \in U$, uniqueness of matrix representations (Theorem 5.4.1) gives $[T \circ S]_{\alpha}^{\gamma} = [T]_{\beta}^{\gamma} [S]_{\alpha}^{\beta}$. $\square$

Key Result: Matrix Multiplication = Composition

Composition of linear transformations corresponds exactly to matrix multiplication of their representations:

$$[T \circ S]_{\alpha}^{\gamma} = [T]_{\beta}^{\gamma} [S]_{\alpha}^{\beta}.$$

The subscript/superscript labels “chain”: the superscript of $[S]$ equals the subscript of $[T]$. The outermost labels give the basis pair for $T \circ S$.

More generally, $[T]_{\gamma}^{\delta} [S]_{\beta}^{\gamma} [R]_{\alpha}^{\beta} = [T \circ S \circ R]_{\alpha}^{\delta}$.

6.2.2 Worked Example: Verifying the Composition Theorem Directly

Example 6.2.2. Define $S \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ and $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}), \mathbb{R}^2)$ by

$$S(P) \stackrel{\text{df}}{=} (3 + X) \cdot P' + 2P, \quad T(a + bX + cX^2) \stackrel{\text{df}}{=} (a + c, a - b).$$

Let β and γ denote the standard ordered bases for $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^2$, respectively. We verify directly that $[T]_{\beta}^{\gamma} [S]_{\beta}^{\beta} = [T \circ S]_{\beta}^{\gamma}$.

Computing $[S]_{\beta}^{\beta}$.

$$\begin{aligned} S(1) &= (3 + X) \cdot 0 + 2 \cdot 1 = 2, \\ S(X) &= (3 + X) \cdot 1 + 2X = 3 + 3X, \\ S(X^2) &= (3 + X) \cdot 2X + 2X^2 = 6X + 4X^2. \end{aligned}$$

Hence $[S]_{\beta}^{\beta} = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{bmatrix}$.

Computing $[T]_{\beta}^{\gamma}$.

$$T(1) = (1, 1), \quad T(X) = (0, -1), \quad T(X^2) = (1, 0).$$

Hence $[T]_{\beta}^{\gamma} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$.

Product $[T]_{\beta}^{\gamma}[S]_{\beta}^{\beta}$.

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 & 0 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 0 & -6 \end{bmatrix}.$$

Computing $[T \circ S]_{\beta}^{\gamma}$ directly.

$$(T \circ S)(1) = T(2) = (2, 2),$$

$$(T \circ S)(X) = T(3 + 3X) = (3, 0),$$

$$(T \circ S)(X^2) = T(6X + 4X^2) = (4, -6).$$

So $[T \circ S]_{\beta}^{\gamma} = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 0 & -6 \end{bmatrix}$, confirming Theorem 6.2.1. $\square$

6.3 Injectivity, Surjectivity, and Invertibility of Linear Transformations

6.3.1 Definitions

Definition 6.3.1 (Injectivity, surjectivity, bijectivity). Let V and W be vector spaces over the same field, and let $T \in \mathcal{L}(V, W)$.

- T is **injective** (one-to-one) if $\forall \mathbf{v}, \tilde{\mathbf{v}} \in V : T(\mathbf{v}) = T(\tilde{\mathbf{v}}) \Rightarrow \mathbf{v} = \tilde{\mathbf{v}}$. Equivalently, different inputs yield different outputs.
- T is **surjective onto** W if $\forall \mathbf{w} \in W, \exists \mathbf{v} \in V : \mathbf{w} = T(\mathbf{v})$. Equivalently, every W -vector is the output of some V -vector input.
- T is **bijective onto** W if it is both injective and surjective onto W ; equivalently, every W -vector is the output of *exactly one* V -vector input.

6.3.2 The Null-Space Test for Injectivity

For general functions, checking injectivity requires verifying the definition directly, which can be tedious. Linear transformations admit a far more convenient criterion.

Theorem 6.3.2 (The Null-Space Test for Injectivity). Let V and W be vector spaces over the same field, and let $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is injective} \iff \text{Null}(T) = \{\mathbf{0}_V\}.$$

Proof. ($\Rightarrow$) Suppose T is injective. Then $\mathbf{0}_V$ is the only vector mapping to $\mathbf{0}_W$ (since $T(\mathbf{0}_V) = \mathbf{0}_W$ and injectivity forces uniqueness). Hence $\text{Null}(T) = \{\mathbf{0}_V\}$.

($\Leftarrow$) Suppose $\text{Null}(T) = \{\mathbf{0}_V\}$. Let $\mathbf{v}, \tilde{\mathbf{v}} \in V$ satisfy $T(\mathbf{v}) = T(\tilde{\mathbf{v}})$. By linearity, $T(\mathbf{v} - \tilde{\mathbf{v}}) = \mathbf{0}_W$, so $\mathbf{v} - \tilde{\mathbf{v}} \in \text{Null}(T) = \{\mathbf{0}_V\}$, giving $\mathbf{v} = \tilde{\mathbf{v}}$. Hence T is injective. $\square$

Key Result: Null-Space Test for Injectivity

T is injective $\iff \text{Null}(T) = \{\mathbf{0}_V\} \iff \text{Nullity}(T) = 0$.

6.3.3 Injective Linear Transformations Preserve Linear Independence

Theorem 6.3.3 (Injective linear transformations preserve linear independence). *Let V and W be vector spaces over a field $\mathbb{F}$, let $T \in \mathcal{L}(V, W)$ be injective, and let $S \subseteq V$. Then S is a linearly independent subset of V if and only if $T[S] := \{T(\mathbf{v}) : \mathbf{v} \in S\}$ is a linearly independent subset of W .*

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$ be distinct.

($\Rightarrow$) Suppose S is linearly independent. Let $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ satisfy $\sum_i \lambda_i T(\mathbf{v}_i) = \mathbf{0}_W$. By linearity, $T(\sum_i \lambda_i \mathbf{v}_i) = \mathbf{0}_W$, so $\sum_i \lambda_i \mathbf{v}_i \in \text{Null}(T) = \{\mathbf{0}_V\}$. Since S is linearly independent, $\lambda_1 = \dots = \lambda_n = 0$. Hence $T[S]$ is linearly independent.

($\Leftarrow$) Suppose $T[S]$ is linearly independent. Let $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ satisfy $\sum_i \lambda_i \mathbf{v}_i = \mathbf{0}_V$. Applying T and using linearity gives $\sum_i \lambda_i T(\mathbf{v}_i) = T(\mathbf{0}_V) = \mathbf{0}_W$. Since $T[S]$ is linearly independent, $\lambda_1 = \dots = \lambda_n = 0$. Hence S is linearly independent. $\square$

Key Result: Injective Maps Preserve Linear Independence Both Ways

If T is injective, then $S \subseteq V$ is linearly independent $\iff T[S]$ is linearly independent.
The preservation is *bidirectional*.

6.3.4 Surjective Linear Transformations Preserve the Spanning Property

Theorem 6.3.4 (Surjective linear transformations preserve the spanning property). *Let V and W be vector spaces over the same field, let $T \in \mathcal{L}(V, W)$ be surjective onto W , and let $S \subseteq V$. If $\text{Span}(S) = V$, then $\text{Span}(T[S]) = W$.*

Warning: Spanning is Not Preserved Backwards

The converse of Theorem 6.3.4 is *false* in general: even if $\text{Span}(T[S]) = W$, it need not follow that $\text{Span}(S) = V$. (Construct a counterexample using a projection map.)

6.3.5 Checking Injectivity and Surjectivity via Matrix Representations

Theorem 6.3.5 (Checking injectivity and surjectivity using matrix representations). *Let V and W be finite-dimensional vector spaces over the same field. Let β and γ be ordered bases for V and W , respectively. Let $T \in \mathcal{L}(V, W)$. Then:*

1. T is injective if and only if the RREF of $[T]_{\beta}^{\gamma}$ has a pivot in every column.
2. T is surjective if and only if the RREF of $[T]_{\beta}^{\gamma}$ has a pivot in every row.

Exam Tip: Injectivity and Surjectivity from the RREF

Compute $[T]_{\beta}^{\gamma}$ and row-reduce.

- **Pivot in every column** $\Rightarrow$ injective (no free variables $\Rightarrow$ trivial null space).
- **Pivot in every row** $\Rightarrow$ surjective (every row equation is satisfiable).
- If $[T]_{\beta}^{\gamma}$ is square ($\dim V = \dim W$), both conditions are equivalent to the matrix being invertible.

6.3.6 Worked Example: Injectivity, Surjectivity, Null Space, and Range

Example 6.3.6. Define $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}), \mathcal{P}_3(\mathbb{R}))$ by

$$T(P) \stackrel{\text{df}}{=} X \cdot P + P',$$

and let β, γ be the standard ordered bases for $\mathcal{P}_2(\mathbb{R})$ and $\mathcal{P}_3(\mathbb{R})$.

1. Computing $[T]_{\beta}^{\gamma}$.

$$T(1) = X, \quad T(X) = X^2 + 1, \quad T(X^2) = X^3 + 2X.$$

$$[T]_{\beta}^{\gamma} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

2–3. *Injectivity and surjectivity.* Row-reducing $[T]_{\beta}^{\gamma}$:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad (\text{RREF}).$$

The RREF has a pivot in every column, so T is **injective**. The RREF does *not* have a pivot in every row (the last row is all zeros), so T is **not surjective** onto $\mathcal{P}_3(\mathbb{R})$.

4. *Null space.* Since T is injective, $\text{Null}(T) = \{0_{\mathcal{P}_2(\mathbb{R})}\}$ and $\text{Nullity}(T) = 0$.

5. *Range.* By the Rank-Nullity Theorem (Theorem 5.2.2),

$$\dim(\mathcal{P}_2(\mathbb{R})) = \text{Rank}(T) + \text{Nullity}(T) \implies 3 = \text{Rank}(T) + 0 \implies \text{Rank}(T) = 3.$$

Since $\text{Range}(T) = \text{Span}(\{T(1), T(X), T(X^2)\}) = \text{Span}(\{X, X^2 + 1, X^3 + 2X\})$ and these three polynomials are linearly independent, $\{X, X^2 + 1, X^3 + 2X\}$ is a basis for $\text{Range}(T)$. $\square$

6.3.7 Bijective Linear Transformations Preserve Bases and Dimension

Corollary 6.3.7 (Bijective linear transformations respect bases and dimension). *Let V and W be vector spaces over the same field, and let $T \in \mathcal{L}(V, W)$ be bijective onto W . Then T sends every basis for V to a basis for W , and consequently $\dim(V) = \dim(W)$.*

Corollary 6.3.8 (Different dimensions $\Rightarrow$ no bijective linear transformation). *Let V and W be vector spaces over the same field. If $\dim(V) \neq \dim(W)$, there is no bijective linear transformation from V onto W (and none from W onto V).*

Corollary 6.3.9 (Checking bijectivity using matrix representations). *Under the hypotheses of Theorem 6.3.5, T is bijective onto W if and only if $[T]_{\beta}^{\gamma}$ is a square matrix of full rank, and hence invertible.*

Remark. If $[T]_{\beta}^{\gamma}$ is square, then $\dim(V) = \dim(W)$ automatically. Conversely, if $[T]_{\beta}^{\gamma}$ is not square, then $\dim(V) \neq \dim(W)$ and T cannot be bijective.

Key Result: Bijective Linear Transformations Preserve Dimension

T bijective $\Rightarrow \dim(V) = \dim(W)$. $[T]_{\beta}^{\gamma}$ invertible $\iff T$ is bijective.

6.3.8 Invertibility of Linear Transformations

Definition 6.3.10 (Invertibility). Let V and W be vector spaces over the same field, and let $T \in \mathcal{L}(V, W)$. T is said to be **invertible from V to W** if there exists $S \in \mathcal{L}(W, V)$ such that

$$S \circ T = I_V \quad \text{and} \quad T \circ S = I_W.$$

Such an S is called an **inverse linear transformation of T** .

Theorem 6.3.11 (Inverse linear transformations are unique). *Let $T \in \mathcal{L}(V, W)$ be invertible. Then T has precisely one inverse linear transformation, denoted T^{-1} .*

Proof. Inverse functions are unique; since T^{-1} is an inverse function that is also linear (Theorem 6.3.12 below), uniqueness follows. $\square$

Theorem 6.3.12 (Bijectivity and invertibility coincide for linear transformations). *Let V and W be vector spaces over a field $\mathbb{F}$, and let $T \in \mathcal{L}(V, W)$. Then T is bijective onto W if and only if T is invertible from V to W .*

Proof. ($\Leftarrow$) By definition, if T is invertible then $T \circ S = I_W$ and $S \circ T = I_V$ for some $S \in \mathcal{L}(W, V)$; these conditions guarantee that T is bijective.

($\Rightarrow$) Suppose T is bijective onto W . The set-theoretic inverse $T^{-1} : W \rightarrow V$ exists and is a bijection. We must show $T^{-1} \in \mathcal{L}(W, V)$. Let $\mathbf{w}, \tilde{\mathbf{w}} \in W$ and $\lambda \in \mathbb{F}$. Set $\mathbf{v} \stackrel{\text{df}}{=} T^{-1}(\mathbf{w})$ and $\tilde{\mathbf{v}} \stackrel{\text{df}}{=} T^{-1}(\tilde{\mathbf{w}})$, so $T(\mathbf{v}) = \mathbf{w}$ and $T(\tilde{\mathbf{v}}) = \tilde{\mathbf{w}}$. By linearity of T ,

$$T(\mathbf{v} + \lambda \tilde{\mathbf{v}}) = \mathbf{w} + \lambda \tilde{\mathbf{w}}.$$

Applying T^{-1} to both sides:

$$T^{-1}(\mathbf{w} + \lambda \tilde{\mathbf{w}}) = \mathbf{v} + \lambda \tilde{\mathbf{v}} = T^{-1}(\mathbf{w}) + \lambda T^{-1}(\tilde{\mathbf{w}}).$$

Hence T^{-1} is linear, i.e. $T^{-1} \in \mathcal{L}(W, V)$. $\square$

Key Result: Bijective $\iff$ Invertible (for Linear Transformations)

For $T \in \mathcal{L}(V, W)$:

$$T \text{ bijective onto } W \iff T \text{ invertible from } V \text{ to } W.$$

Moreover, $T^{-1} \in \mathcal{L}(W, V)$ (the inverse is automatically linear).

6.3.9 Worked Example: Finding an Invertible Linear Transformation and Its Inverse

Example 6.3.13. Define $T \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathbb{R}^{2,2})$ by

$$T(P) \stackrel{\text{df}}{=} \begin{bmatrix} P(0) & P(1) \\ P(2) & P(3) \end{bmatrix}.$$

Let β and γ denote the standard ordered bases for $\mathcal{P}_3(\mathbb{R})$ and $\mathbb{R}^{2,2}$, respectively.

Step 1: Matrix representation.

$$T(1) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad T(X) = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}, \quad T(X^2) = \begin{bmatrix} 0 & 1 \\ 4 & 9 \end{bmatrix}, \quad T(X^3) = \begin{bmatrix} 0 & 1 \\ 8 & 27 \end{bmatrix}.$$

$$[T]_{\beta}^{\gamma} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{bmatrix}.$$

Step 2: Show T is invertible. Row-reducing the augmented matrix $[[T]_{\beta}^{\gamma} \mid I_4]$ yields RREF equal to I_4 , confirming that $[T]_{\beta}^{\gamma}$ is invertible, so T is bijective (Corollary 6.3.9) and hence invertible.

Step 3: The inverse. The inverse matrix is

$$\left([T]_{\beta}^{\gamma}\right)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{11}{6} & 3 & -\frac{3}{2} & \frac{1}{3} \\ 1 & -\frac{5}{2} & 2 & -\frac{1}{2} \\ -\frac{1}{6} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{6} \end{bmatrix}.$$

By Theorem 6.2.1, $[T^{-1}]_{\gamma}^{\beta} = \left([T]_{\beta}^{\gamma}\right)^{-1}$. Reading off the columns gives, for all $a, b, c, d \in \mathbb{R}$:

$$\begin{aligned} T^{-1}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) &= a \cdot 1 + \left(-\frac{11}{6}a + 3b - \frac{3}{2}c + \frac{1}{3}d\right) X \\ &\quad + \left(a - \frac{5}{2}b + 2c - \frac{1}{2}d\right) X^2 + \left(-\frac{1}{6}a + \frac{1}{2}b - \frac{1}{2}c + \frac{1}{6}d\right) X^3. \quad \square \end{aligned}$$

Exam Tip: Finding T^{-1} via the Inverse Matrix

If $T \in \mathcal{L}(V, W)$ is bijective (hence invertible), compute $[T^{-1}]_{\gamma}^{\beta}$ by inverting the matrix $[T]_{\beta}^{\gamma}$. Then read off the explicit formula for T^{-1} from the columns of $[T^{-1}]_{\gamma}^{\beta}$, using the β -basis vectors as coefficient targets.

6.3.10 Isomorphic Vector Spaces

Definition 6.3.14 (Isomorphic vector spaces). Let V and W be vector spaces over the same field. V is said to be **isomorphic to** W if any one (and hence all) of the following equivalent statements hold:

1. There exists a bijective linear transformation from V onto W .
2. There exists an invertible linear transformation from V to W .
3. $\dim(V) = \dim(W)$ (in the finite-dimensional case).

We write $V \cong W$ and say V and W are **isomorphic**. Isomorphism is an equivalence relation.

Key Result: Isomorphism and Dimension

For finite-dimensional vector spaces over the same field:

$$V \cong W \iff \dim(V) = \dim(W).$$

In particular, every n -dimensional vector space over $\mathbb{F}$ is isomorphic to $\mathbb{F}^n$.

6.3.11 The Fundamental Theorem of Finite-Dimensional Vector Spaces

The following theorem unifies everything we have learned about matrix representations. It asserts that the “concrete” world of matrix multiplication is isomorphic to the “abstract” world of linear transformations, and that the coordinate maps $[\bullet]_{\alpha}$, $[\bullet]_{\beta}$, $[\bullet]_{\gamma}$ are the bridges between the two worlds.

Theorem 6.3.15 (Fundamental Theorem of Finite-Dimensional Vector Spaces). *Let U , V , and W be finite-dimensional vector spaces over the same field $\mathbb{F}$. Let α , β , and γ be ordered bases for U , V , and W , respectively. Let $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$.*

Then the following diagram commutes (“ $\equiv$ ” indicates an isomorphism):

$$\begin{array}{ccccc} U & \xrightarrow{S} & V & \xrightarrow{T} & W \\ \downarrow [\bullet]_\alpha \equiv & & & & \downarrow \equiv [\bullet]_\gamma \\ \mathbb{F}^{\dim(U),1} & \xrightarrow{\cdot[S]_\alpha^\beta} & \mathbb{F}^{\dim(V),1} & \xrightarrow{\cdot[T]_\beta^\gamma} & \mathbb{F}^{\dim(W),1} \end{array}$$

and the bottom row (from $\mathbb{F}^{\dim(U),1}$ all the way to $\mathbb{F}^{\dim(W),1}$) is multiplication by $[T \circ S]_\alpha^\gamma$.

Furthermore, the following three maps are all isomorphisms:

1. $[\bullet]_\alpha^\beta: \mathcal{L}(U, V) \rightarrow \mathbb{F}^{\dim(V), \dim(U)}$.
2. $[\bullet]_\beta^\gamma: \mathcal{L}(V, W) \rightarrow \mathbb{F}^{\dim(W), \dim(V)}$.
3. $[\bullet]_\alpha^\gamma: \mathcal{L}(U, W) \rightarrow \mathbb{F}^{\dim(W), \dim(U)}$.

Key Result: The Fundamental Theorem

For finite-dimensional vector spaces, the matrix-representation map $[\bullet]_\alpha^\beta: \mathcal{L}(U, V) \rightarrow \mathbb{F}^{\dim(V), \dim(U)}$ is an *isomorphism*. In particular:

$$[T \circ S]_\alpha^\gamma = [T]_\beta^\gamma [S]_\alpha^\beta.$$

Thus composition of linear maps corresponds *exactly* to matrix multiplication.

Chapter 7

Change of Coordinates

7.1 Change of Coordinates

We know from Theorem 5.4.1 that every linear transformation $T \in \mathcal{L}(V, W)$ can be represented by a matrix $[T]_{\beta}^{\gamma}$ once ordered bases β for V and γ for W are chosen. A natural question is: how does this matrix change if we switch to a different pair of bases? The theory of *change-of-coordinate matrices* answers this question precisely.

7.1.1 Matrix Representation of an Inverse

We begin by connecting the invertibility of T to the invertibility of its matrix representation. Inversion of invertible linear transformations corresponds to inversion of invertible matrices.

Theorem 7.1.1 (Matrix representation of an inverse linear transformation). *Let V and W be finite-dimensional vector spaces over a field $\mathbb{F}$. Let β and γ be ordered bases for V and W , respectively. Let $T \in \mathcal{L}(V, W)$.*

Then T is invertible from V to W if and only if $[T]_{\beta}^{\gamma}$ is an invertible square $\mathbb{F}$ -matrix, in which case

$$[T^{-1}]_{\gamma}^{\beta} = \left([T]_{\beta}^{\gamma}\right)^{-1}.$$

Proof. ($\Rightarrow$) Suppose T is invertible, so $\dim(V) = \dim(W)$. Let $n = \dim(V) = \dim(W)$. Since $T^{-1} \circ T = I_V$ and $T \circ T^{-1} = I_W$, applying the Matrix Representation Theorem 5.4.1 and the Composition Theorem 6.2.1 gives

$$\mathbf{I}_n = [I_V]_{\beta}^{\beta} = [T^{-1} \circ T]_{\beta}^{\beta} = [T^{-1}]_{\gamma}^{\beta} [T]_{\beta}^{\gamma},$$

$$\mathbf{I}_n = [I_W]_{\gamma}^{\gamma} = [T \circ T^{-1}]_{\gamma}^{\gamma} = [T]_{\beta}^{\gamma} [T^{-1}]_{\gamma}^{\beta}.$$

Hence $[T]_{\beta}^{\gamma}$ is an invertible $n \times n$ matrix and $[T^{-1}]_{\gamma}^{\beta} = ([T]_{\beta}^{\gamma})^{-1}$.

($\Leftarrow$) Suppose $[T]_{\beta}^{\gamma}$ is an invertible square matrix, so $\dim(V) = \dim(W) = n$. Let $\mathbf{A} \in \mathbb{F}^{n,n}$ satisfy $[T]_{\beta}^{\gamma} \mathbf{A} = \mathbf{I}_n = \mathbf{A} [T]_{\beta}^{\gamma}$. By the Fundamental Theorem (Theorem 6.3.15) there exists $S \in \mathcal{L}(W, V)$ with $[S]_{\gamma}^{\beta} = \mathbf{A}$, and then

$$[S \circ T]_{\beta}^{\beta} = [S]_{\gamma}^{\beta} [T]_{\beta}^{\gamma} = \mathbf{A} [T]_{\beta}^{\gamma} = \mathbf{I}_n = [I_V]_{\beta}^{\beta},$$

$$[T \circ S]_{\gamma}^{\gamma} = [T]_{\beta}^{\gamma} [S]_{\gamma}^{\beta} = [T]_{\beta}^{\gamma} \mathbf{A} = \mathbf{I}_n = [I_W]_{\gamma}^{\gamma}.$$

Hence $S \circ T = I_V$ and $T \circ S = I_W$, so T is invertible from V to W . □

Key Result: Invertibility via the Matrix

$T \in \mathcal{L}(V, W)$ is invertible $\iff [T]_{\beta}^{\gamma}$ is an invertible square matrix. Moreover, the matrix of the inverse is the inverse of the matrix:

$$[T^{-1}]_{\gamma}^{\beta} = \left([T]_{\beta}^{\gamma}\right)^{-1}.$$

7.1.2 The Change-of-Coordinate Matrix

Definition 7.1.2 (Change-of-coordinate matrix). Let V be a finite-dimensional vector space. Let β and γ be ordered bases for V . The **change-of-coordinate matrix from β to γ** is defined to be the square matrix

$$[I_V]_{\beta}^{\gamma}.$$

The name is justified by the following key property: for any $\mathbf{v} \in V$,

$$[I_V]_{\beta}^{\gamma} [\mathbf{v}]_{\beta} = [I_V(\mathbf{v})]_{\gamma} = [\mathbf{v}]_{\gamma}.$$

In other words, multiplying the β -coordinate column of $\mathbf{v}$ by $[I_V]_{\beta}^{\gamma}$ converts it into the γ -coordinate column of $\mathbf{v}$.

Remark 7.1.3. An alternative notation for the change-of-coordinate matrix from β to γ is $P_{\gamma \leftarrow \beta}$.

Example 7.1.4. Let $\alpha = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ and $\beta = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ be ordered bases for a vector space V satisfying

$$\mathbf{b}_1 = 2\mathbf{a}_1 - \mathbf{a}_2 + \mathbf{a}_3, \quad \mathbf{b}_2 = 3\mathbf{a}_2 + \mathbf{a}_3, \quad \mathbf{b}_3 = -3\mathbf{a}_1 + 2\mathbf{a}_3.$$

1. Compute $[I_V]_{\beta}^{\alpha}$.
2. Compute $[\mathbf{b}_1 - 2\mathbf{b}_2 + 2\mathbf{b}_3]_{\alpha}$.

Solution.

(1) Using the formula from Theorem 5.4.2, the columns of $[I_V]_{\beta}^{\alpha}$ are the α -coordinates of $I_V(\mathbf{b}_j) = \mathbf{b}_j$:

$$[I_V]_{\beta}^{\alpha} = \begin{bmatrix} [\mathbf{b}_1]_{\alpha} & [\mathbf{b}_2]_{\alpha} & [\mathbf{b}_3]_{\alpha} \end{bmatrix} = \begin{bmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{bmatrix}.$$

(2) Since $\mathbf{b}_1 - 2\mathbf{b}_2 + 2\mathbf{b}_3$ has β -coordinates $(1, -2, 2)^T$, we use the change-of-coordinate matrix to convert:

$$[\mathbf{b}_1 - 2\mathbf{b}_2 + 2\mathbf{b}_3]_{\alpha} = [I_V]_{\beta}^{\alpha} [\mathbf{b}_1 - 2\mathbf{b}_2 + 2\mathbf{b}_3]_{\beta} = \begin{bmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ -7 \\ 3 \end{bmatrix}. \quad \square$$

7.1.3 Change of Matrix Representations

The following theorem answers the central question of this section: given $[T]_{\beta}^{\beta}$ with respect to one basis, how do we find $[T]_{\gamma}^{\gamma}$ with respect to another?

Theorem 7.1.5 (Change of matrix representations under a change of coordinates). *Let V be a finite-dimensional vector space. Let β and γ be ordered bases for V . Let $T \in \mathcal{L}(V)$. Then*

$$[T]_{\gamma}^{\gamma} = [I_V]_{\beta}^{\gamma} [T]_{\beta}^{\beta} [I_V]_{\gamma}^{\beta}.$$

Proof. Apply the Composition Theorem (Theorem 6.2.1) twice:

$$[T]_{\gamma}^{\gamma} = [I_V \circ T \circ I_V]_{\gamma}^{\gamma} = [I_V]_{\beta}^{\gamma} [T]_{\beta}^{\beta} \dots$$

More precisely, write $T = I_V \circ T \circ I_V$ and note that the identity on V with domain basis γ and codomain basis β has matrix $[I_V]_{\gamma}^{\beta}$. Then

$$[T]_{\gamma}^{\gamma} = [I_V]_{\beta}^{\gamma} [T]_{\beta}^{\beta} [I_V]_{\gamma}^{\beta}. \quad \square$$

$\square$

Key Result: Changing Basis for an Operator

For $T \in \mathcal{L}(V)$ and two ordered bases β, γ of V :

$$[T]_{\gamma}^{\gamma} = [I_V]_{\beta}^{\gamma} [T]_{\beta}^{\beta} [I_V]_{\gamma}^{\beta} = P^{-1} [T]_{\beta}^{\beta} P,$$

where $P = [I_V]_{\gamma}^{\beta}$. Two matrices related in this way are called **similar**.

7.1.4 Computing Change-of-Coordinate Matrices

When β and γ are non-standard bases it may be hard to read off $[I_V]_{\beta}^{\gamma}$ or $[I_V]_{\gamma}^{\beta}$ directly. A standard basis σ for V makes this easy.

Using a standard basis. If σ is any standard ordered basis for V and $\beta = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, $\gamma = (\mathbf{c}_1, \dots, \mathbf{c}_n)$, then

$$[I_V]_{\beta}^{\sigma} = [\mathbf{b}_1]_{\sigma} \cdots [\mathbf{b}_n]_{\sigma}, \quad [I_V]_{\gamma}^{\sigma} = [\mathbf{c}_1]_{\sigma} \cdots [\mathbf{c}_n]_{\sigma}.$$

Both matrices are obtained by simply writing the basis vectors as columns in the standard coordinates. Then:

$$\begin{aligned} [I_V]_{\beta}^{\gamma} &= [I_V]_{\sigma}^{\gamma} [I_V]_{\beta}^{\sigma} = ([I_V]_{\gamma}^{\sigma})^{-1} [I_V]_{\beta}^{\sigma}, \\ [I_V]_{\gamma}^{\beta} &= ([I_V]_{\beta}^{\sigma})^{-1} [I_V]_{\gamma}^{\sigma}. \end{aligned}$$

Row-reduction shortcut. Both change-of-coordinate matrices can be computed simultaneously using row reduction:

$$\begin{aligned} \left[[I_V]_{\gamma}^{\sigma} \mid [I_V]_{\beta}^{\sigma} \right] &\xrightarrow{\text{row reduce}} \left[\mathbf{I}_n \mid [I_V]_{\beta}^{\gamma} \right], \\ \left[[I_V]_{\beta}^{\sigma} \mid [I_V]_{\gamma}^{\sigma} \right] &\xrightarrow{\text{row reduce}} \left[\mathbf{I}_n \mid [I_V]_{\gamma}^{\beta} \right]. \end{aligned}$$

Exam Tip: Always Put σ on the Left

To find $[I_V]_{\beta}^{\gamma}$, form the augmented matrix with the *target* basis γ (as columns in σ -coordinates) on the left and the *source* basis β (as columns in σ -coordinates) on the right. Row-reduce until the left block is $\mathbf{I}_n$; the right block is then $[I_V]_{\beta}^{\gamma}$.

Example 7.1.6. Define ordered bases β and γ for $\mathbb{R}^2$ by

$$\beta \stackrel{\text{df}}{=} ((-9, 1), (-5, -1)) \quad \text{and} \quad \gamma \stackrel{\text{df}}{=} ((1, -4), (3, -5)).$$

Compute $[I_{\mathbb{R}^2}]_{\beta}^{\gamma}$ and $[I_{\mathbb{R}^2}]_{\gamma}^{\beta}$.

Solution. Let σ be the standard basis for $\mathbb{R}^2$. Using the row-reduction shortcut to find $[I_{\mathbb{R}^2}]_{\beta}^{\gamma}$, we form $[[I_{\mathbb{R}^2}]_{\gamma}^{\sigma} \mid [I_{\mathbb{R}^2}]_{\beta}^{\sigma}]$ and row-reduce:

$$\begin{aligned} \left[\begin{array}{cc|cc} 1 & 3 & -9 & -5 \\ -4 & -5 & 1 & -1 \end{array} \right] &\xrightarrow{R_2 \mapsto R_2 + 4R_1} \left[\begin{array}{cc|cc} 1 & 3 & -9 & -5 \\ 0 & 7 & -35 & -21 \end{array} \right] \xrightarrow{R_2 \mapsto \frac{1}{7}R_2} \left[\begin{array}{cc|cc} 1 & 3 & -9 & -5 \\ 0 & 1 & -5 & -3 \end{array} \right] \\ &\xrightarrow{R_1 \mapsto R_1 - 3R_2} \left[\begin{array}{cc|cc} 1 & 0 & 6 & 4 \\ 0 & 1 & -5 & -3 \end{array} \right]. \end{aligned}$$

Hence

$$[I_{\mathbb{R}^2}]_{\beta}^{\gamma} = \begin{bmatrix} 6 & 4 \\ -5 & -3 \end{bmatrix}, \quad [I_{\mathbb{R}^2}]_{\gamma}^{\beta} = ([I_{\mathbb{R}^2}]_{\beta}^{\gamma})^{-1} = \begin{bmatrix} 6 & 4 \\ -5 & -3 \end{bmatrix}^{-1} = \begin{bmatrix} -\frac{3}{2} & -2 \\ \frac{5}{2} & 3 \end{bmatrix}. \quad \square$$

Example 7.1.7. Define $T \in \mathcal{L}(\mathbb{R}^3)$ by

$$T(x, y, z) \stackrel{\text{df}}{=} (2x + y, 5y + 3z, 8z), \quad \forall x, y, z \in \mathbb{R}.$$

Let $\beta = ((1, 0, 0), (1, 3, 0), (1, 6, 6))$ be an ordered basis for $\mathbb{R}^3$.

1. Compute $[T]_{\beta}^{\beta}$.
2. Compute $T^4(0, -3, 6)$.

Solution. Let σ denote the standard ordered basis for $\mathbb{R}^3$.

Step 1: Standard matrix. Applying T to each standard basis vector:

$$[T]_{\sigma}^{\sigma} = \begin{bmatrix} [T(1, 0, 0)]_{\sigma} & [T(0, 1, 0)]_{\sigma} & [T(0, 0, 1)]_{\sigma} \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{bmatrix}.$$

Step 2: Change-of-coordinate matrices. The β -basis vectors written in standard coordinates give

$$[I_{\mathbb{R}^3}]_{\beta}^{\sigma} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 6 \end{bmatrix}, \quad [I_{\mathbb{R}^3}]_{\sigma}^{\beta} = ([I_{\mathbb{R}^3}]_{\beta}^{\sigma})^{-1} = \begin{bmatrix} 1 & -\frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{6} \end{bmatrix}.$$

Step 3: $[T]_{\beta}^{\beta}$ via Theorem 7.1.5.

$$\begin{aligned} [T]_{\beta}^{\beta} &= [I_{\mathbb{R}^3}]_{\sigma}^{\beta} [T]_{\sigma}^{\sigma} [I_{\mathbb{R}^3}]_{\beta}^{\sigma} \\ &= \begin{bmatrix} 1 & -\frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{bmatrix}. \end{aligned}$$

Remarkably, $[T]_{\beta}^{\beta}$ is a diagonal matrix!

Step 4: Computing $T^4(0, -3, 6)$. First, convert $(0, -3, 6)$ to β -coordinates:

$$[(0, -3, 6)]_{\beta} = [I_{\mathbb{R}^3}]_{\sigma}^{\beta} [(0, -3, 6)]_{\sigma} = \begin{bmatrix} 1 & -\frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 0 \\ -3 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

Check: $(0, -3, 6) = 2(1, 0, 0) + (-3)(1, 3, 0) + 1(1, 6, 6)$. ✓

Since $[T]_{\beta}^{\beta} = \text{diag}(2, 5, 8)$, we have $[T^4]_{\beta}^{\beta} = \text{diag}(2^4, 5^4, 8^4) = \text{diag}(16, 625, 4096)$. Thus

$$[T^4(0, -3, 6)]_{\beta} = [T^4]_{\beta}^{\beta} [(0, -3, 6)]_{\beta} = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 625 & 0 \\ 0 & 0 & 4096 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 32 \\ -1875 \\ 4096 \end{bmatrix}.$$

Converting back to standard coordinates:

$$\begin{aligned} T^4(0, -3, 6) &= 32(1, 0, 0) + (-1875)(1, 3, 0) + 4096(1, 6, 6) \\ &= (2253, 18951, 24576). \quad \square \end{aligned}$$

Exam Tip: Diagonal Matrix Powers

If you can find a basis β for which $[T]_{\beta}^{\beta}$ is *diagonal*, computing T^k becomes trivial: simply raise each diagonal entry to the k -th power. This is the key motivation for diagonalisation, a topic in later weeks.

Chapter 8

Similarity, Eigenvalues, and Eigenvectors

8.1 Similarity of Square Matrices

Having studied how the matrix representation of a linear transformation changes under a change of ordered basis (Theorem 7.1.5), we now formalise the relationship between the resulting matrices as an equivalence relation.

8.1.1 Definition of Similarity

Definition 8.1.1 (Similarity of square matrices). Let $\mathbb{F}$ be a field and $n \in \mathbb{N}$. Let $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{n,n}$. The *similarity relation* on $\mathbb{F}^{n,n}$, denoted “ $\sim$ ”, is defined by

$$\forall \mathbf{A}, \mathbf{B} \in \mathbb{F}^{n,n} : \mathbf{A} \sim \mathbf{B} \iff \exists \text{ invertible } \mathbf{Q} \in \mathbb{F}^{n,n} : \mathbf{B} = \mathbf{Q}^{-1} \mathbf{A} \mathbf{Q}.$$

We read “ $\mathbf{A} \sim \mathbf{B}$ ” as “ $\mathbf{A}$ is similar to $\mathbf{B}$ ”.

Theorem 8.1.2 (Similarity is an equivalence relation). Let $\mathbb{F}$ be a field and $n \in \mathbb{N}$. The similarity relation $\sim$ on $\mathbb{F}^{n,n}$ is an equivalence relation on $\mathbb{F}^{n,n}$.

Remark 8.1.3. If $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{n,n}$ satisfy $\mathbf{A} \sim \mathbf{B}$, we may simply say that $\mathbf{A}$ and $\mathbf{B}$ are *similar*.

Key Result: Similarity is an Equivalence Relation

The relation $\mathbf{A} \sim \mathbf{B} \iff \exists \text{ invertible } \mathbf{Q} \text{ such that } \mathbf{B} = \mathbf{Q}^{-1} \mathbf{A} \mathbf{Q}$ is reflexive, symmetric, and transitive on $\mathbb{F}^{n,n}$.

8.1.2 Basic Properties of Similar Matrices

Theorem 8.1.4 (Determinant and trace are similarity invariants). Let $\mathbb{F}$ be a field, $n \in \mathbb{N}$, and $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{n,n}$. Then:

1. If $\mathbf{A} \sim \mathbf{B}$, then $\det(\mathbf{A}) = \det(\mathbf{B})$.
2. If $\mathbf{A} \sim \mathbf{B}$, then $\text{trace}(\mathbf{A}) = \text{trace}(\mathbf{B})$.

Both statements follow from standard properties of the determinant and the trace.

Warning: Similarity Is Stronger Than Equal Det/Trace

The converse of Theorem 8.1.4 does *not* hold in general: two matrices can share the same determinant and trace without being similar. Equal determinant and equal trace are necessary but not sufficient conditions for similarity.

8.1.3 Similarity of Matrix Representations

The following theorem connects the abstract notion of similarity to the change-of-coordinates theory developed in Chapter 7.

Theorem 8.1.5 (Matrix representations under different ordered bases are similar). *Let V be a finite-dimensional vector space, let β and γ be ordered bases for V , and let $T \in \mathcal{L}(V)$. Then*

$$[T]_{\beta}^{\beta} \sim [T]_{\gamma}^{\gamma}.$$

Explicitly, using the change-of-coordinates formula from Theorem 7.1.5, there exists an invertible $\mathbf{Q} \in \mathbb{F}^{n,n}$ such that

$$[T]_{\gamma}^{\gamma} = \mathbf{Q}^{-1} [T]_{\beta}^{\beta} \mathbf{Q}, \quad \text{where } \mathbf{Q} = [I_V]_{\gamma}^{\beta}.$$

Proof. By Theorem 7.1.5, with $\sigma = \beta$:

$$[T]_{\gamma}^{\gamma} = [I_V]_{\beta}^{\gamma} [T]_{\beta}^{\beta} [I_V]_{\gamma}^{\beta} = ([I_V]_{\gamma}^{\beta})^{-1} [T]_{\beta}^{\beta} [I_V]_{\gamma}^{\beta},$$

since $([I_V]_{\gamma}^{\beta})^{-1} = [I_V]_{\beta}^{\gamma}$. Setting $\mathbf{Q} = [I_V]_{\gamma}^{\beta}$ gives the result. $\square$

Key Result: Basis-Independence of Eigenvalues (Preview)

Because $[T]_{\beta}^{\beta} \sim [T]_{\gamma}^{\gamma}$ for any two ordered bases, and because $\det(\mathbf{A}) = \det(\mathbf{B})$ whenever $\mathbf{A} \sim \mathbf{B}$, it will follow (see Section 8.2) that the eigenvalues of T do not depend on the choice of basis used to compute them.

8.2 Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors have far-reaching applications, including data analytics (principal component analysis), Google's PageRank algorithm, quantum mechanics, computer animation, and material science. The terminology comes from the German word *eigen*, meaning “characteristic”.

8.2.1 Definition and Characterisation of Eigenvalues

Definition 8.2.1 (Eigenvalue). Let V be a vector space over a field $\mathbb{F}$, and let $T \in \mathcal{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T if any one of the following three equivalent conditions holds:

1. $\text{Null}(T - \lambda \cdot I_V) \neq \{\mathbf{0}_V\}$.
2. $\{\mathbf{0}_V\} \subsetneq \text{Null}(T - \lambda \cdot I_V)$.
3. $T - \lambda \cdot I_V$ is not injective.

Here $(T - \lambda \cdot I_V)(\mathbf{x}) \stackrel{\text{df}}{=} T(\mathbf{x}) - \lambda \cdot \mathbf{x}$.

Remark 8.2.2. An eigenvalue of a linear transformation is *field-specific*: λ must belong to the base field $\mathbb{F}$ of V .

Theorem 8.2.3 (Zero eigenvalue $\iff$ non-injectivity). *Let V be a vector space over $\mathbb{F}$ and $T \in \mathcal{L}(V)$. Then $0_{\mathbb{F}}$ is an eigenvalue of T if and only if T is not injective.*

Proof. $0_{\mathbb{F}}$ is an eigenvalue iff $T - 0_{\mathbb{F}} \cdot I_V = T$ is not injective, which is the stated condition. $\square$

Corollary 8.2.4 (Non-injectivity implies the existence of an eigenvalue). *If $T \in \mathcal{L}(V)$ is not injective, then T automatically has an eigenvalue, namely $0_{\mathbb{F}}$ (though there may be more).*

Example 8.2.5. Let V be a non-trivial vector space over $\mathbb{F}$.

- **Zero map $0_{V \rightarrow V}$:** For any $\lambda \in \mathbb{F}$, $\text{Null}(0 - \lambda I_V) = \text{Null}(-\lambda I_V)$. If V is trivial this is $\{0_V\}$ (no eigenvalue); if V is non-trivial and $\lambda = 0_{\mathbb{F}}$ then $\text{Null}(0) = V \neq \{0_V\}$. Hence the *only* eigenvalue of the zero map on a non-trivial V is $0_{\mathbb{F}}$.
- **Identity map I_V :** Similarly, λ is an eigenvalue of I_V iff $\text{Null}((1_{\mathbb{F}} - \lambda)I_V) \neq \{0_V\}$, which happens exactly when $\lambda = 1_{\mathbb{F}}$ (for non-trivial V). Hence the only eigenvalue of I_V is $1_{\mathbb{F}}$.

8.2.2 Computing Eigenvalues: The Characteristic Polynomial

Let V be finite-dimensional with $\dim(V) = n$, let β be any ordered basis, and let $T \in \mathcal{L}(V)$. Using the fact that matrix representations are linear and that $[I_V]_{\beta} = \mathbf{I}_n$, one can show:

$$\lambda \text{ is an eigenvalue of } T \iff [T]_{\beta}^{\beta} - \lambda \cdot \mathbf{I}_n \text{ is not invertible} \iff \det([T]_{\beta}^{\beta} - \lambda \cdot \mathbf{I}_n) = 0.$$

This motivates the following definition.

Definition 8.2.6 (Characteristic polynomial). Let V be a finite-dimensional vector space over $\mathbb{F}$ with $\dim(V) = n$, let β be any ordered basis for V , and let $T \in \mathcal{L}(V)$. The *characteristic polynomial function* of T , denoted p_T , is defined by

$$\forall \lambda \in \mathbb{F} : \quad p_T(\lambda) \stackrel{\text{df}}{=} \det([T]_{\beta}^{\beta} - \lambda \cdot \mathbf{I}_n).$$

It is a polynomial function of degree n with $\mathbb{F}$ -coefficients, and is *independent of the choice of ordered basis β* by Theorem 8.1.4 and Theorem 8.1.5.

Corollary 8.2.7 (Eigenvalues are $\mathbb{F}$ -roots of the characteristic polynomial). *Let V be a finite-dimensional vector space over $\mathbb{F}$ with $\dim(V) = n$, and let $T \in \mathcal{L}(V)$. The eigenvalues of T are precisely the $\mathbb{F}$ -roots of p_T , so T has at most n distinct eigenvalues.*

Exam Tip: Choice of Basis for Computing Eigenvalues

Since the characteristic polynomial is basis-independent, always choose the standard ordered basis σ when computing p_T — it gives the most direct matrix representation.

Example 8.2.8 (Rotation by 90 over $\mathbb{R}$ vs $\mathbb{C}$). Define $T \in \mathcal{L}(\mathbb{R}^{2,1})$ by

$$\forall \mathbf{v} \in \mathbb{R}^{2,1} : \quad T(\mathbf{v}) \stackrel{\text{df}}{=} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{v}.$$

Over $\mathbb{R}$: Letting σ denote the standard basis, $[T]_{\sigma}^{\sigma} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, so

$$p_T(\lambda) = \det \begin{bmatrix} -\lambda & 1 \\ -1 & -\lambda \end{bmatrix} = \lambda^2 + 1.$$

Since p_T has no real roots, T has *no eigenvalues* over $\mathbb{R}$.

Over $\mathbb{C}$: The same matrix representation holds over $\mathbb{C}$, giving $p_T(\lambda) = \lambda^2 + 1$, which factors over $\mathbb{C}$ as $(\lambda - i)(\lambda + i)$. Hence T has precisely two eigenvalues: $\pm i$.

Warning: Eigenvalues Are Field-Specific

As seen in Example 8.2.8, whether a linear transformation has eigenvalues depends on the base field. A transformation with no real eigenvalues may acquire eigenvalues over $\mathbb{C}$.

8.2.3 Definition and Properties of Eigenvectors

Definition 8.2.9 (Eigenvector). Let V be a vector space over $\mathbb{F}$ and $T \in \mathcal{L}(V)$. An *eigenvector* of T is a vector $\mathbf{v} \in V$ with $\mathbf{v} \neq \mathbf{0}_V$ for which there exists a scalar $\lambda \in \mathbb{F}$ such that

$$T(\mathbf{v}) = \lambda \cdot \mathbf{v}.$$

An eigenvector is thus a non-zero vector that undergoes pure *scaling* under T .

The scalar λ in the above definition is *uniquely determined* by $\mathbf{v}$ (the eigenvalue for $\mathbf{v}$): if also $T(\mathbf{v}) = \mu \cdot \mathbf{v}$, then $(\lambda - \mu)\mathbf{v} = \mathbf{0}_V$, and since $\mathbf{v} \neq \mathbf{0}_V$ we get $\lambda = \mu$. We call λ the *eigenvalue* of T for $\mathbf{v}$.

The relationship between eigenvectors and eigenvalues is symmetric: if λ is an eigenvalue of T , then any non-zero $\mathbf{v} \in \text{Null}(T - \lambda I_V)$ is an eigenvector of T for λ . Conversely, every eigenvector of T determines a unique eigenvalue.

Example 8.2.10 (Verifying an eigenvector). Define $T \in \mathcal{L}(\mathbb{R}^{3,1})$ by

$$T(\mathbf{v}) \stackrel{\text{df}}{=} \begin{bmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{bmatrix} \mathbf{v}.$$

We verify that $\mathbf{v} = [16, 4, 1]^\top$ is an eigenvector:

$$T(\mathbf{v}) = \begin{bmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix} = 2 \cdot \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix}.$$

Hence $\mathbf{v}$ is an eigenvector of T with eigenvalue 2.

8.2.4 Eigenspaces

Definition 8.2.11 (Eigenspace). Let V be a vector space over $\mathbb{F}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$ an eigenvalue of T . The *eigenspace* of T for λ , denoted $E_{T,\lambda}$, is defined by

$$E_{T,\lambda} \stackrel{\text{df}}{=} \text{Null}(T - \lambda \cdot I_V) \neq \{\mathbf{0}_V\}.$$

$E_{T,\lambda}$ is always a vector subspace of V . The set of all eigenvectors of T for λ is precisely $E_{T,\lambda} \setminus \{\mathbf{0}_V\}$.

Example 8.2.12 (Finding eigenvalues and eigenspaces). Define $T \in \mathcal{L}(\mathbb{R}^{2,2})$ by

$$\forall a, b, c, d \in \mathbb{R} : \quad T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) \stackrel{\text{df}}{=} \begin{bmatrix} c & d \\ a & b \end{bmatrix}.$$

Step 1: Standard matrix. Using the standard ordered basis σ for $\mathbb{R}^{2,2}$, one computes

$$[T]_\sigma^\sigma = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{4,4}.$$

Step 2: Characteristic polynomial. Expanding the 4×4 determinant (by cofactor expansion along the first row, using the checkerboard sign pattern):

$$\begin{aligned} p_T(\lambda) &= \det\left(\begin{bmatrix} -\lambda & 0 & 1 & 0 \\ 0 & -\lambda & 0 & 1 \\ 1 & 0 & -\lambda & 0 \\ 0 & 1 & 0 & -\lambda \end{bmatrix}\right) \\ &= \lambda^2(\lambda^2 - 1) - (\lambda^2 - 1) = (\lambda^2 - 1)(\lambda^2 - 1) = (\lambda^2 - 1)^2 = (\lambda + 1)^2(\lambda - 1)^2. \end{aligned}$$

Hence T has exactly two eigenvalues: $\lambda = 1$ and $\lambda = -1$.

Step 3: Eigenspace $E_{T,-1}$. Row-reducing $[T]_{\mathcal{C}}^{\sigma} - (-1)\mathbf{I}_4$:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \mapsto R_3 - R_1, R_4 \mapsto R_4 - R_2} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The null space is spanned by $\{[-1, 0, 1, 0]^{\top}, [0, -1, 0, 1]^{\top}\}$, so

$$E_{T,-1} = \text{Span}\left(\left\{\begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}\right\}\right).$$

Step 4: Eigenspace $E_{T,1}$. Row-reducing $[T]_{\mathcal{C}}^{\sigma} - \mathbf{I}_4$:

$$\begin{bmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \xrightarrow{R_3 + R_1, R_4 + R_2} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The null space is spanned by $\{[1, 0, 1, 0]^{\top}, [0, 1, 0, 1]^{\top}\}$, so

$$E_{T,1} = \text{Span}\left(\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}\right\}\right). \quad \square$$

8.2.5 Linear Independence of Eigenvectors for Distinct Eigenvalues

Theorem 8.2.13 (Eigenvectors for distinct eigenvalues are linearly independent). *Let V be a vector space over $\mathbb{F}$ (not necessarily finite-dimensional), $T \in \mathcal{L}(V)$, and let $\lambda_1, \dots, \lambda_n$ be distinct eigenvalues of T . Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be eigenvectors of T for $\lambda_1, \dots, \lambda_n$ respectively. Then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly independent subset of V .*

Proof. Since $\mathbf{v}_1 \neq \mathbf{0}_V$, the singleton $\{\mathbf{v}_1\}$ is linearly independent. Let $m \leq n$ be the largest integer for which $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent; we show $m = n$ by contradiction. Suppose $m < n$, and let $\mu_1, \dots, \mu_m, \mu_{m+1} \in \mathbb{F}$ satisfy

$$\mathbf{0}_V = \mu_1 \mathbf{v}_1 + \dots + \mu_m \mathbf{v}_m + \mu_{m+1} \mathbf{v}_{m+1}. \quad (\spadesuit)$$

Multiplying $(\spadesuit)$ by λ_{m+1} and also applying T to $(\spadesuit)$ (using $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i$), then subtracting, yields

$$\mathbf{0}_V = \mu_1(\lambda_1 - \lambda_{m+1})\mathbf{v}_1 + \dots + \mu_m(\lambda_m - \lambda_{m+1})\mathbf{v}_m.$$

Since $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent and all $\lambda_i - \lambda_{m+1} \neq 0_{\mathbb{F}}$ (eigenvalues are distinct), we get $\mu_1 = \dots = \mu_m = 0_{\mathbb{F}}$. Then $(\spadesuit)$ gives $\mu_{m+1} \mathbf{v}_{m+1} = \mathbf{0}_V$, so $\mu_{m+1} = 0_{\mathbb{F}}$ since $\mathbf{v}_{m+1} \neq \mathbf{0}_V$. Hence $\{\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{v}_{m+1}\}$ is linearly independent, contradicting the maximality of m . Therefore $m = n$. $\square$

Key Result: Distinct Eigenvalues Give Linearly Independent Eigenvectors

If T has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$ with corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent. In particular, for $T \in \mathcal{L}(V)$ with $\dim(V) = n$, having n distinct eigenvalues guarantees a basis of eigenvectors (the key condition for diagonalisability).

8.2.6 The Direct Sum of Eigenspaces

Theorem 8.2.14 (A sum of distinct eigenspaces is a direct sum). *Let V be a vector space over $\mathbb{F}$, $T \in \mathcal{L}(V)$, and let $\lambda_1, \dots, \lambda_n$ be distinct eigenvalues of T . Then*

$$E_{T,\lambda_1} + \dots + E_{T,\lambda_n} = E_{T,\lambda_1} \oplus \dots \oplus E_{T,\lambda_n}.$$

Proof. Pick $\mathbf{v}_i \in E_{T,\lambda_i}$ for each i , and suppose $\mathbf{v}_1 + \dots + \mathbf{v}_n = \mathbf{0}_V$. We must show every $\mathbf{v}_i = \mathbf{0}_V$. Remove all zero summands; if any non-zero summands remain, they form a non-trivial linear combination of zero using eigenvectors for distinct eigenvalues, contradicting Theorem 8.2.13. Hence every $\mathbf{v}_i = \mathbf{0}_V$, which is precisely the condition for a direct sum. $\square$

Remark 8.2.15. Theorem 8.2.14 is a key ingredient in the theory of *diagonalisation*: a linear transformation T on an n -dimensional space is diagonalisable if and only if $V = E_{T,\lambda_1} \oplus \dots \oplus E_{T,\lambda_k}$ for its distinct eigenvalues $\lambda_1, \dots, \lambda_k$.

Chapter 9

Diagonalization

9.1 Further Results on Eigenspaces

Building on Theorem 8.2.14, we establish a bound on the total dimension contributed by eigenspaces, which is the last ingredient needed before tackling diagonalizability.

Theorem 9.1.1 (Sum of eigenspace dimensions). *Let V be a finite-dimensional vector space and $T \in \mathcal{L}(V)$. Then*

$$\sum_{\lambda \text{ eigenvalue of } T} \dim(E_{T,\lambda}) \leq \dim(V).$$

(If T has no eigenvalues the sum is empty and equals 0 by convention.)

Proof. Since V is finite-dimensional, T has finitely many eigenvalues. By Theorem 8.2.14 the sum of all eigenspaces is in fact a direct sum:

$$\sum_{\lambda} E_{T,\lambda} = \bigoplus_{\lambda} E_{T,\lambda}.$$

Using the dimension formula for a direct sum (Axler, Theorem 3.94: $\dim(W_1 \oplus \cdots \oplus W_n) = \dim W_1 + \cdots + \dim W_n$), we obtain

$$\sum_{\lambda} \dim(E_{T,\lambda}) = \dim\left(\bigoplus_{\lambda} E_{T,\lambda}\right) \leq \dim(V),$$

the inequality holding because $\bigoplus_{\lambda} E_{T,\lambda}$ is a vector subspace of V . □

Key Result: Eigenspace Dimension Bound

For any $T \in \mathcal{L}(V)$ on a finite-dimensional space, the sum of the dimensions of all eigenspaces is at most $\dim(V)$. Equality is the criterion for diagonalizability (Theorem 9.2.4).

9.2 Diagonalization

9.2.1 Definition

Definition 9.2.1 (Diagonalizability). Let V be a vector space (not necessarily finite-dimensional) and $T \in \mathcal{L}(V)$. T is said to be *diagonalizable* if and only if V has a basis consisting entirely of eigenvectors of T .

9.2.2 Diagonalizability and Diagonal Matrix Representations

Theorem 9.2.2 (Diagonalizability and diagonal matrix representations). *Let V be a finite-dimensional vector space over $\mathbb{F}$ with $\dim(V) = n$, and let $T \in \mathcal{L}(V)$. The following two statements are equivalent:*

1. T is diagonalizable.

2. There exists an ordered basis β for V such that $[T]_\beta^\beta \in \mathbb{F}^{n,n}$ is a diagonal matrix.

Remark. In statement (2), β is not required to be the standard basis; and β necessarily consists of eigenvectors of T .

Proof. ($\Rightarrow$) Suppose T is diagonalizable, so there is an ordered basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of eigenvectors. Let λ_i denote the eigenvalue of T for $\mathbf{v}_i$. Then $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i$, so $[T(\mathbf{v}_i)]_\beta = \lambda_i [\mathbf{v}_i]_\beta = \lambda_i \mathbf{e}_i$. Therefore

$$[T]_\beta^\beta = [[T(\mathbf{v}_1)]_\beta \cdots [T(\mathbf{v}_n)]_\beta] = \begin{bmatrix} \lambda_1 & \cdots & 0_{\mathbb{F}} \\ \vdots & \ddots & \vdots \\ 0_{\mathbb{F}} & \cdots & \lambda_n \end{bmatrix},$$

which is diagonal. (Repeated eigenvalues are permitted since $\mathbf{v}_1, \dots, \mathbf{v}_n$ are distinct basis vectors, even if some λ_i coincide.)

($\Leftarrow$) Suppose there is an ordered basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ such that $[T]_\beta^\beta = \text{diag}(\lambda_1, \dots, \lambda_n)$. For each i , the i -th column of $[T]_\beta^\beta$ is $\lambda_i \mathbf{e}_i$, so

$$[T(\mathbf{v}_i)]_\beta = [T]_\beta^\beta [\mathbf{v}_i]_\beta = \lambda_i \mathbf{e}_i = \lambda_i [\mathbf{v}_i]_\beta = [\lambda_i \mathbf{v}_i]_\beta.$$

Since the coordinate map is injective, $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i$, i.e. $\mathbf{v}_i$ is an eigenvector of T with eigenvalue λ_i . Hence β consists of eigenvectors and T is diagonalizable. $\square$

Remark 9.2.3. Theorem 9.2.2 tells us that diagonalizability is *equivalent* to the existence of a diagonal matrix representation, but it does not tell us *how* to determine whether such a representation exists. The Fundamental Theorem of Diagonalizability below provides the computational answer.

9.2.3 The Fundamental Theorem of Diagonalizability

Theorem 9.2.4 (Fundamental Theorem of Diagonalizability). *Let V be a finite-dimensional vector space and $T \in \mathcal{L}(V)$. The following three statements are equivalent:*

1. T is diagonalizable.

2. $V = \sum_{\lambda \text{ eigenvalue of } T} E_{T,\lambda}$ (this sum is automatically a direct sum).

3. $\dim(V) = \sum_{\lambda \text{ eigenvalue of } T} \dim(E_{T,\lambda})$.

Proof. **Statements (2) and (3) are equivalent.** Since the sum $\sum_{\lambda} E_{T,\lambda}$ is a direct sum (Theorem 8.2.14), its dimension equals $\sum_{\lambda} \dim(E_{T,\lambda})$. As this direct sum is a subspace of V , it equals V if and only if its dimension equals $\dim(V)$, establishing (2) $\iff$ (3).

Statement (1) implies Statement (2). Suppose T is diagonalizable with eigenvector basis B for V . For each eigenvalue λ define $B_\lambda = B \cap E_{T,\lambda}$. Then $B = \bigcup_{\lambda} B_\lambda$, and

$$V = \text{Span}(B) = \sum_{\lambda} \text{Span}(B_\lambda) \subseteq \sum_{\lambda} E_{T,\lambda} \subseteq V.$$

Both inclusions are therefore equalities, giving $V = \sum_{\lambda} E_{T,\lambda}$.

Statement (2) implies Statement (1). Suppose $V = \sum_{\lambda} E_{T,\lambda}$. For each eigenvalue λ let B_λ be a basis for $E_{T,\lambda}$, and set $B = \bigcup_{\lambda} B_\lambda$. Then $\text{Span}(B) = \sum_{\lambda} E_{T,\lambda} = V$, so B spans V and consists of eigenvectors. B therefore contains a basis $\tilde{B}$ for V (a spanning set contains a basis). In fact $\tilde{B} = B$, though we need not verify this. Hence T is diagonalizable. $\square$

Key Result: Fundamental Theorem of Diagonalizability

T is diagonalizable if and only if $\dim(V) = \sum_{\lambda} \dim(E_{T,\lambda})$. Equivalently, V decomposes as the direct sum of all eigenspaces of T .

9.2.4 The Diagonalization Procedure

Given a linear transformation $T \in \mathcal{L}(V)$ on a finite-dimensional space over $\mathbb{F}$, diagonalizability is determined and a diagonalizing basis constructed as follows.

1. **Find eigenvalues.** Compute the characteristic polynomial $p_T(\lambda)$ (see Definition 8.2.6) and find its $\mathbb{F}$ -roots.
2. **Find eigenspaces.** For each eigenvalue λ , compute $E_{T,\lambda} = \text{Null}([T]_{\sigma}^{\sigma} - \lambda I)$ via row reduction, where σ is any chosen ordered basis.
3. **Compute d .** Set $d = \sum_{\lambda \text{ eigenvalue}} \dim(E_{T,\lambda})$.
 - If $d < \dim(V)$: T is **not** diagonalizable. Stop.
 - If $d = \dim(V)$: T is diagonalizable. Continue.
4. **Build the basis.** For each eigenvalue λ , derive a basis B_{λ} for $E_{T,\lambda}$. Let $B = \bigcup_{\lambda} B_{\lambda}$; then B is a basis for V .
5. **Order B into β .** Choose any ordering $\beta = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ of B , and let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues. Then

$$[T]_{\beta}^{\beta} = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix}.$$

(Eigenvalues may repeat; the diagonal entries depend on the chosen ordering of B .)

6. **Matrix powers.** If V has a standard ordered basis σ , then

$$[T]_{\beta}^{\beta} = [[\mathbf{b}_1]_{\sigma} \cdots [\mathbf{b}_n]_{\sigma}]^{-1} [T]_{\sigma}^{\sigma} [[\mathbf{b}_1]_{\sigma} \cdots [\mathbf{b}_n]_{\sigma}],$$

and for all $n \in \mathbb{N}$,

$$([T]_{\beta}^{\beta})^n = [[\mathbf{b}_1]_{\sigma} \cdots [\mathbf{b}_n]_{\sigma}]^{-1} [T^n]_{\sigma}^{\sigma} [[\mathbf{b}_1]_{\sigma} \cdots [\mathbf{b}_n]_{\sigma}].$$

Exam Tip: Checking Diagonalizability

The quickest check is to compute $d = \sum_{\lambda} \dim(E_{T,\lambda})$ and compare with $\dim(V)$. If $d = \dim(V)$, diagonalizable; if $d < \dim(V)$, not. Always compute eigenspaces explicitly via row reduction — do not guess dimensions from the characteristic polynomial alone.

9.3 Worked Examples

Example 9.3.1 (Transpose operator on $\mathbb{R}^{2,2}$). Define $T \in \mathcal{L}(\mathbb{R}^{2,2})$ by $T \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c & d \\ a & b \end{pmatrix}$. The eigenvalues of T are ± 1 , determined from the characteristic polynomial. The eigenspaces are:

$$E_{T,1} = \text{Span} \left(\left\{ \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\} \right), \quad E_{T,-1} = \text{Span} \left(\left\{ \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} \right\} \right).$$

Then $\dim(E_{T,-1}) + \dim(E_{T,1}) = 2 + 2 = 4 = \dim(\mathbb{R}^{2,2})$, so T is **diagonalizable**.

Example 9.3.2 ($T(x, y, z) = (6x + 3y + 4z, 6y + 2z, 7z)$ on $\mathbb{R}^3$). The standard-basis matrix representation is the upper-triangular matrix $[T]_\sigma^\sigma = \begin{pmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{pmatrix}$. The characteristic polynomial is

$$p_T(\lambda) = \det([T]_\sigma^\sigma - \lambda I_3) = (6 - \lambda)^2(7 - \lambda) = -(\lambda - 6)^2(\lambda - 7),$$

so T has eigenvalues 6 and 7. Row-reducing $[T]_\sigma^\sigma - 6I_3$ and $[T]_\sigma^\sigma - 7I_3$ gives:

$$E_{T,6} = \text{Span}(\{(1, 0, 0)\}), \quad E_{T,7} = \text{Span}(\{(10, 2, 1)\}).$$

Therefore $\dim(E_{T,6}) + \dim(E_{T,7}) = 1 + 1 = 2 < 3 = \dim(\mathbb{R}^3)$, so T is **not diagonalizable**. The eigenvalue $\lambda = 6$ has algebraic multiplicity 2 but geometric multiplicity 1, preventing a full eigenvector basis.

Example 9.3.3 ($D + D^2$ on $\mathcal{P}_3(\mathbb{R})$). Let D denote the derivative operator on $\mathcal{P}_3(\mathbb{R})$, the space of polynomials of degree at most 3. With respect to the standard ordered basis $\sigma = (1, X, X^2, X^3)$,

$$[D + D^2]_\sigma^\sigma = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

and $p_T(\lambda) = \det([D + D^2]_\sigma^\sigma - \lambda I_4) = (-\lambda)^4 = \lambda^4$. So $D + D^2$ has exactly one eigenvalue: 0. Row-reducing $[D + D^2]_\sigma^\sigma$ to RREF gives

$$E_{D+D^2,0} = \text{Span}(\{1\}),$$

with $\dim(E_{D+D^2,0}) = 1 < 4 = \dim(\mathcal{P}_3(\mathbb{R}))$. Therefore $D + D^2$ is **not diagonalizable**.

Example 9.3.4 ($T(a + bX + cX^2) = c + bX + aX^2$ on $\mathcal{P}_2(\mathbb{R})$). The standard-basis matrix is $[T]_\sigma^\sigma = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. The characteristic polynomial factors as $p_T(\lambda) = -(\lambda + 1)(\lambda - 1)^2$, giving eigenvalues ± 1 . Row reduction yields:

$$E_{T,-1} = \text{Span}\left(\left\{\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right\}\right), \quad E_{T,1} = \text{Span}\left(\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right\}\right).$$

Then $\dim(E_{T,-1}) + \dim(E_{T,1}) = 1 + 2 = 3 = \dim(\mathcal{P}_2(\mathbb{R}))$, so T is **diagonalizable**. In coordinates, the eigenvalue -1 corresponds to the polynomial $-1 + X^2$ and the eigenvalue 1 corresponds to $1 + X^2$ and X , so the diagonalizing basis is $\beta = (-1 + X^2, 1 + X^2, X)$ with $[T]_\beta^\beta = \text{diag}(-1, 1, 1)$.

To compute $T^{2025}(1 + 2X + 3X^2)$, express $1 + 2X + 3X^2$ in terms of the eigenvectors. Writing $1 + 2X + 3X^2 = \alpha(-1 + X^2) + \beta(1 + X^2) + \gamma X$ and solving gives $\alpha = -1$, $\beta = 2$, $\gamma = 2$. Then

$$T^{2025}(1 + 2X + 3X^2) = (-1)^{2025}(-1 + X^2) + 1^{2025} \cdot 2(1 + X^2) + 1^{2025} \cdot 2X = 3 + 2X + X^2.$$

9.4 A Sufficient Criterion for Diagonalizability

In some cases, diagonalizability can be established purely by counting distinct eigenvalues.

Theorem 9.4.1 (Sufficient criterion: n distinct eigenvalues). *Let V be a finite-dimensional vector space with $\dim(V) = n$ and $T \in \mathcal{L}(V)$. If T has n distinct eigenvalues, then T is diagonalizable.*

Proof. Suppose T has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$. For each i , $E_{T, \lambda_i} \neq \{\mathbf{0}_V\}$, so $\dim(E_{T, \lambda_i}) \geq 1$. Therefore

$$\sum_{i=1}^n \dim(E_{T, \lambda_i}) \geq \sum_{i=1}^n 1 = n.$$

By Theorem 9.1.1 the sum is also $\leq n$, so $\sum_i \dim(E_{T, \lambda_i}) = n = \dim(V)$. By the Fundamental Theorem of Diagonalizability (Theorem 9.2.4), T is diagonalizable. $\square$

Warning: Converse is False

Having fewer than n distinct eigenvalues does **not** imply T is non-diagonalizable. For example, the identity I_V has only one eigenvalue ($\lambda = 1$) yet is trivially diagonalizable. The Fundamental Theorem must be used when the number of distinct eigenvalues is less than $\dim(V)$.

9.5 Geometric and Algebraic Multiplicity

Definition 9.5.1 (Geometric multiplicity). Let V be a finite-dimensional vector space, $T \in \mathcal{L}(V)$, and λ an eigenvalue of T . The *geometric multiplicity* of λ for T is $\dim(E_{T, \lambda})$.

Definition 9.5.2 (Algebraic multiplicity). The *algebraic multiplicity* of an eigenvalue λ_* of T in p_T is the highest power of the linear factor $(\cdot - \lambda_*)$ that divides p_T (see Definition 8.2.6).

Theorem 9.5.3 (Multiplicity characterization of diagonalizability). *Let V be finite-dimensional and $T \in \mathcal{L}(V)$. The following two statements are equivalent:*

1. T is diagonalizable.
2. p_T factors completely into linear factors over $\mathbb{F}$, and for every eigenvalue λ_* of T , the geometric multiplicity of λ_* equals its algebraic multiplicity in p_T .

Remark 9.5.4. Theorem 9.5.3 provides a clean algebraic criterion: compute p_T , check it splits into linear factors, and verify that each eigenspace has the dimension predicted by the corresponding factor's exponent. The examples in Section 9.3 illustrate both the diagonalizable case (multiplicities match) and the non-diagonalizable case (a factor's exponent exceeds the eigenspace dimension).

Key Result: Multiplicity Criterion

T is diagonalizable if and only if p_T splits into linear factors over $\mathbb{F}$ and geometric multiplicity = algebraic multiplicity for every eigenvalue.

Chapter 10

Diagonalization in Practice and Inner Product Spaces

10.1 Diagonalization Applied: A Worked Problem

The following example demonstrates how the theoretical results from Chapter 9—in particular the Fundamental Theorem of Diagonalizability (Theorem 9.2.4) and the sufficient criterion (Theorem 9.4.1)—can be combined with similarity to compute otherwise intractable matrix expressions.

Example 10.1.1 (Computing $\det\left(\sum_{k=1}^n \mathbf{A}^k\right)$ via diagonalization). Let $\mathbf{A} \in \mathbb{R}^{3,3}$ satisfy the property that $\mathbf{A} + \mathbf{I}_3$, $\mathbf{A} - \mathbf{I}_3$, and $\mathbf{A} - 2\mathbf{I}_3$ are all *not* invertible. Derive, for any $n \in \mathbb{N}$, an explicit formula for $\det\left(\sum_{k=1}^n \mathbf{A}^k\right)$.

Step 1: Identify the eigenvalues. Recall from Definition 8.2.1 that λ is an eigenvalue of the linear map $T_{\mathbf{A}} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $\vec{x} \mapsto \mathbf{A}\vec{x}$ if and only if $T_{\mathbf{A}} - \lambda\mathbf{I}$ is not injective, equivalently $\mathbf{A} - \lambda\mathbf{I}_3$ is not invertible (in finite dimensions). Therefore:

- $\mathbf{A} + \mathbf{I}_3 = \mathbf{A} - (-1)\mathbf{I}_3$ not invertible $\Rightarrow -1$ is an eigenvalue.
- $\mathbf{A} - \mathbf{I}_3 = \mathbf{A} - 1 \cdot \mathbf{I}_3$ not invertible $\Rightarrow 1$ is an eigenvalue.
- $\mathbf{A} - 2\mathbf{I}_3$ not invertible $\Rightarrow 2$ is an eigenvalue.

Since the characteristic polynomial $p_{\mathbf{A}}$ is cubic (degree equal to $\dim = 3$) and has at least three distinct real roots $-1, 1, 2$, these are *all* the eigenvalues.

Step 2: Diagonalize $\mathbf{A}$. $\mathbf{A} \in \mathbb{R}^{3,3}$ has three distinct eigenvalues, so by Theorem 9.4.1 it is diagonalizable. Hence $\mathbf{A}$ is similar (Definition 8.1.1) to

$$\mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

meaning there exists an invertible $\mathbf{Q} \in \mathbb{R}^{3,3}$ such that $\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{D}$.

Step 3: Express $\mathbf{A}^k$. For all $k \in \mathbb{N}$,

$$\mathbf{D}^k = (\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q})^k = \underbrace{(\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}) \cdots (\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q})}_{k \text{ times}} = \mathbf{Q}^{-1}\mathbf{A}^k\mathbf{Q},$$

so $\mathbf{A}^k = \mathbf{Q}\mathbf{D}^k\mathbf{Q}^{-1}$. Since $\mathbf{D}$ is diagonal,

$$\mathbf{D}^k = \begin{pmatrix} (-1)^k & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^k \end{pmatrix},$$

and thus

$$\mathbf{A}^k = \mathbf{Q} \begin{pmatrix} (-1)^k & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^k \end{pmatrix} \mathbf{Q}^{-1}.$$

Step 4: Sum and take the determinant. For all $n \in \mathbb{N}$,

$$\sum_{k=1}^n \mathbf{A}^k = \sum_{k=1}^n \mathbf{Q} \mathbf{D}^k \mathbf{Q}^{-1} = \mathbf{Q} \left(\sum_{k=1}^n \mathbf{D}^k \right) \mathbf{Q}^{-1} = \mathbf{Q} \mathbf{S}_n \mathbf{Q}^{-1},$$

where

$$\mathbf{S}_n = \begin{pmatrix} \sum_{k=1}^n (-1)^k & 0 & 0 \\ 0 & \sum_{k=1}^n 1 & 0 \\ 0 & 0 & \sum_{k=1}^n 2^k \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}[1 - (-1)^n] & 0 & 0 \\ 0 & n & 0 \\ 0 & 0 & 2(2^n - 1) \end{pmatrix}.$$

Since $\sum_{k=1}^n \mathbf{A}^k = \mathbf{Q} \mathbf{S}_n \mathbf{Q}^{-1}$, these two matrices are similar and therefore share the same determinant (Theorem 8.1.4). Using $\det(\mathbf{Q} \mathbf{S}_n \mathbf{Q}^{-1}) = \det(\mathbf{Q}) \det(\mathbf{S}_n) \det(\mathbf{Q}^{-1}) = \det(\mathbf{S}_n)$, and the fact that the determinant of a diagonal matrix is the product of its diagonal entries,

$$\det \left(\sum_{k=1}^n \mathbf{A}^k \right) = -n[1 - (-1)^n](2^n - 1).$$

Exam Tip: Diagonalization for Matrix Powers and Sums

When asked to compute $f(\mathbf{A})$ for a polynomial or power-series expression f applied to a diagonalizable matrix $\mathbf{A}$: (1) identify eigenvalues and confirm diagonalizability, (2) write $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^{-1}$, (3) compute $f(\mathbf{D})$ entry-wise (since $\mathbf{D}$ is diagonal), and (4) reassemble via $f(\mathbf{A}) = \mathbf{Q} f(\mathbf{D}) \mathbf{Q}^{-1}$. For determinants, recall that $\det(\mathbf{Q} \mathbf{D} \mathbf{Q}^{-1}) = \det(\mathbf{D})$, so you never need to know $\mathbf{Q}$ explicitly.

10.2 Inner Product Spaces

Up to this point the course has studied vector spaces as purely algebraic objects, with no notion of angle, length, or distance. Inner products provide the bridge to geometry.

10.2.1 Inner Products on Real Vector Spaces

Definition 10.2.1 (Inner/scalar product on a real vector space). Let V be a real vector space (not necessarily finite-dimensional). An *inner product* (or *scalar product*) on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

satisfying the following three properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $\lambda \in \mathbb{R}$:

1. **Linearity in the second argument:** $\langle \mathbf{u}, \mathbf{v} + \lambda \cdot \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \lambda \langle \mathbf{u}, \mathbf{w} \rangle$.
2. **Symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
3. **Positive-definiteness:** $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{0}_V$.

The ordered pair $(V, \langle \cdot, \cdot \rangle)$ is then called a *real inner product space*.

Remark 10.2.2. Some textbooks place the linearity axiom in the *first* argument instead of the second; both conventions define the same class of objects for real spaces, since symmetry implies linearity in one argument forces linearity in the other.

Theorem 10.2.3 (Linearity in the first argument). *Let $\langle \cdot, \cdot \rangle$ be an inner product on a real vector space V . Then $\langle \cdot, \cdot \rangle$ is also linear in its first argument: for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{R}$,*

$$\langle \mathbf{u} + \lambda \cdot \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \lambda \langle \mathbf{v}, \mathbf{w} \rangle.$$

Proof. Using symmetry (axiom 2) then linearity in the second argument (axiom 1) then symmetry again:

$$\langle \mathbf{u} + \lambda \cdot \mathbf{v}, \mathbf{w} \rangle \stackrel{(2)}{=} \langle \mathbf{w}, \mathbf{u} + \lambda \cdot \mathbf{v} \rangle \stackrel{(1)}{=} \langle \mathbf{w}, \mathbf{u} \rangle + \lambda \langle \mathbf{w}, \mathbf{v} \rangle \stackrel{(2)}{=} \langle \mathbf{u}, \mathbf{w} \rangle + \lambda \langle \mathbf{v}, \mathbf{w} \rangle. \quad \square$$

Remark 10.2.4. An inner product on a real vector space is therefore *bilinear*: linear in each argument separately. This is a direct consequence of symmetry.

10.2.2 Examples of Real Inner Products

Example 10.2.5 (Standard inner product on $\mathbb{R}^n$). Let $n \in \mathbb{N}$. The *standard inner product* (also called the *dot product*) on $\mathbb{R}^n$, denoted $\bullet$, is defined by

$$\forall x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R} : (x_1, \dots, x_n) \bullet (y_1, \dots, y_n) \stackrel{\text{df}}{=} x_1 y_1 + \dots + x_n y_n.$$

One verifies the three axioms directly:

1. *Linearity in the second argument:*

$$\begin{aligned} (x_1, \dots, x_n) \bullet [(y_1, \dots, y_n) + \lambda(z_1, \dots, z_n)] &= \sum_{i=1}^n x_i(y_i + \lambda z_i) \\ &= \sum_{i=1}^n x_i y_i + \lambda \sum_{i=1}^n x_i z_i \\ &= (x_1, \dots, x_n) \bullet (y_1, \dots, y_n) \\ &\quad + \lambda (x_1, \dots, x_n) \bullet (z_1, \dots, z_n). \end{aligned}$$

2. *Symmetry:* $(x_1, \dots, x_n) \bullet (y_1, \dots, y_n) = \sum x_i y_i = \sum y_i x_i = (y_1, \dots, y_n) \bullet (x_1, \dots, x_n)$.

3. *Positive-definiteness:* $(x_1, \dots, x_n) \bullet (x_1, \dots, x_n) = x_1^2 + \dots + x_n^2 \geq 0$, with equality iff $x_i = 0$ for all i .

Example 10.2.6 (Standard inner product on $C[a, b]_{\mathbb{R}}$). Let $a < b$ be real numbers and let $C[a, b]_{\mathbb{R}} \stackrel{\text{df}}{=} \{f \in \mathbb{R}^{[a, b]} \mid f \text{ is continuous}\}$ (continuous real-valued functions on $[a, b]$, viewed as a vector subspace of $\mathbb{R}^{[a, b]}$ — see Example 1.2.11). The *standard inner product* on $C[a, b]_{\mathbb{R}}$ is

$$\forall f, g \in C[a, b]_{\mathbb{R}} : \langle f, g \rangle_{C[a, b]_{\mathbb{R}}} \stackrel{\text{df}}{=} \int_a^b f(x) g(x) dx.$$

Linearity in the second argument and symmetry follow from standard integral rules. Positive-definiteness holds because $\langle f, f \rangle = \int_a^b [f(x)]^2 dx \geq 0$, with equality iff $f \equiv 0$ on $[a, b]$ (by continuity and Calculus I).

Example 10.2.7 (Frobenius inner product on $\mathbb{R}^{n,n}$). Let $n \in \mathbb{N}$. The *Frobenius inner product* on $\mathbb{R}^{n,n}$, denoted $\langle \cdot, \cdot \rangle_F$, is

$$\forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{n,n} : \quad \langle \mathbf{A}, \mathbf{B} \rangle_F \stackrel{\text{df}}{=} \text{trace}(\mathbf{A}^\top \mathbf{B}).$$

1. *Linearity in the second argument:* Using the linearity of trace and transpose,

$$\langle \mathbf{A}, \mathbf{B} + \lambda \mathbf{C} \rangle_F = \text{trace}(\mathbf{A}^\top (\mathbf{B} + \lambda \mathbf{C})) = \text{trace}(\mathbf{A}^\top \mathbf{B}) + \lambda \text{trace}(\mathbf{A}^\top \mathbf{C}) = \langle \mathbf{A}, \mathbf{B} \rangle_F + \lambda \langle \mathbf{A}, \mathbf{C} \rangle_F.$$

2. *Symmetry:* Using $\text{trace}(\mathbf{M}) = \text{trace}(\mathbf{M}^\top)$ for any square $\mathbf{M}$,

$$\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{trace}(\mathbf{A}^\top \mathbf{B}) = \text{trace}((\mathbf{A}^\top \mathbf{B})^\top) = \text{trace}(\mathbf{B}^\top \mathbf{A}) = \langle \mathbf{B}, \mathbf{A} \rangle_F.$$

3. *Positive-definiteness:* Expanding,

$$\langle \mathbf{A}, \mathbf{A} \rangle_F = \text{trace}(\mathbf{A}^\top \mathbf{A}) = \sum_{i,j=1}^n (A[j, i])^2 \geq 0,$$

with equality iff $A[j, i] = 0$ for all i, j , i.e. iff $\mathbf{A} = \mathbf{0}_{n,n}$.

Example 10.2.8 (A matrix-induced inner product on $\mathbb{R}^{3,1}$). Define $\langle \cdot, \cdot \rangle : \mathbb{R}^{3,1} \times \mathbb{R}^{3,1} \rightarrow \mathbb{R}$ by

$$\forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^{3,1} : \quad \langle \mathbf{u}, \mathbf{v} \rangle \stackrel{\text{df}}{=} \mathbf{u}^\top \begin{pmatrix} 11 & 6 & 0 \\ 6 & 4 & -2 \\ 0 & -2 & 6 \end{pmatrix} \mathbf{v}.$$

Let $\mathbf{A}$ denote the 3×3 matrix above; note $\mathbf{A}^\top = \mathbf{A}$ (symmetric matrix).

1. *Linearity in the second argument:* $\langle \mathbf{u}, \mathbf{v} + \lambda \mathbf{w} \rangle = \mathbf{u}^\top \mathbf{A}(\mathbf{v} + \lambda \mathbf{w}) = \mathbf{u}^\top \mathbf{A} \mathbf{v} + \lambda \mathbf{u}^\top \mathbf{A} \mathbf{w} = \langle \mathbf{u}, \mathbf{v} \rangle + \lambda \langle \mathbf{u}, \mathbf{w} \rangle$.
2. *Symmetry:* Since 1×1 matrices are their own transpose and $\mathbf{A}^\top = \mathbf{A}$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{A} \mathbf{v} = (\mathbf{u}^\top \mathbf{A} \mathbf{v})^\top = \mathbf{v}^\top \mathbf{A}^\top \mathbf{u}^{\top\top} = \mathbf{v}^\top \mathbf{A} \mathbf{u} = \langle \mathbf{v}, \mathbf{u} \rangle.$$

3. *Positive-definiteness:* For $(x, y, z)^\top \in \mathbb{R}^{3,1}$, expanding gives

$$\begin{aligned} \langle (x, y, z)^\top, (x, y, z)^\top \rangle &= 11x^2 + 4y^2 + 6z^2 + 12xy - 4yz \\ &= 11\left(x + \frac{6}{11}y\right)^2 + 6\left(z - \frac{1}{3}y\right)^2 + \frac{2}{33}y^2 \geq 0, \end{aligned}$$

with equality iff $x + \frac{6}{11}y = 0$, $z - \frac{1}{3}y = 0$, and $y = 0$, i.e. iff $x = y = z = 0$.

Hence $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{R}^{3,1}$.

Remark 10.2.9. More generally, if $\mathbf{A} \in \mathbb{R}^{n,n}$ is symmetric and *positive definite* (meaning $\mathbf{u}^\top \mathbf{A} \mathbf{u} > 0$ for all non-zero $\mathbf{u}$), then $\langle \mathbf{u}, \mathbf{v} \rangle \stackrel{\text{df}}{=} \mathbf{u}^\top \mathbf{A} \mathbf{v}$ defines an inner product on $\mathbb{R}^{n,1}$.

10.2.3 Inner Products on Complex Vector Spaces

On a complex vector space the symmetry axiom must be replaced, since there is no reason for $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{C}$ to equal $\langle \mathbf{v}, \mathbf{u} \rangle$ in general.

Definition 10.2.10 (Inner/scalar product on a complex vector space). Let V be a complex vector space (not necessarily finite-dimensional). An *inner product* on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$$

satisfying, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $\lambda \in \mathbb{C}$:

1. **Linearity in the second argument:** $\langle \mathbf{u}, \mathbf{v} + \lambda \cdot \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \lambda \langle \mathbf{u}, \mathbf{w} \rangle$.
2. **Conjugate symmetry:** $\overline{\langle \mathbf{u}, \mathbf{v} \rangle} = \langle \mathbf{v}, \mathbf{u} \rangle$.
3. **Positive-definiteness:** $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ (hence $\langle \mathbf{u}, \mathbf{u} \rangle \in \mathbb{R}$), with equality iff $\mathbf{u} = \mathbf{0}_V$.

The ordered pair $(V, \langle \cdot, \cdot \rangle)$ is then a *complex inner product space*.

Remark 10.2.11. Conjugate symmetry for a real inner product specialises to ordinary symmetry, since $\bar{r} = r$ for $r \in \mathbb{R}$. Thus real inner product spaces are a special case. Note also that axiom 1 combined with conjugate symmetry forces $\langle \cdot, \cdot \rangle$ to be *conjugate-linear* (anti-linear) in the first argument.

Theorem 10.2.12 (Conjugate-linearity in the first argument). *Let $\langle \cdot, \cdot \rangle$ be an inner product on a complex vector space V . Then for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{C}$,*

$$\langle \mathbf{u} + \lambda \cdot \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \bar{\lambda} \langle \mathbf{v}, \mathbf{w} \rangle.$$

Proof. Applying conjugate symmetry, then linearity in the second argument, then conjugate symmetry again:

$$\begin{aligned} \langle \mathbf{u} + \lambda \cdot \mathbf{v}, \mathbf{w} \rangle &\stackrel{(2)}{=} \overline{\langle \mathbf{w}, \mathbf{u} + \lambda \cdot \mathbf{v} \rangle} \stackrel{(1)}{=} \overline{\langle \mathbf{w}, \mathbf{u} \rangle + \lambda \langle \mathbf{w}, \mathbf{v} \rangle} = \overline{\langle \mathbf{w}, \mathbf{u} \rangle} + \bar{\lambda} \overline{\langle \mathbf{w}, \mathbf{v} \rangle} \\ &\stackrel{(2)}{=} \langle \mathbf{u}, \mathbf{w} \rangle + \bar{\lambda} \langle \mathbf{v}, \mathbf{w} \rangle. \end{aligned} \quad \square$$

10.2.4 Examples of Complex Inner Products

Example 10.2.13 (Standard inner product on $\mathbb{C}^n$). Let $n \in \mathbb{N}$. The *standard inner product* on $\mathbb{C}^n$ is

$$\forall w_1, \dots, w_n, z_1, \dots, z_n \in \mathbb{C} : (w_1, \dots, w_n) \bullet (z_1, \dots, z_n) \stackrel{\text{df}}{=} \bar{w}_1 z_1 + \dots + \bar{w}_n z_n.$$

The conjugate is placed on the *first* argument so that $(w_1, \dots, w_n) \bullet (w_1, \dots, w_n) = |w_1|^2 + \dots + |w_n|^2 \geq 0$, ensuring positive-definiteness.

Remark 10.2.14. Given $z \in \mathbb{C}$, note $z^2 \notin \mathbb{R}$ in general, but $|z|^2 = z\bar{z} \in \mathbb{R}_{\geq 0}$. This is why the standard inner product on $\mathbb{C}^n$ involves conjugation.

Example 10.2.15 (Standard inner product on $C[a, b]_{\mathbb{C}}$). Let $C[a, b]_{\mathbb{C}} \stackrel{\text{df}}{=} \{f \in \mathbb{C}^{[a, b]} \mid f \text{ is continuous}\}$. The standard inner product is

$$\langle f, g \rangle_{C[a, b]_{\mathbb{C}}} \stackrel{\text{df}}{=} \int_a^b \overline{f(x)} g(x) dx.$$

Verification of the axioms is left as an exercise.

Example 10.2.16 (Frobenius inner product on $\mathbb{C}^{n, n}$). Let $n \in \mathbb{N}$. The *Frobenius inner product* on $\mathbb{C}^{n, n}$ is

$$\forall \mathbf{A}, \mathbf{B} \in \mathbb{C}^{n, n} : \langle \mathbf{A}, \mathbf{B} \rangle_F \stackrel{\text{df}}{=} \text{trace}(\mathbf{A}^\dagger \mathbf{B}),$$

where $\mathbf{A}^\dagger$ denotes the *conjugate transpose* (also called the *Hermitian adjoint*): $\mathbf{A}^\dagger[j, i] \stackrel{\text{df}}{=} \overline{\mathbf{A}[i, j]}$. One verifies:

1. *Linearity in the second argument:* Identical calculation to the real Frobenius case (Example 10.2.7), with $^\top$ replaced by † .
2. *Conjugate symmetry:* Using $\text{trace}(\mathbf{M}^\dagger) = \overline{\text{trace}(\mathbf{M})}$,

$$\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{trace}(\mathbf{A}^\dagger \mathbf{B}) = \text{trace}((\mathbf{A}^\dagger \mathbf{B})^\dagger)^* = \overline{\text{trace}(\mathbf{B}^\dagger \mathbf{A})} = \overline{\langle \mathbf{B}, \mathbf{A} \rangle_F}.$$

3. *Positive-definiteness*: $\langle \mathbf{A}, \mathbf{A} \rangle_F = \sum_{i,j} |\mathbf{A}[j, i]|^2 \geq 0$, with equality iff $\mathbf{A} = \mathbf{0}_{n,n}$.

Example 10.2.17 (Weighted inner product on $\mathbb{C}^n$). Let $n \in \mathbb{N}$ and $c_1, \dots, c_n > 0$ be positive real numbers. The function $\langle \cdot, \cdot \rangle : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ defined by

$$\langle (w_1, \dots, w_n), (z_1, \dots, z_n) \rangle \stackrel{\text{df}}{=} \sum_{k=1}^n c_k \overline{w_k} z_k$$

is an inner product on $\mathbb{C}^n$.

1. *Linearity in the second argument*: $\sum_{k=1}^n c_k \overline{w_k} (y_k + \lambda z_k) = \sum_{k=1}^n c_k \overline{w_k} y_k + \lambda \sum_{k=1}^n c_k \overline{w_k} z_k$.

2. *Conjugate symmetry*: Since $c_k \in \mathbb{R}$ and $\overline{\overline{c_k}} = c_k$,

$$\overline{\langle (w_1, \dots, w_n), (z_1, \dots, z_n) \rangle} = \overline{\sum_{k=1}^n c_k \overline{w_k} z_k} = \sum_{k=1}^n c_k w_k \overline{z_k} = \langle (z_1, \dots, z_n), (w_1, \dots, w_n) \rangle.$$

3. *Positive-definiteness*: $\langle \mathbf{w}, \mathbf{w} \rangle = \sum_{k=1}^n c_k |w_k|^2 \geq 0$, with equality iff $c_k |w_k|^2 = 0$ for all k ; since $c_k > 0$, this holds iff $w_k = 0$ for all k .

When all $c_k = 1$ this reduces to the standard inner product on $\mathbb{C}^n$ (Example 10.2.13).

Key Result: Axioms of an Inner Product

To verify that $\langle \cdot, \cdot \rangle$ is an inner product check three properties: (1) linearity in the second argument, (2) symmetry (real) or conjugate symmetry (complex), and (3) positive-definiteness. Linearity in the first argument then follows automatically.

10.3 The Norm Associated to an Inner Product

Inner products enable us to assign lengths to vectors.

Definition 10.3.1 (Norm associated to an inner product). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space (real or complex). The *norm associated to* $\langle \cdot, \cdot \rangle$ is the function $\| \cdot \| : V \rightarrow \mathbb{R}_{\geq 0}$ defined by

$$\forall \mathbf{u} \in V : \quad \|\mathbf{u}\| \stackrel{\text{df}}{=} \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}.$$

For any $\mathbf{u} \in V$, the *length* of $\mathbf{u}$ is defined to be $\|\mathbf{u}\|$.

A norm on a vector space V must satisfy three properties:

1. **Positive-definiteness**: $\|\mathbf{u}\| \geq 0$, with equality iff $\mathbf{u} = \mathbf{0}_V$. (Follows immediately from axiom 3 of the inner product.)
2. **Absolute homogeneity**: $\|\lambda \cdot \mathbf{u}\| = |\lambda| \|\mathbf{u}\|$ for all $\mathbf{u} \in V$ and scalars λ . (Follows from bilinearity/sesquilinearity of the inner product.)
3. **Triangle inequality**: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v} \in V$. (Requires the Cauchy–Schwarz inequality; see Theorem 10.3.6.)

Example 10.3.2 (Pythagorean norm on $\mathbb{R}^n$ and $\mathbb{C}^n$). The norm associated to the standard inner product on $\mathbb{R}^n$ is the *Pythagorean norm* $\| \cdot \|_n$:

$$\|(x_1, \dots, x_n)\|_n = \sqrt{x_1^2 + \dots + x_n^2}.$$

For $\mathbb{C}^n$ with the standard inner product:

$$\|(z_1, \dots, z_n)\|_n = \sqrt{|z_1|^2 + \dots + |z_n|^2}.$$

10.3.1 The Cauchy–Schwarz Inequality

Theorem 10.3.3 (Cauchy–Schwarz inequality). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space (real or complex). Then for all $\mathbf{u}, \mathbf{v} \in V$,*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Proof. Denote the base field by $\mathbb{F}$ (either $\mathbb{R}$ or $\mathbb{C}$).

Choice of phase scalar. Pick $\alpha \in \mathbb{F}$ with $|\alpha| = 1$ such that $\alpha \langle \mathbf{u}, \mathbf{v} \rangle = |\langle \mathbf{u}, \mathbf{v} \rangle|$. (If $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, take $\alpha = 1$; otherwise take $\alpha = |\langle \mathbf{u}, \mathbf{v} \rangle| / \langle \mathbf{u}, \mathbf{v} \rangle$.)

A non-negative quadratic. Define $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ by $f(t) \stackrel{\text{df}}{=} \langle \mathbf{u} + t\alpha \cdot \mathbf{v}, \mathbf{u} + t\alpha \cdot \mathbf{v} \rangle$. Expanding using (bi/sesqui)linearity and the choice of α :

$$\begin{aligned} f(t) &= \|\mathbf{u}\|^2 + t\alpha \langle \mathbf{u}, \mathbf{v} \rangle + t\overline{\alpha \langle \mathbf{u}, \mathbf{v} \rangle} + t^2 |\alpha|^2 \|\mathbf{v}\|^2 \\ &= \|\mathbf{u}\|^2 + t|\langle \mathbf{u}, \mathbf{v} \rangle| + t|\langle \mathbf{u}, \mathbf{v} \rangle| + t^2 \|\mathbf{v}\|^2 \\ &= (\|\mathbf{v}\|^2)t^2 + (2|\langle \mathbf{u}, \mathbf{v} \rangle|)t + \|\mathbf{u}\|^2. \end{aligned}$$

Non-negativity forces a non-positive discriminant. Since $f(t) \geq 0$ for all $t \in \mathbb{R}$, the discriminant of this quadratic (in t) satisfies $\Delta \leq 0$:

$$(2|\langle \mathbf{u}, \mathbf{v} \rangle|)^2 - 4\|\mathbf{v}\|^2 \|\mathbf{u}\|^2 \leq 0.$$

Rearranging gives $|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2$, and taking square roots (both sides are non-negative):

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|. \quad \square$$

Corollary 10.3.4 (Standard Cauchy–Schwarz for $\mathbb{R}^n$). *For all $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$,*

$$|x_1 y_1 + \dots + x_n y_n| \leq \sqrt{x_1^2 + \dots + x_n^2} \sqrt{y_1^2 + \dots + y_n^2}.$$

Corollary 10.3.5 (Standard Cauchy–Schwarz for $\mathbb{C}^n$). *For all $w_1, \dots, w_n, z_1, \dots, z_n \in \mathbb{C}$,*

$$|\overline{w_1} z_1 + \dots + \overline{w_n} z_n| \leq \sqrt{|w_1|^2 + \dots + |w_n|^2} \sqrt{|z_1|^2 + \dots + |z_n|^2}.$$

10.3.2 The Triangle Inequality

Theorem 10.3.6 (Triangle inequality for norms from inner products). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. For all $\mathbf{u}, \mathbf{v} \in V$,*

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Proof. Expanding $\|\mathbf{u} + \mathbf{v}\|^2$ using the inner product axioms:

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \|\mathbf{v}\|^2 \\ &= \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \|\mathbf{v}\|^2 \\ &= \|\mathbf{u}\|^2 + 2\Re(\langle \mathbf{u}, \mathbf{v} \rangle) + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2|\langle \mathbf{u}, \mathbf{v} \rangle| + \|\mathbf{v}\|^2 \quad (\text{since } \Re(z) \leq |z| \text{ for all } z \in \mathbb{C}) \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| + \|\mathbf{v}\|^2 \quad (\text{Cauchy–Schwarz, Theorem 10.3.3}) \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2. \end{aligned}$$

Taking square roots of both sides gives the result. $\square$

Key Result: Inner Product Geometry

Every inner product $\langle \cdot, \cdot \rangle$ on V induces a norm $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$ satisfying:

- Cauchy–Schwarz: $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$,
- Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Chapter 11

Orthogonality, Orthonormal Bases, and Gram–Schmidt

The preceding chapter introduced inner products and the norms they induce. We now use the inner product to define a notion of *perpendicularity* between vectors, study the special bases that respect this geometry, and derive a systematic procedure for constructing them.

11.1 Orthogonality

11.1.1 Orthogonal Vectors and Subsets

Definition 11.1.1 (Orthogonality). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let $\mathbf{u}, \mathbf{v} \in V$. We say $\mathbf{u}$ is *orthogonal* to $\mathbf{v}$, written $\mathbf{u} \perp \mathbf{v}$, if and only if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Remark 11.1.2. By symmetry (or conjugate symmetry), $\mathbf{u} \perp \mathbf{v}$ if and only if $\mathbf{v} \perp \mathbf{u}$, so we may simply say “ $\mathbf{u}$ and $\mathbf{v}$ are orthogonal” without specifying direction. For example, in $(\mathbb{R}^3, \bullet)$: $(1, 0, 0) \bullet (0, 1, 0) = 0$, so $(1, 0, 0) \perp (0, 1, 0)$.

Definition 11.1.3 (Orthogonal subset). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. An *orthogonal subset* of V is a subset $S \subseteq V$ such that

$$\forall \mathbf{u}, \mathbf{v} \in S : \quad \mathbf{u} \neq \mathbf{v} \implies \mathbf{u} \perp \mathbf{v}.$$

In other words, any two *distinct* elements of S are orthogonal to each other.

Example 11.1.4 (An orthogonal subset of $\mathbb{R}^3$). Consider the real inner product space $(\mathbb{R}^3, \bullet)$. We show that $S = \{(1, 1, 0), (1, -1, 1), (-1, 1, 2)\}$ is an orthogonal subset. It suffices to check that every pair of distinct elements has zero dot product:

$$\begin{aligned} (1, 1, 0) \bullet (1, -1, 1) &= (1)(1) + (1)(-1) + (0)(1) = 1 - 1 + 0 = 0, \\ (1, 1, 0) \bullet (-1, 1, 2) &= (1)(-1) + (1)(1) + (0)(2) = -1 + 1 + 0 = 0, \\ (1, -1, 1) \bullet (-1, 1, 2) &= (1)(-1) + (-1)(1) + (1)(2) = -1 - 1 + 2 = 0. \end{aligned}$$

Hence S is an orthogonal subset of $\mathbb{R}^3$.

11.1.2 Orthogonal Sets are Linearly Independent

Theorem 11.1.5 (An orthogonal set of non-zero vectors is linearly independent). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let S be an orthogonal subset of V consisting of non-zero vectors. Then S is a linearly independent subset of V .*

Proof. Let $n \in \mathbb{N}$ and let $\mathbf{u}_1, \dots, \mathbf{u}_n$ be n distinct vectors in S . Suppose $\lambda_1, \dots, \lambda_n$ are scalars satisfying $\mathbf{0}_V = \lambda_1 \cdot \mathbf{u}_1 + \dots + \lambda_n \cdot \mathbf{u}_n$. For each $i \in \{1, \dots, n\}$, take the inner product of both

sides with $\mathbf{u}_i$ in the first argument:

$$\begin{aligned} 0 &= \langle \mathbf{u}_i, \mathbf{0}_V \rangle = \left\langle \mathbf{u}_i, \lambda_1 \cdot \mathbf{u}_1 + \cdots + \lambda_n \cdot \mathbf{u}_n \right\rangle \\ &= \lambda_1 \langle \mathbf{u}_i, \mathbf{u}_1 \rangle + \cdots + \lambda_n \langle \mathbf{u}_i, \mathbf{u}_n \rangle \quad (\text{linearity in the second argument}) \\ &= \lambda_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle, \end{aligned}$$

where the last step uses $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $j \neq i$ (orthogonality). Since $\mathbf{u}_i \neq \mathbf{0}_V$, positive-definiteness gives $\langle \mathbf{u}_i, \mathbf{u}_i \rangle > 0$, and hence $\lambda_i = 0$. Since i was arbitrary, $\lambda_1 = \cdots = \lambda_n = 0$, so S is linearly independent. $\square$

Key Result: Orthogonal $\Rightarrow$ Linearly Independent

Any orthogonal subset of non-zero vectors is linearly independent. In a finite-dimensional space of dimension n , an orthogonal set of n non-zero vectors is therefore automatically a basis.

11.2 Orthonormal Bases

11.2.1 Definition and First Examples

Definition 11.2.1 (Orthonormal basis). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. An *orthonormal basis* for $(V, \langle \cdot, \cdot \rangle)$ is a set β satisfying:

1. β is a basis for V ,
2. β is an orthogonal subset of V (distinct elements are orthogonal), and
3. $\|\mathbf{u}\| = 1$ for all $\mathbf{u} \in \beta$.

Conditions (2) and (3) together may be stated compactly as: for all $\mathbf{u}, \mathbf{v} \in \beta$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \begin{cases} 1 & \text{if } \mathbf{u} = \mathbf{v}, \\ 0 & \text{if } \mathbf{u} \neq \mathbf{v}. \end{cases}$$

Example 11.2.2 (Standard basis for $\mathbb{R}^n$). The standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$ is an orthonormal basis for $(\mathbb{R}^n, \bullet)$. Indeed, $\mathbf{e}_i \bullet \mathbf{e}_j = \delta_{ij}$ (the Kronecker delta) by direct inspection.

Example 11.2.3 (Standard basis for $\mathbb{C}^n$). The standard basis for $\mathbb{C}^n$ is an orthonormal basis for $(\mathbb{C}^n, \bullet)$, since $\mathbf{e}_i \bullet \mathbf{e}_j = \overline{(\mathbf{e}_i)_j} = \delta_{ij}$ for all i, j .

Example 11.2.4 (Standard basis for $\mathbb{R}^{n,n}$ under the Frobenius inner product). Fix $n \in \mathbb{N}$. Recall (Example 10.2.7) the Frobenius inner product $\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{trace}(\mathbf{A}^\top \mathbf{B})$ on $\mathbb{R}^{n,n}$. The standard basis $\{E^{ab}\}_{a,b=1}^n$ for $\mathbb{R}^{n,n}$, where E^{ab} is the matrix with a 1 in position (a, b) and zeros elsewhere, is an orthonormal basis for $(\mathbb{R}^{n,n}, \langle \cdot, \cdot \rangle_F)$.

To see this, let $a, b, c, d \in \{1, \dots, n\}$. Then

$$\langle E^{ab}, E^{cd} \rangle_F = \text{trace}((E^{ab})^\top E^{cd}) = \sum_{i,j=1}^n E^{ab}[j, i] E^{cd}[j, i] = \begin{cases} 1 & \text{if } (a, b) = (c, d), \\ 0 & \text{otherwise,} \end{cases}$$

since $E^{ab}[j, i] = 1$ only when $(j, i) = (a, b)$ and $E^{cd}[j, i] = 1$ only when $(j, i) = (c, d)$. Furthermore, $\|E^{ab}\|_F = \sqrt{\langle E^{ab}, E^{ab} \rangle_F} = 1$ for all a, b . Hence $\{E^{ab}\}_{a,b=1}^n$ is an orthonormal basis.

11.2.2 Fourier Coefficients

One of the most powerful features of orthonormal bases is that coordinates can be computed directly via inner products — no linear system needs to be solved.

Definition 11.2.5 (Fourier coefficients). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space with orthonormal basis β , and let $\mathbf{v} \in V$. The β -Fourier coefficients of $\mathbf{v}$ with respect to $(V, \langle \cdot, \cdot \rangle)$ are the coefficients of the basis vectors in β when $\mathbf{v}$ is expressed as a linear combination of them, i.e. the scalars $\lambda_{\mathbf{u}}$ defined by

$$\mathbf{v} = \sum_{\mathbf{u} \in \beta} \lambda_{\mathbf{u}} \cdot \mathbf{u}.$$

Theorem 11.2.6 (Formula for Fourier coefficients). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space with orthonormal basis β , and let $\mathbf{v} \in V$. Then

$$\mathbf{v} = \sum_{\mathbf{u} \in \beta} \langle \mathbf{u}, \mathbf{v} \rangle \cdot \mathbf{u},$$

so the β -Fourier coefficient of $\mathbf{v}$ corresponding to $\mathbf{u} \in \beta$ is $\langle \mathbf{u}, \mathbf{v} \rangle$.

Proof. Since β is a basis for V , we may write $\mathbf{v} = \sum_{\mathbf{u} \in \beta} \lambda_{\mathbf{u}} \cdot \mathbf{u}$ for unique scalars $\{\lambda_{\mathbf{u}}\}$. (Although β may be infinite, only finitely many $\lambda_{\mathbf{u}}$ are non-zero.) Fix $\tilde{\mathbf{u}} \in \beta$; then by linearity in the second argument and orthonormality:

$$\langle \tilde{\mathbf{u}}, \mathbf{v} \rangle = \left\langle \tilde{\mathbf{u}}, \sum_{\mathbf{u} \in \beta} \lambda_{\mathbf{u}} \cdot \mathbf{u} \right\rangle = \sum_{\mathbf{u} \in \beta} \lambda_{\mathbf{u}} \langle \tilde{\mathbf{u}}, \mathbf{u} \rangle = \lambda_{\tilde{\mathbf{u}}} \langle \tilde{\mathbf{u}}, \tilde{\mathbf{u}} \rangle = \lambda_{\tilde{\mathbf{u}}} \cdot 1 = \lambda_{\tilde{\mathbf{u}}},$$

where we used $\langle \tilde{\mathbf{u}}, \mathbf{u} \rangle = 0$ for $\mathbf{u} \neq \tilde{\mathbf{u}}$ and $\langle \tilde{\mathbf{u}}, \tilde{\mathbf{u}} \rangle = \|\tilde{\mathbf{u}}\|^2 = 1$. Hence the Fourier coefficient for $\tilde{\mathbf{u}}$ is $\langle \tilde{\mathbf{u}}, \mathbf{v} \rangle$. $\square$

Exam Tip: Computing Fourier Coefficients

Given an orthonormal basis $\beta = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and a vector $\mathbf{v}$, the coordinate of $\mathbf{v}$ relative to $\mathbf{b}_i$ is simply $\langle \mathbf{b}_i, \mathbf{v} \rangle$ — no row reduction needed. This is the key advantage of orthonormal over arbitrary bases.

Example 11.2.7 (Orthonormal basis and Fourier coefficients for $\mathcal{P}_2(\mathbb{R})$). Define an inner product on $\mathcal{P}_2(\mathbb{R})$ by $\langle P, Q \rangle \stackrel{\text{df}}{=} \int_{-1}^1 P Q$. Consider the set

$$\beta \stackrel{\text{df}}{=} \left\{ \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} X, \sqrt{\frac{5}{8}} (3X^2 - 1) \right\}.$$

Part 1: Verify β is an orthonormal basis. We check that each element has norm 1 and that distinct elements are orthogonal.

- $\left\| \frac{1}{\sqrt{2}} \right\|^2 = \int_{-1}^1 \frac{1}{2} dx = 1$. ✓
- $\left\| \sqrt{\frac{3}{2}} X \right\|^2 = \frac{3}{2} \int_{-1}^1 X^2 dx = \frac{3}{2} \cdot \frac{2}{3} = 1$. ✓
- $\left\| \sqrt{\frac{5}{8}} (3X^2 - 1) \right\|^2 = \frac{5}{8} \int_{-1}^1 (3X^2 - 1)^2 dx = \frac{5}{8} \int_{-1}^1 (9X^4 - 6X^2 + 1) dx = \frac{5}{8} \cdot \left[\frac{9}{5} X^5 - 2X^3 + X \right]_{-1}^1 = \frac{5}{8} \cdot \frac{16}{5} = 2 \cdot \frac{1}{1} = 1$. ✓

- $\left\langle \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} X \right\rangle = \int_{-1}^1 \frac{\sqrt{3}}{2} X \, dx = 0$ (odd function on $[-1, 1]$). ✓
- $\left\langle \frac{1}{\sqrt{2}}, \sqrt{\frac{5}{8}}(3X^2 - 1) \right\rangle = \frac{\sqrt{5}}{4} \int_{-1}^1 (3X^2 - 1) \, dx = \frac{\sqrt{5}}{4} [X^3 - X]_{-1}^1 = 0$. ✓
- $\left\langle \sqrt{\frac{3}{2}} X, \sqrt{\frac{5}{8}}(3X^2 - 1) \right\rangle = \sqrt{\frac{15}{16}} \int_{-1}^1 X(3X^2 - 1) \, dx = 0$ (odd function on $[-1, 1]$). ✓

Since β consists of three non-zero pairwise orthogonal vectors in the 3-dimensional space $\mathcal{P}_2(\mathbb{R})$, Theorem 11.1.5 shows β is linearly independent, hence a basis. Combined with the unit-norm checks, β is an orthonormal basis.

Part 2: Compute the β -Fourier coefficients of $1 + 4X^2$. By Theorem 11.2.6, the Fourier coefficient for each $\mathbf{b} \in \beta$ is $\langle \mathbf{b}, 1 + 4X^2 \rangle$:

$$\left\langle \frac{1}{\sqrt{2}}, 1 + 4X^2 \right\rangle = \int_{-1}^1 \frac{1 + 4X^2}{\sqrt{2}} \, dx = \frac{1}{\sqrt{2}} \left[X + \frac{4}{3}X^3 \right]_{-1}^1 = \frac{1}{\sqrt{2}} \cdot \frac{14}{3} = \frac{7\sqrt{2}}{3}.$$

$$\left\langle \sqrt{\frac{3}{2}} X, 1 + 4X^2 \right\rangle = \sqrt{\frac{3}{2}} \int_{-1}^1 X(1 + 4X^2) \, dx = 0 \quad (\text{odd integrand}).$$

$$\begin{aligned} \left\langle \sqrt{\frac{5}{8}}(3X^2 - 1), 1 + 4X^2 \right\rangle &= \sqrt{\frac{5}{8}} \int_{-1}^1 (3X^2 - 1)(1 + 4X^2) \, dx \\ &= \sqrt{\frac{5}{8}} \int_{-1}^1 (12X^4 - X^2 - 1) \, dx \\ &= \sqrt{\frac{5}{8}} \cdot \left[\frac{12}{5}X^5 - \frac{1}{3}X^3 - X \right]_{-1}^1 = \sqrt{\frac{5}{8}} \cdot \frac{32}{15} = \frac{8\sqrt{10}}{15}. \end{aligned}$$

Therefore

$$1 + 4X^2 = \frac{7\sqrt{2}}{3} \cdot \frac{1}{\sqrt{2}} + 0 \cdot \sqrt{\frac{3}{2}} X + \frac{8\sqrt{10}}{15} \cdot \sqrt{\frac{5}{8}}(3X^2 - 1) = \frac{7}{3} + \frac{4}{3}(3X^2 - 1) = 1 + 4X^2. \quad \checkmark$$

11.2.3 Orthogonal Projection onto a Vector

Definition 11.2.8 (Orthogonal projection). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space, and let $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{u} \neq \mathbf{0}_V$. The *orthogonal projection of $\mathbf{v}$ onto $\mathbf{u}$* , denoted $\text{proj}_{\mathbf{u}}(\mathbf{v})$, is defined by

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) \stackrel{\text{df}}{=} \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \cdot \mathbf{u}.$$

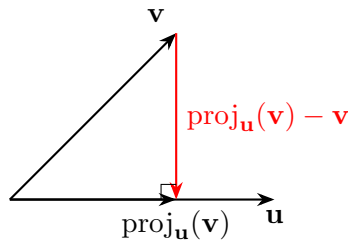


Figure 11.1: The orthogonal projection of $\mathbf{v}$ onto $\mathbf{u}$. The residual $\mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ is perpendicular to $\mathbf{u}$.

Proposition 11.2.9 (The residual is orthogonal to the base vector). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{u} \neq \mathbf{0}_V$. Then $\mathbf{u} \perp (\mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v}))$.*

Proof.

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v}) \rangle &= \left\langle \mathbf{u}, \mathbf{v} - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \cdot \mathbf{u} \right\rangle \\ &= \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \langle \mathbf{u}, \mathbf{u} \rangle \\ &= \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle = 0. \end{aligned}$$

□

Remark 11.2.10. Proposition 11.2.9 is illustrated in Figure 11.1: $\text{proj}_{\mathbf{u}}(\mathbf{v})$ is the component of $\mathbf{v}$ parallel to $\mathbf{u}$, and $\mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ is the component of $\mathbf{v}$ perpendicular to $\mathbf{u}$.

11.3 The Gram–Schmidt Orthogonalization Process

Theorem 11.3.1 (The Gram–Schmidt Orthogonalization Process). *Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space with $\dim(V) = n \geq 1$, and let $\beta = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be an ordered basis for V . Define a sequence $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ of V -vectors recursively by*

$$\begin{aligned} \mathbf{v}_1 &\stackrel{\text{df}}{=} \mathbf{b}_1, \\ \mathbf{v}_2 &\stackrel{\text{df}}{=} \mathbf{b}_2 - \text{proj}_{\mathbf{v}_1}(\mathbf{b}_2), \\ &\vdots \\ \mathbf{v}_k &\stackrel{\text{df}}{=} \mathbf{b}_k - \text{proj}_{\mathbf{v}_1}(\mathbf{b}_k) - \dots - \text{proj}_{\mathbf{v}_{k-1}}(\mathbf{b}_k), \quad k = 2, \dots, n. \end{aligned}$$

Equivalently, expanding the projection formula (Definition 11.2.8):

$$\mathbf{v}_k = \mathbf{b}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{v}_j, \mathbf{b}_k \rangle}{\langle \mathbf{v}_j, \mathbf{v}_j \rangle} \cdot \mathbf{v}_j, \quad k = 2, \dots, n.$$

Then

$$\left\{ \frac{1}{\|\mathbf{v}_1\|} \cdot \mathbf{v}_1, \dots, \frac{1}{\|\mathbf{v}_n\|} \cdot \mathbf{v}_n \right\}$$

is an orthonormal basis for $(V, \langle \cdot, \cdot \rangle)$.

Proof. For each $k \in \{1, \dots, n\}$, let $P(k)$ denote the joint statement:

$$\begin{cases} \{\mathbf{v}_1, \dots, \mathbf{v}_k\} \text{ is an orthogonal set of } k \text{ distinct non-zero vectors,} \\ \text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\}) = \text{Span}(\{\mathbf{b}_1, \dots, \mathbf{b}_k\}). \end{cases}$$

We prove $P(k)$ for all $k \in \{1, \dots, n\}$ by induction.

Base case ($k = 1$): $\mathbf{v}_1 = \mathbf{b}_1 \neq \mathbf{0}_V$ (since β is a basis) and $\text{Span}(\{\mathbf{v}_1\}) = \text{Span}(\{\mathbf{b}_1\})$, so $P(1)$ holds.

Inductive step: Suppose $k \in \{1, \dots, n-1\}$ and $P(k)$ holds. We verify $P(k+1)$.

Step 1 ($\mathbf{v}_{k+1} \neq \mathbf{0}_V$). If $\mathbf{v}_{k+1} = \mathbf{0}_V$, then

$$\mathbf{b}_{k+1} = \sum_{j=1}^k \frac{\langle \mathbf{v}_j, \mathbf{b}_{k+1} \rangle}{\langle \mathbf{v}_j, \mathbf{v}_j \rangle} \cdot \mathbf{v}_j \in \text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\}) = \text{Span}(\{\mathbf{b}_1, \dots, \mathbf{b}_k\}),$$

contradicting the linear independence of $\{\mathbf{b}_1, \dots, \mathbf{b}_{k+1}\}$. Hence $\mathbf{v}_{k+1} \neq \mathbf{0}_V$.

Step 2 (Span condition). Since $\mathbf{v}_{k+1} = \mathbf{b}_{k+1} - \sum_{j=1}^k c_j \mathbf{v}_j$ for some scalars c_j , we have $\mathbf{v}_{k+1} \in \text{Span}(\{\mathbf{b}_1, \dots, \mathbf{b}_{k+1}\})$. Similarly, $\mathbf{b}_{k+1} = \mathbf{v}_{k+1} + \sum_{j=1}^k c_j \mathbf{v}_j \in \text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_{k+1}\})$. Together with $P(k)$, this gives $\text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_{k+1}\}) = \text{Span}(\{\mathbf{b}_1, \dots, \mathbf{b}_{k+1}\})$.

Step 3 (Orthogonality). For each $i \in \{1, \dots, k\}$, using linearity in the second argument and $P(k)$:

$$\begin{aligned}\langle \mathbf{v}_i, \mathbf{v}_{k+1} \rangle &= \left\langle \mathbf{v}_i, \mathbf{b}_{k+1} - \sum_{j=1}^k \frac{\langle \mathbf{v}_j, \mathbf{b}_{k+1} \rangle}{\langle \mathbf{v}_j, \mathbf{v}_j \rangle} \cdot \mathbf{v}_j \right\rangle \\ &= \langle \mathbf{v}_i, \mathbf{b}_{k+1} \rangle - \sum_{j=1}^k \frac{\langle \mathbf{v}_j, \mathbf{b}_{k+1} \rangle}{\langle \mathbf{v}_j, \mathbf{v}_j \rangle} \langle \mathbf{v}_i, \mathbf{v}_j \rangle \\ &= \langle \mathbf{v}_i, \mathbf{b}_{k+1} \rangle - \frac{\langle \mathbf{v}_i, \mathbf{b}_{k+1} \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \langle \mathbf{v}_i, \mathbf{v}_i \rangle = 0,\end{aligned}$$

where we used $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $j \neq i$ (from $P(k)$). Hence $P(k+1)$ holds.

By induction, $P(k)$ holds for all k . In particular $P(n)$ says $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthogonal set of n non-zero vectors spanning V . By Theorem 11.1.5 this set is linearly independent, hence a basis. Dividing each $\mathbf{v}_k$ by its norm $\|\mathbf{v}_k\|$ yields a unit vector, producing an orthonormal basis. $\square$

Key Result: Gram–Schmidt Algorithm

Starting from any ordered basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$, recursively set $\mathbf{v}_k = \mathbf{b}_k - \sum_{j < k} \text{proj}_{\mathbf{v}_j}(\mathbf{b}_k)$, then normalize: $\mathbf{e}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$. The result $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis.

Example 11.3.2 (Gram–Schmidt on $\mathcal{P}_2(\mathbb{R})$). Let $\langle P, Q \rangle \stackrel{\text{df}}{=} \int_0^1 P Q$ be an inner product on $\mathcal{P}_2(\mathbb{R})$. Apply Gram–Schmidt to the standard ordered basis $\sigma = (1, X, X^2)$ to obtain an orthonormal basis.

Step 1. Set $\mathbf{v}_1 = 1$. Then $\|\mathbf{v}_1\|^2 = \int_0^1 1 \, dx = 1$, so $\|\mathbf{v}_1\| = 1$ and $\mathbf{e}_1 = 1$.

Step 2. Compute $\mathbf{v}_2 = X - \text{proj}_{\mathbf{v}_1}(X)$:

$$\langle \mathbf{v}_1, X \rangle = \int_0^1 X \, dx = \frac{1}{2}, \quad \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 1,$$

so $\text{proj}_{\mathbf{v}_1}(X) = \frac{1}{2} \cdot 1 = \frac{1}{2}$ and $\mathbf{v}_2 = X - \frac{1}{2}$.

$$\|\mathbf{v}_2\|^2 = \int_0^1 \left(X - \frac{1}{2}\right)^2 dx = \int_0^1 \left(X^2 - X + \frac{1}{4}\right) dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}, \quad \text{so } \|\mathbf{v}_2\| = \frac{1}{2\sqrt{3}}.$$

Hence $\mathbf{e}_2 = 2\sqrt{3}\left(X - \frac{1}{2}\right) = \sqrt{3}(2X - 1)$.

Step 3. Compute $\mathbf{v}_3 = X^2 - \text{proj}_{\mathbf{v}_1}(X^2) - \text{proj}_{\mathbf{v}_2}(X^2)$:

$$\langle \mathbf{v}_1, X^2 \rangle = \int_0^1 X^2 \, dx = \frac{1}{3}, \quad \text{so } \text{proj}_{\mathbf{v}_1}(X^2) = \frac{1}{3}.$$

$$\langle \mathbf{v}_2, X^2 \rangle = \int_0^1 \left(X - \frac{1}{2}\right) X^2 \, dx = \int_0^1 \left(X^3 - \frac{X^2}{2}\right) dx = \frac{1}{4} - \frac{1}{6} = \frac{1}{12},$$

$$\text{so } \text{proj}_{\mathbf{v}_2}(X^2) = \frac{1/12}{1/12} \cdot \left(X - \frac{1}{2}\right) = X - \frac{1}{2}.$$

Therefore

$$\mathbf{v}_3 = X^2 - \frac{1}{3} - \left(X - \frac{1}{2}\right) = X^2 - X + \frac{1}{6}.$$

$$\|\mathbf{v}_3\|^2 = \int_0^1 \left(X^2 - X + \frac{1}{6}\right)^2 dx = \int_0^1 \left(X^4 - 2X^3 + \frac{4}{3}X^2 - \frac{X}{3} + \frac{1}{36}\right) dx = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.$$

Hence $\|\mathbf{v}_3\| = \frac{1}{6\sqrt{5}}$ and $\mathbf{e}_3 = 6\sqrt{5} \left(X^2 - X + \frac{1}{6}\right) = \sqrt{5}(6X^2 - 6X + 1)$.

Conclusion. An orthonormal basis for $(\mathcal{P}_2(\mathbb{R}), \langle \cdot, \cdot \rangle)$ is

$$\{1, \sqrt{3}(2X - 1), \sqrt{5}(6X^2 - 6X + 1)\}.$$

Exam Tip: Gram–Schmidt Workflow

Always compute and simplify $\langle \mathbf{v}_j, \mathbf{b}_{k+1} \rangle$ and $\langle \mathbf{v}_j, \mathbf{v}_j \rangle$ separately before forming the projection. Errors in $\|\mathbf{v}_j\|^2$ propagate into all later steps. Check orthogonality of intermediate vectors before normalizing.

Chapter 12

Orthogonal Complements and Applications

The previous chapter built up the geometry of an inner product space: orthogonal sets, orthonormal bases, Fourier coefficients, projection onto a single vector, and the Gram–Schmidt process for producing an orthonormal basis. We now use these tools to study the *orthogonal complement* of a subset A of V — the set of all vectors perpendicular to everything in A . When A is a finite-dimensional subspace W , the complement $W^\perp$ pairs with W to decompose V as the direct sum $W \oplus W^\perp$, and the resulting projection P_W solves the geometric problem of finding the closest point in W to a given vector. We close the chapter — and the course — with two applications: the Schrödinger equation as an eigenvalue problem on an infinite-dimensional inner product space, and the method of linear least-squares as an exercise in orthogonal projection onto a column space.

12.1 Orthogonal Complements

12.1.1 Definition and First Examples

Definition 12.1.1 (Orthogonal complement). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space (not necessarily finite-dimensional), and let A be a *non-empty* subset of V . The *orthogonal complement* of A with respect to $(V, \langle \cdot, \cdot \rangle)$, denoted $A^\perp$, is the set

$$A^\perp \stackrel{\text{df}}{=} \{ \mathbf{v} \in V \mid \forall \mathbf{u} \in A : \mathbf{u} \perp \mathbf{v} \} = \{ \mathbf{v} \in V \mid \forall \mathbf{u} \in A : \langle \mathbf{u}, \mathbf{v} \rangle = 0 \}.$$

That is, $A^\perp$ is the set of all V -vectors orthogonal (in the sense of Definition 11.1.1) to every vector in A .

Remark 12.1.2. A is *not* required to be a vector subspace of V . The complement $A^\perp$ makes sense for any non-empty subset — a single polynomial, a finite list of vectors, an arbitrary set — and as we are about to see, $A^\perp$ is always a vector subspace of V regardless of how A looks.

12.1.2 Orthogonal Complements are Vector Subspaces

Theorem 12.1.3 (Orthogonal complements are vector subspaces). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let A be a non-empty subset of V . Then $A^\perp$ is a vector subspace of V .*

Proof. We apply the Subspace Test (Key Result on page 7).

1. *Non-emptiness:* For every $\mathbf{u} \in A$ we have $\langle \mathbf{u}, \mathbf{0}_V \rangle = 0$ (a property of inner products), so $\mathbf{0}_V \in A^\perp$ and $A^\perp \neq \emptyset$.

2. *Closure:* Let $\mathbf{v}, \mathbf{w} \in A^\perp$ and let λ be a scalar. For every $\mathbf{u} \in A$,

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle &= \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle = 0 + 0 = 0, \\ \langle \mathbf{u}, \lambda \cdot \mathbf{v} \rangle &= \lambda \langle \mathbf{u}, \mathbf{v} \rangle = \lambda \cdot 0 = 0, \end{aligned}$$

so $\mathbf{v} + \mathbf{w} \in A^\perp$ and $\lambda \cdot \mathbf{v} \in A^\perp$.

Hence $A^\perp$ is closed under addition and scalar multiplication, and is therefore a vector subspace of V . $\square$

Key Result: $A^\perp$ is Always a Subspace

For *any* non-empty subset A of an inner product space V , the orthogonal complement $A^\perp$ is a vector subspace of V — even when A itself is just a single vector or an unstructured set. In particular, $\mathbf{0}_V \in A^\perp$ always.

12.1.3 Properties of Orthogonal Complements

Theorem 12.1.4 (Properties of orthogonal complements). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. The following hold:*

1. $\{\mathbf{0}_V\}^\perp = V$.
2. $V^\perp = \{\mathbf{0}_V\}$.
3. For every non-empty subset A of V , $A \cap A^\perp \subseteq \{\mathbf{0}_V\}$.
4. For all non-empty subsets A, B of V , $A \subseteq B \implies B^\perp \subseteq A^\perp$.
5. For every non-empty subset A of V , $A^\perp = \text{Span}(A)^\perp$.

Proof.

(1) By definition, $\{\mathbf{0}_V\}^\perp = \{\mathbf{v} \in V \mid \langle \mathbf{0}_V, \mathbf{v} \rangle = 0\} = V$, since $\langle \mathbf{0}_V, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in V$.

(2) By definition, $V^\perp = \{\mathbf{v} \in V \mid \forall \mathbf{u} \in V : \langle \mathbf{u}, \mathbf{v} \rangle = 0\}$. Let $\mathbf{v} \in V^\perp$ be arbitrary. Then for all $\mathbf{u} \in V$ we have $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. Taking $\mathbf{u} = \mathbf{v}$ gives $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, which by positive-definiteness forces $\mathbf{v} = \mathbf{0}_V$. Hence $V^\perp \subseteq \{\mathbf{0}_V\}$. The reverse inclusion is immediate from (1) (or directly: $\langle \mathbf{u}, \mathbf{0}_V \rangle = 0$ for all $\mathbf{u}$). Therefore $V^\perp = \{\mathbf{0}_V\}$.

(3) If $A \cap A^\perp = \emptyset$, then $A \cap A^\perp \subseteq \{\mathbf{0}_V\}$ trivially. Otherwise let $\mathbf{v} \in A \cap A^\perp$. Since $\mathbf{v} \in A^\perp$, we have $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ for all $\mathbf{u} \in A$. Taking $\mathbf{u} = \mathbf{v} \in A$ yields $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}_V$ by positive-definiteness. Hence $A \cap A^\perp \subseteq \{\mathbf{0}_V\}$.

(4) Suppose $A \subseteq B$ and let $\mathbf{v} \in B^\perp$. Then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ for all $\mathbf{u} \in B$; in particular this holds for all $\mathbf{u} \in A$, so $\mathbf{v} \in A^\perp$. Hence $B^\perp \subseteq A^\perp$.

(5) Since $A \subseteq \text{Span}(A)$, statement (4) gives $\text{Span}(A)^\perp \subseteq A^\perp$. Conversely, let $\mathbf{v} \in A^\perp$; we must show $\langle \mathbf{w}, \mathbf{v} \rangle = 0$ for every $\mathbf{w} \in \text{Span}(A)$. Write $\mathbf{w} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n$ for some $\mathbf{u}_1, \dots, \mathbf{u}_n \in A$ and scalars $\lambda_1, \dots, \lambda_n$. Then, using conjugate-linearity in the first argument,

$$\langle \mathbf{w}, \mathbf{v} \rangle = \left\langle \sum_{k=1}^n \lambda_k \mathbf{u}_k, \mathbf{v} \right\rangle = \sum_{k=1}^n \overline{\lambda_k} \langle \mathbf{u}_k, \mathbf{v} \rangle = \sum_{k=1}^n \overline{\lambda_k} \cdot 0 = 0,$$

so $\mathbf{v} \in \text{Span}(A)^\perp$. Hence $A^\perp \subseteq \text{Span}(A)^\perp$, and therefore $A^\perp = \text{Span}(A)^\perp$. $\square$

Remark 12.1.5. Property (3) is the orthogonal analogue of the direct-sum criterion (Definition 2.3.1): a vector that is both *in* a set and *perpendicular to it* can only be the zero vector. Property (5) is the workhorse: it converts any complement-of-a-set problem into a complement-of-a-subspace problem.

12.1.4 Computing Orthogonal Complements

The following theorem, combined with property (5) above, reduces the computation of $A^\perp$ to a finite system of linear equations whenever $\text{Span}(A)$ is finite-dimensional.

Theorem 12.1.6 (Computing the orthogonal complement of a subspace). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space (not necessarily finite-dimensional), let W be a vector subspace of V , and let B be a basis for W . Then*

$$W^\perp = \{ \mathbf{v} \in V \mid \forall \mathbf{b} \in B : \langle \mathbf{b}, \mathbf{v} \rangle = 0 \}.$$

Proof. ($\subseteq$) If $\mathbf{v} \in W^\perp$ then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ for all $\mathbf{u} \in W$; since $B \subseteq W$, this in particular gives $\langle \mathbf{b}, \mathbf{v} \rangle = 0$ for all $\mathbf{b} \in B$.

($\supseteq$) Conversely, suppose $\mathbf{v} \in V$ satisfies $\langle \mathbf{b}, \mathbf{v} \rangle = 0$ for every $\mathbf{b} \in B$. For an arbitrary $\mathbf{w} \in W$, write $\mathbf{w} = \sum_{\mathbf{b} \in B} \lambda_{\mathbf{b}} \cdot \mathbf{b}$ (a finite sum, since basis representations are finite). Then

$$\langle \mathbf{w}, \mathbf{v} \rangle = \left\langle \sum_{\mathbf{b} \in B} \lambda_{\mathbf{b}} \cdot \mathbf{b}, \mathbf{v} \right\rangle = \sum_{\mathbf{b} \in B} \overline{\lambda_{\mathbf{b}}} \langle \mathbf{b}, \mathbf{v} \rangle = \sum_{\mathbf{b} \in B} \overline{\lambda_{\mathbf{b}}} \cdot 0 = 0,$$

so $\mathbf{v} \in W^\perp$. □

Procedure: Finding $A^\perp$ for a Non-Empty Set A

Let A be a non-empty subset of V .

1. Replace A by $\text{Span}(A)$ using property (5): $A^\perp = \text{Span}(A)^\perp$.
2. Pick a basis B for $\text{Span}(A)$. In practice, take B to be any *maximal linearly independent* subset of A — if V is finite-dimensional, “maximal” just means “largest”.
3. Solve the system of equations $\forall \mathbf{b} \in B : \langle \mathbf{b}, \mathbf{v} \rangle = 0$ for $\mathbf{v} \in V$. The solution set is $A^\perp$.

If B is finite, the system in step 3 is also finite — one equation per basis vector.

12.1.5 Worked Example: A Singleton Subset of $\mathcal{P}_2(\mathbb{R})$

Example 12.1.7 (Orthogonal complement of $\{1 + X + X^2\}$). Consider the real inner product space $(\mathcal{P}_2(\mathbb{R}), \langle \cdot, \cdot \rangle)$ where

$$\forall P, Q \in \mathcal{P}_2(\mathbb{R}) : \quad \langle P, Q \rangle \stackrel{\text{df}}{=} \int_{-1}^1 P(x)Q(x) \, dx.$$

We derive a basis for $\{1 + X + X^2\}^\perp$.

Step 1 (set up the equation). A general element of $\mathcal{P}_2(\mathbb{R})$ has the form $a + bX + cX^2$. We compute $\langle 1 + X + X^2, a + bX + cX^2 \rangle$ and set it to zero:

$$\int_{-1}^1 (1 + x + x^2)(a + bx + cx^2) \, dx = \int_{-1}^1 [a + (a + b)x + (a + b + c)x^2 + (b + c)x^3 + cx^4] \, dx.$$

The odd-power terms integrate to zero on $[-1, 1]$, leaving

$$2a + \frac{2}{3}(a + b + c) + \frac{2}{5}c = 0.$$

Multiplying by 15 and simplifying,

$$30a + 10(a + b + c) + 6c = 0 \iff 40a + 10b + 16c = 0 \iff a = -\frac{1}{4}b - \frac{2}{5}c.$$

Step 2 (parametrise the solution set). Substituting back,

$$\begin{aligned}
\{1 + X + X^2\}^\perp &= \{a + bX + cX^2 \mid a, b, c \in \mathbb{R}, a = -\frac{1}{4}b - \frac{2}{5}c\} \\
&= \{(-\frac{1}{4}b - \frac{2}{5}c) + bX + cX^2 \mid b, c \in \mathbb{R}\} \\
&= \{b(-\frac{1}{4} + X) + c(-\frac{2}{5} + X^2) \mid b, c \in \mathbb{R}\} \\
&= \text{Span}(\{-\frac{1}{4} + X, -\frac{2}{5} + X^2\}) \\
&= \text{Span}(\{-1 + 4X, -2 + 5X^2\}) \quad (\text{rescaling each generator}).
\end{aligned}$$

Step 3 (verify linear independence). Neither $-1 + 4X$ nor $-2 + 5X^2$ is a scalar multiple of the other, so $\{-1 + 4X, -2 + 5X^2\}$ is linearly independent and hence a basis for $\{1 + X + X^2\}^\perp$.

Sanity check. $\dim(\text{Span}(\{1 + X + X^2\})) = 1$ and $\dim(\{1 + X + X^2\}^\perp) = 2$, summing to $3 = \dim(\mathcal{P}_2(\mathbb{R}))$ — consistent with the direct-sum decomposition we are about to prove.

12.2 Direct-Sum Decomposition and Orthogonal Projection

12.2.1 The Decomposition $V = W \oplus W^\perp$

Theorem 12.2.1 (A subspace and its orthogonal complement form a direct sum). *Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space and let W be a vector subspace of V . Then*

$$V = W \oplus W^\perp,$$

and consequently $\dim(V) = \dim(W) + \dim(W^\perp)$.

Proof. By the Gram–Schmidt Orthogonalization Process (Theorem 11.3.1), the subspace W admits an orthonormal basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ where $m = \dim(W)$. Let $\mathbf{v} \in V$ be arbitrary, and define

$$\mathbf{p} \stackrel{\text{df}}{=} \sum_{k=1}^m \langle \mathbf{b}_k, \mathbf{v} \rangle \cdot \mathbf{b}_k \quad \text{and} \quad \mathbf{q} \stackrel{\text{df}}{=} \mathbf{v} - \mathbf{p}.$$

The coefficients $\langle \mathbf{b}_k, \mathbf{v} \rangle$ are the B -Fourier coefficients of $\mathbf{v}$ (Definition 11.2.5). Plainly $\mathbf{p} \in \text{Span}(B) = W$, and $\mathbf{v} = \mathbf{p} + \mathbf{q}$ by construction.

Claim: $\mathbf{q} \in W^\perp$. For any $\mathbf{w} \in W$ we may write $\mathbf{w} = \sum_{j=1}^m \lambda_j \mathbf{b}_j$, and then

$$\begin{aligned}
\langle \mathbf{w}, \mathbf{q} \rangle &= \langle \mathbf{w}, \mathbf{v} \rangle - \langle \mathbf{w}, \mathbf{p} \rangle \\
&= \left\langle \sum_{j=1}^m \lambda_j \mathbf{b}_j, \mathbf{v} \right\rangle - \left\langle \sum_{j=1}^m \lambda_j \mathbf{b}_j, \sum_{k=1}^m \langle \mathbf{b}_k, \mathbf{v} \rangle \mathbf{b}_k \right\rangle \\
&= \sum_{j=1}^m \overline{\lambda_j} \langle \mathbf{b}_j, \mathbf{v} \rangle - \sum_{j,k=1}^m \overline{\lambda_j} \langle \mathbf{b}_k, \mathbf{v} \rangle \langle \mathbf{b}_j, \mathbf{b}_k \rangle.
\end{aligned}$$

Orthonormality of B gives $\langle \mathbf{b}_j, \mathbf{b}_k \rangle = \delta_{jk}$, so the double sum collapses to $\sum_{j=1}^m \overline{\lambda_j} \langle \mathbf{b}_j, \mathbf{v} \rangle$, which cancels the first term. Hence $\langle \mathbf{w}, \mathbf{q} \rangle = 0$ for all $\mathbf{w} \in W$, i.e. $\mathbf{q} \in W^\perp$.

We have shown $V = W + W^\perp$. By property (3) of Theorem 12.1.4, $W \cap W^\perp \subseteq \{\mathbf{0}_V\}$, and the reverse inclusion is immediate (the zero vector lies in every subspace). Hence $W \cap W^\perp = \{\mathbf{0}_V\}$, and the sum is direct: $V = W \oplus W^\perp$. The dimension formula follows from the dimension formula for direct sums. $\square$

Warning: Finite-Dimensionality is Essential

Theorem 12.2.1 can fail when V is infinite-dimensional. The proof relied on Gram–Schmidt producing a finite orthonormal basis for W ; without finite dimension, $W^\perp$ may be smaller than expected and $W + W^\perp$ may fail to exhaust V . (The remedy in the infinite-dimensional setting requires *closed* subspaces and Hilbert space machinery, beyond the scope of MH1201.)

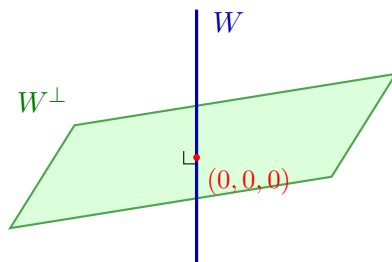


Figure 12.1: Geometric picture of $\mathbb{R}^3 = W \oplus W^\perp$ with $\dim(W) = 1$ and $\dim(W^\perp) = 2$. Every vector in $\mathbb{R}^3$ decomposes uniquely as a sum of a vector along the line W and a vector in the plane $W^\perp$.

12.2.2 The Orthogonal Projection Operator

The unique decomposition $\mathbf{v} = \mathbf{p} + \mathbf{q}$ from Theorem 12.2.1 extracts a well-defined W -component $\mathbf{p}$ from each $\mathbf{v} \in V$. The map sending $\mathbf{v}$ to that component is the orthogonal projection onto W .

Definition 12.2.2 (Orthogonal projection operator onto a subspace). Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space and let W be a vector subspace of V . The *orthogonal projection operator onto W* , denoted P_W , is the linear transformation $P_W : V \rightarrow W$ that maps each $\mathbf{v} \in V$ to its W -component in the decomposition $V = W \oplus W^\perp$.

Remark 12.2.3. Definition 12.2.2 generalizes the projection of a vector onto another vector (Definition 11.2.8): if $W = \text{Span}(\{\mathbf{u}\})$ for some non-zero $\mathbf{u}$, then $P_W(\mathbf{v}) = \text{proj}_{\mathbf{u}}(\mathbf{v})$. The new definition extends the same idea to subspaces of arbitrary (finite) dimension. If $\{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ is an orthonormal basis for W , the proof of Theorem 12.2.1 gives the explicit formula

$$P_W(\mathbf{v}) = \sum_{k=1}^m \langle \mathbf{b}_k, \mathbf{v} \rangle \cdot \mathbf{b}_k.$$

Theorem 12.2.4 (The shortest distance from a point to a subspace). Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space, let W be a vector subspace of V , and let $\mathbf{v} \in V$.

1. There is precisely one $\mathbf{w} \in W$ closest to $\mathbf{v}$ in the norm induced by the inner product, namely $\mathbf{w} = P_W(\mathbf{v})$. This is also the unique $\mathbf{w} \in W$ for which $\mathbf{v} - \mathbf{w} \in W^\perp$.
2. Consequently, the shortest distance from $\mathbf{v}$ to W is $\|\mathbf{v} - P_W(\mathbf{v})\|$. If $\mathbf{v}$ already lies in W then $P_W(\mathbf{v}) = \mathbf{v}$ and the distance is 0.

The proof, which is not examinable in MH1201, follows from a Pythagoras-style argument: for any $\mathbf{w} \in W$, write $\mathbf{v} - \mathbf{w} = (\mathbf{v} - P_W(\mathbf{v})) + (P_W(\mathbf{v}) - \mathbf{w})$, note that the first summand lies in $W^\perp$ and the second in W , and apply the Pythagorean identity $\|\mathbf{a} + \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2$ when $\mathbf{a} \perp \mathbf{b}$.

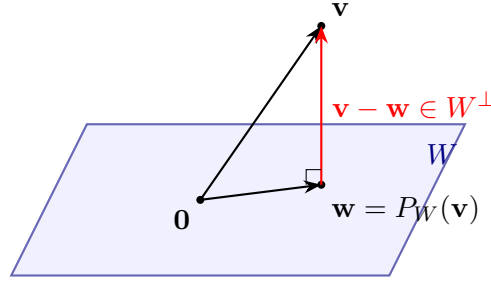


Figure 12.2: The orthogonal projection $P_W(\mathbf{v})$ is the unique point in W closest to $\mathbf{v}$. The residual $\mathbf{v} - P_W(\mathbf{v})$ is orthogonal to every vector in W .

Key Result: Closest Point in a Subspace

For a finite-dimensional subspace W of an inner product space V , the unique point of W closest to a given $\mathbf{v} \in V$ is the orthogonal projection $P_W(\mathbf{v})$, characterised by the perpendicularity condition $\mathbf{v} - P_W(\mathbf{v}) \in W^\perp$. This is the geometric fact that drives both Fourier-style approximation (Chapter 11) and the linear least-squares method of Section 12.3.

12.2.3 Worked Example: A Matrix-Induced Inner Product on $\mathbb{R}^{4,1}$

Example 12.2.5 (Decomposition with respect to a matrix-induced inner product). Define $\langle \cdot, \cdot \rangle : \mathbb{R}^{4,1} \times \mathbb{R}^{4,1} \rightarrow \mathbb{R}$ by

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^{4,1} : \quad \langle \mathbf{x}, \mathbf{y} \rangle \stackrel{\text{df}}{=} \mathbf{x}^\top \mathbf{A} \mathbf{y}, \quad \mathbf{A} \stackrel{\text{df}}{=} \begin{bmatrix} 4 & 0 & 2 & 0 \\ 0 & 5 & 0 & 2 \\ 2 & 0 & 6 & 0 \\ 0 & 2 & 0 & 7 \end{bmatrix}.$$

Let $W = \text{Span}(\{\mathbf{e}_1, \mathbf{e}_3\})$ where $\mathbf{e}_1 = (1, 0, 0, 0)^\top$ and $\mathbf{e}_3 = (0, 0, 1, 0)^\top$.

1. $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{R}^{4,1}$. Note that $\mathbf{A}^\top = \mathbf{A}$.

Linearity in the second argument. For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^{4,1}$ and $\lambda \in \mathbb{R}$,

$$\langle \mathbf{u}, \mathbf{v} + \lambda \mathbf{w} \rangle = \mathbf{u}^\top \mathbf{A}(\mathbf{v} + \lambda \mathbf{w}) = \mathbf{u}^\top \mathbf{A} \mathbf{v} + \lambda \mathbf{u}^\top \mathbf{A} \mathbf{w} = \langle \mathbf{u}, \mathbf{v} \rangle + \lambda \langle \mathbf{u}, \mathbf{w} \rangle.$$

Symmetry. For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{4,1}$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{A} \mathbf{v} = (\mathbf{u}^\top \mathbf{A} \mathbf{v})^\top = \mathbf{v}^\top \mathbf{A}^\top \mathbf{u} = \mathbf{v}^\top \mathbf{A} \mathbf{u} = \langle \mathbf{v}, \mathbf{u} \rangle,$$

using that $\mathbf{u}^\top \mathbf{A} \mathbf{v}$ is a 1×1 matrix and is therefore self-transpose.

Positive-definiteness. For $\mathbf{x} = (w, x, y, z)^\top$,

$$\langle \mathbf{x}, \mathbf{x} \rangle = 4w^2 + 5x^2 + 6y^2 + 7z^2 + 4wy + 4xz.$$

Completing squares gives

$$\langle \mathbf{x}, \mathbf{x} \rangle = (2w + y)^2 + (2x + z)^2 + x^2 + 5y^2 + 6z^2 \geq 0,$$

with equality iff $2w + y = 2x + z = x = y = z = 0$, i.e. iff $\mathbf{x} = \mathbf{0}$. Hence $\langle \cdot, \cdot \rangle$ is an inner product.

2. **Basis for $W^\perp$.** By Theorem 12.1.6, $\mathbf{x} = (w, x, y, z)^\top \in W^\perp$ iff $\langle \mathbf{e}_1, \mathbf{x} \rangle = 0$ and $\langle \mathbf{e}_3, \mathbf{x} \rangle = 0$. Computing $\mathbf{e}_1^\top \mathbf{A} = (4, 0, 2, 0)$ and $\mathbf{e}_3^\top \mathbf{A} = (2, 0, 6, 0)$, the system becomes

$$4w + 2y = 0, \quad 2w + 6y = 0 \iff w = y = 0.$$

Hence

$$W^\perp = \{(0, x, 0, z)^\top \mid x, z \in \mathbb{R}\} = \text{Span}(\{\mathbf{e}_2, \mathbf{e}_4\}),$$

and $\{\mathbf{e}_2, \mathbf{e}_4\}$ — linearly independent — is a basis for $W^\perp$.

3. **Decomposing $(1, 2, 3, 4)^\top$.** Since W is spanned by $\mathbf{e}_1, \mathbf{e}_3$ and $W^\perp$ by $\mathbf{e}_2, \mathbf{e}_4$, we read off

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}}_{\mathbf{p} \in W} + \underbrace{\begin{bmatrix} 0 \\ 2 \\ 0 \\ 4 \end{bmatrix}}_{\mathbf{q} \in W^\perp}.$$

That is, $\mathbf{p} = (1, 0, 3, 0)^\top$ and $\mathbf{q} = (0, 2, 0, 4)^\top$. Note that $\mathbf{p} = P_W((1, 2, 3, 4)^\top)$.

Remark 12.2.6. The decomposition in part 3 worked out cleanly because the W -basis $\{\mathbf{e}_1, \mathbf{e}_3\}$ and the $W^\perp$ -basis $\{\mathbf{e}_2, \mathbf{e}_4\}$ together form the standard basis of $\mathbb{R}^{4,1}$. In a different inner product or with a different W , finding the W -component of a vector typically requires the Fourier-coefficient formula $P_W(\mathbf{v}) = \sum_k \langle \mathbf{b}_k, \mathbf{v} \rangle \mathbf{b}_k$ applied to an *orthonormal* basis B of W .

12.2.4 Worked Example: Upper-Triangular Matrices

Example 12.2.7 (Orthogonal complement of upper-triangular matrices). Fix $n \in \mathbb{N}$ and consider the real inner product space $(\mathbb{R}^{n,n}, \langle \cdot, \cdot \rangle_F)$ where $\langle \cdot, \cdot \rangle_F$ is the Frobenius inner product (Example 10.2.7). Let

$$U \stackrel{\text{df}}{=} \{\mathbf{A} \in \mathbb{R}^{n,n} \mid \mathbf{A} \text{ is upper-triangular}\}.$$

We derive a basis for $U^\perp$.

U is a vector subspace of $\mathbb{R}^{n,n}$ with basis $\{\mathbf{E}^{a,b} \mid a, b \in \{1, \dots, n\}, a \leq b\}$, where $\mathbf{E}^{a,b}$ is the standard basis matrix with a 1 in position (a, b) and zeros elsewhere. (For $n = 4$ this basis has 10 elements: $\mathbf{E}^{1,1}, \mathbf{E}^{1,2}, \mathbf{E}^{1,3}, \mathbf{E}^{1,4}, \mathbf{E}^{2,2}, \mathbf{E}^{2,3}, \mathbf{E}^{2,4}, \mathbf{E}^{3,3}, \mathbf{E}^{3,4}, \mathbf{E}^{4,4}$.)

By Theorem 12.1.6,

$$U^\perp = \{\mathbf{B} \in \mathbb{R}^{n,n} \mid \forall a, b \in \{1, \dots, n\} \text{ s.t. } a \leq b : \langle \mathbf{E}^{a,b}, \mathbf{B} \rangle_F = 0\}.$$

We compute the inner product. By definition $\langle \mathbf{E}^{a,b}, \mathbf{B} \rangle_F = \text{trace}((\mathbf{E}^{a,b})^\top \mathbf{B})$, and expanding the trace and the matrix product,

$$\langle \mathbf{E}^{a,b}, \mathbf{B} \rangle_F = \sum_{i=1}^n \sum_{j=1}^n \mathbf{E}^{a,b}[j, i] \mathbf{B}[j, i].$$

But $\mathbf{E}^{a,b}[j, i] = 1$ iff $j = a$ and $i = b$ (and 0 otherwise), so the double sum collapses to the single entry $\mathbf{B}[a, b]$:

$$\langle \mathbf{E}^{a,b}, \mathbf{B} \rangle_F = \mathbf{B}[a, b].$$

Hence

$$U^\perp = \{\mathbf{B} \in \mathbb{R}^{n,n} \mid \forall a \leq b : \mathbf{B}[a, b] = 0\},$$

which is precisely the subspace of *strictly lower-triangular* matrices. A basis for $U^\perp$ is therefore

$$\{\mathbf{E}^{c,d} \mid c, d \in \{1, \dots, n\}, c > d\}.$$

For $n = 4$ this gives the six matrices $\{\mathbf{E}^{2,1}, \mathbf{E}^{3,1}, \mathbf{E}^{3,2}, \mathbf{E}^{4,1}, \mathbf{E}^{4,2}, \mathbf{E}^{4,3}\}$, matching the count $\dim(\mathbb{R}^{4,4}) - \dim(U) = 16 - 10 = 6$ predicted by Theorem 12.2.1.

12.3 Some Applications of Linear Algebra

We close the course with two applications of the eigenvalue/orthogonality machinery developed in Chapters 11–12. The first is a famous problem from quantum mechanics; the second, a data-fitting technique used everywhere from statistics to machine learning.

12.3.1 Quantum Mechanics: The Schrödinger Equation

In the early development of quantum mechanics, one of the central challenges was to predict the spectral frequencies of the hydrogen atom from a coherent theoretical framework. Erwin Schrödinger resolved this with what is now called the *Time-Independent Schrödinger Equation*, an **eigenvalue–eigenvector equation** on the (infinite-dimensional) complex inner product space of wave-functions.

For the hydrogen atom, the equation reads

$$\underbrace{\left[-\frac{h^2}{8\pi^2 \left(\frac{m_e m_p}{m_e + m_p} \right)} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) - \frac{1}{4\pi\epsilon_0} \frac{e^2}{\sqrt{x^2 + y^2 + z^2}} \right]}_{\text{the Schrödinger operator for the hydrogen atom}} \psi = E \cdot \psi,$$

where m_e, m_p are the electron and proton masses, e is the elementary charge, ϵ_0 is the vacuum permittivity, h is Planck’s constant, $E \in \mathbb{R}$ is the energy eigenvalue, and ψ is a non-zero eigenvector—a wave-function—in the complex vector space of square-integrable wave-functions.

Solving this equation produces the discrete sequence of eigenvalues

$$\forall n \in \mathbb{N} : \quad E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n^2}.$$

These match the experimentally measured energy levels of the hydrogen atom essentially exactly—a striking confirmation that the algebraic framework of eigenvalues, eigenvectors, and inner product spaces describes physical reality.

Remark 12.3.1. The Schrödinger operator is a linear operator on an infinite-dimensional complex inner product space, so the eigenvalue theory of Chapter 8 applies in spirit but not literally: the matrix-based proofs (e.g. via the characteristic polynomial) do not work, and one must work directly with the operator equation. Theorem 12.2.1 also fails in general for such spaces—see the warning above. Nonetheless, the linear-algebraic vocabulary (*eigenvalue*, *eigenvector*, *inner product*, *orthogonality*) remains exactly the right one.

12.3.2 Statistics: Linear Least-Squares Approximation

Given n distinct data points $(x_1, y_1), \dots, (x_n, y_n)$ in $\mathbb{R}^2$, we want to find the linear function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = mx + c$, that “best fits” the data in the sense of minimizing the sum of squared vertical errors:

$$E_f \stackrel{\text{df}}{=} \sum_{k=1}^n |y_k - f(x_k)|^2.$$

Any f that minimizes E_f is called a *linear least-squares approximation* (LLS approximation) to the data points, and E_f itself is called the *linear least-squares error* of f .

Geometrically, the picture is a scatter plot with a candidate fit-line $f(x)$; the vertical distances $d_k = |y_k - f(x_k)|$ from each data point to the line are the *residuals*, and $E_f = \sum_k d_k^2$.

Reformulation as a projection problem. Set

$$\mathbf{y} \stackrel{\text{df}}{=} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^{n,1}, \quad \mathbf{A} \stackrel{\text{df}}{=} \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \in \mathbb{R}^{n,2}.$$

Then $\mathbf{A} \begin{bmatrix} m \\ c \end{bmatrix} = (mx_1 + c, \dots, mx_n + c)^\top$ is the vector whose k th entry is $f(x_k)$, and so

$$E_f = \sum_{k=1}^n |y_k - mx_k - c|^2 = \left\| \mathbf{y} - \mathbf{A} \begin{bmatrix} m \\ c \end{bmatrix} \right\|^2.$$

Define $W \stackrel{\text{df}}{=} \{\mathbf{A}\mathbf{b} \mid \mathbf{b} \in \mathbb{R}^{2,1}\}$, the column space of $\mathbf{A}$ — a vector subspace of $\mathbb{R}^{n,1}$. Minimizing E_f over (m, c) is precisely the problem of finding the point in W closest to $\mathbf{y}$, which by Theorem 12.2.4 is the orthogonal projection $P_W(\mathbf{y})$.

Theorem 12.3.2 (Three equivalent characterisations of an LLS solution). *With $\mathbf{y}, \mathbf{A}, W$ as above, the following statements about $\begin{bmatrix} m \\ c \end{bmatrix} \in \mathbb{R}^{2,1}$ are equivalent:*

1. $\begin{bmatrix} m \\ c \end{bmatrix}$ minimizes E_f .
2. The point in W that $\mathbf{y}$ is closest to is $\mathbf{A} \begin{bmatrix} m \\ c \end{bmatrix}$; equivalently, $\mathbf{y} - \mathbf{A} \begin{bmatrix} m \\ c \end{bmatrix} \in W^\perp$.
3. $\mathbf{A}^\top \mathbf{A} \begin{bmatrix} m \\ c \end{bmatrix} = \mathbf{A}^\top \mathbf{y}$ (the normal equations).

Proof. (1) $\Leftrightarrow$ (2): Since $E_f = \|\mathbf{y} - \mathbf{A}[m, c]^\top\|^2$ and $\mathbf{A}[m, c]^\top$ ranges over all of W as (m, c) ranges over $\mathbb{R}^{2,1}$, minimizing E_f is the same as finding the point of W closest to $\mathbf{y}$. By Theorem 12.2.4, this point is unique and characterised by the residual lying in $W^\perp$.

(2) $\Leftrightarrow$ (3): The condition $\mathbf{y} - \mathbf{A}[m, c]^\top \in W^\perp$ says, by Theorem 12.1.6 applied to the spanning set of columns of $\mathbf{A}$,

$$\forall \mathbf{b} \in \mathbb{R}^{2,1} : \quad \langle \mathbf{A}\mathbf{b}, \mathbf{y} - \mathbf{A}[m, c]^\top \rangle = 0.$$

With the standard dot product on $\mathbb{R}^{n,1}$, this becomes $(\mathbf{A}\mathbf{b})^\top (\mathbf{y} - \mathbf{A}[m, c]^\top) = 0$, i.e. $\mathbf{b}^\top \mathbf{A}^\top (\mathbf{y} - \mathbf{A}[m, c]^\top) = 0$, for every $\mathbf{b} \in \mathbb{R}^{2,1}$. This forces $\mathbf{A}^\top (\mathbf{y} - \mathbf{A}[m, c]^\top) = \mathbf{0}$, which rearranges to $\mathbf{A}^\top \mathbf{A}[m, c]^\top = \mathbf{A}^\top \mathbf{y}$. $\square$

Key Result: The Normal Equations

The linear least-squares fit $f(x) = mx + c$ to data points $(x_1, y_1), \dots, (x_n, y_n)$ is found by solving

$$\mathbf{A}^\top \mathbf{A} \begin{bmatrix} m \\ c \end{bmatrix} = \mathbf{A}^\top \mathbf{y}, \quad \mathbf{A} = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

This is a 2×2 linear system in (m, c) , solvable by any standard method; the solution exists and is unique whenever the x_k are not all equal (so that the columns of $\mathbf{A}$ are linearly independent and $\mathbf{A}^\top \mathbf{A}$ is invertible).

Remark 12.3.3. The same machinery extends without modification to higher-degree polynomial fits, multivariate regression, and more general LLS problems: replace the two-column matrix $\mathbf{A}$ with the appropriate design matrix, and the normal equations $\mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{A}^\top \mathbf{y}$ still characterise the optimal coefficients $\mathbf{x}$. This is the linear-algebraic core of ordinary least squares in statistics.

Course wrap-up. From the field axioms in Chapter 1 to the Schrödinger equation and least-squares fit in Chapter 12, MH1201 has built the language of linear algebra in a single thread:

vector spaces and subspaces; bases and dimension; linear transformations and their matrix representations; eigenvalues, eigenvectors, and diagonalization; inner products, orthogonality, and projection. Each layer added geometric or computational content to the algebraic skeleton, and the final two applications show that the same vocabulary describes both the ground state of the hydrogen atom and the line of best fit through a cloud of data points.